

Abstract

This dissertation employs computational methods to address concepts of economic dynamics, statistical equilibrium and economic disequilibrium, complex emergence, and input-output analysis in relation to economic topics of firm and multi-sector growth, structural change, technological innovation, and uneven development from the critical perspective of classical political economy.

The first chapter contributes a statistical approach to Anwar Shaikh's theory of real competition within a single industry, which is disaggregated by firms under alternative micro-foundations on classical-evolutionary lines, with the aim of extending it to a multi-sector framework. Contra the neoclassical view, it is shown that individual optimization (as in utility or profit maximization) is not necessary to reproduce a relevant variety of stylized facts of economic competition and thus can be theoretically dispensed. Economic equilibrium is re-conceptualized in conversation with the literature in econo-physics as an emergent, macro-level outcome of economic competition in the form of stable statistical distributions of micro-economic variables, instead of the conventional notion as a mechanical single-point, rest state.

The second chapter makes a contribution to ecological economics by employing an extension with technological dynamics of the Flaschel-Semmler deterministic classical model of multi-sector growth in order to investigate the speed of the dynamic substitution of carbon sectors by green sectors under directed technical change. The model relies on the cross-dual adjustment between prices and quantities with respect to supply-demand and revenue-cost imbalances in the form of the laws of excess demand and excess profitability and is calibrated using standard OLS regressions on EU KLEMS and WIOD data. Fiscal policy, especially when the carbon tax is accompanied by a green subsidy, greatly accelerates the phase-in of the green sector, with the largest impact at its earliest stages. Without fiscal policy, it is impossible for any economy to reach the IPCC targets in time.

The third chapter provides a literature review of how different theories of trade approach the issue of uneven development and empirically addresses it by applying the machine-learning method of Gaussian mixture modeling on Italy as a case study in economic integration. The analysis of the complex statistical structure of Italian regional inequality corroborates that it is most importantly characterized by a marked dual polarization into North and South as core and periphery, while revealing a more nuanced bi-modal/tri-modal division. Regional rates of gross value added follow a Laplace distribution, while the distributions of sectoral compensation and gross value added follow a power law. After 160 years of political and monetary unification, there is no evidence for any convergence of Italian regions in most of the data.

**ECONOMIC PRODUCTION AS LIFE: A CLASSICAL APPROACH TO
COMPUTATIONAL SOCIAL SCIENCE**

by

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Darwin has interested us in the history of Nature's Technology, i.e., in the formation of the organs of plants and animals, which organs serve as instruments of production for sustaining life.

Does not the history of the productive organs of man, of organs that are the material basis of all social organisation, deserve equal attention?

Karl Marx, *Capital*, Vol. 1

Als meus estimats pares i germà,
i a la preciosa llum de la meva vida, China i Mateu Jamal.

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Stephen Jay Gould once wrote, “I am, somehow, less interested in the weight and convolutions of Einstein’s brain than in the near certainty that people of equal talent have lived and died in cotton fields and sweatshops.” I am equally certain so many invisible workers of the world, who made my own subsistence and thus the present dissertation possible and yet were erased from history, would have written a much better scholarly work, granted the same opportunity and privilege. My first acknowledgement humbly goes to them.

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Contents

List of Figures	viii
------------------------	-------------

List of Tables	x
-----------------------	----------

1 Economic Life: An Evolutionary Approach to Classical Competition	1
1.1 Introduction	1
1.2 On Theories of Competition	4
1.3 Firm Growth as an Evolutionary Process	7
1.4 A Classical-Evolutionary Model of Economic Competition	9
1.4.1 Compact Description	9
1.4.2 Basic Dynamic Setup	12
1.4.3 Firm Profitability	13
1.4.4 Firm Behavior: Pricing	13
1.4.5 Competitive Values for Prices and Equilibrium	15
1.5 Simulation Results	16
1.6 Conclusions	19
2 The Green Transition: Directed Technical Change Towards Decarbonization	29
2.1 Introduction	29
2.2 A multi-sector growth model of the green transition	31
2.2.1 Constant-Technology Dynamics	31
2.2.2 Technical change with process innovation and extinction	33
2.3 Model Calibration	37
2.3.1 Synthetic Data	37
2.3.2 Estimation using real training data	39
2.3.3 Regressions	40
2.4 The Green Transition	41
2.4.1 Comparative Statics of Tax-Subsidy Policy on sector Growth	44
2.4.2 Simulations	47
2.5 Conclusions	53

3	Uneven Development as the Outcome of Economic Integration: Analysis of the Statistical Structure of Italian Economic Geography	55
3.1	Introduction	55
3.2	Uneven development and theories of trade	57
3.3	Italy's Southern Question	61
3.4	A Gaussian Mixing Model of the Geographical Division of Labor	63
3.5	Statistical Analysis of the Italian Economic Geography	72
3.5.1	GDP per capita relative to the EU average	73
3.5.2	Cluster Analysis of Key Indicators on the Italian Economic Geography	78
3.5.3	Regional Classification	83
3.5.4	Multi-Variate Cluster Analysis	83
3.5.5	Sectoral Cluster Analysis	84
3.6	European Economic Integration	84
3.7	Conclusions	88
	Bibliography	99

List of Figures

1.1	Profit Rate Equalization	5
1.2	Changes in Selling Prices vs Changes in Unit Labor Costs	6
1.3	Simulation with one-off demand shock	21
1.4	Spectrum in profit rates following cost differences	22
1.5	Simulation under Constant Technology 1	23
1.6	Simulation under Constant Technology 2	24
1.7	Distributions of Micro-Economic Variables	25
1.8	Kinked Demand Curve	26
1.9	Profit Rate Distribution of Ensemble of Simulations	26
1.10	Distribution of Growth Rates	27
1.11	Price-Inventory Cycle	27
1.12	Stability of the Pricing Process	28
2.1	Constant-Technology Dynamics for the 2003 US Direct Requirements Matrix	36
2.2	Relative price and quantity changes with respect to excess demand and excess unit profit	38
2.3	Sector Law of Excess Demand for 37 sectors, Germany	41
2.4	Aggregate Law of Excess Demand for 6 countries	42
2.5	Aggregate Law of Excess Profitability for 6 countries	43
2.6	Green vs Carbon Energy Costs	48
2.7	Time Evolution of C19 Sector	49
2.8	Simulation of the Green Transition for sector C19 without fiscal policy, Germany	50
2.9	Simulation of the Green Transition for sector C19 with fiscal policy, Germany	51
2.10	Duration of the Green Transition for sector D, selected countries	52
3.1	Model-Based Clustering for Economic Indicators	66
3.2	Model-Based Clustering for Health and Poverty Indicators	67
3.3	BIC Score For Different Cluster Numbers, Economic Indicators	68
3.4	Geographical Variation: Optimal Number of Clusters	69
3.5	Optimal Number of Clusters	69
3.6	Distribution of BIC Score Differences between Optimal and 1 Number of Clusters	71
3.7	Time Series of Italian Regional Income Data relative to the EU Average and its corresponding frequency distributions	74

3.8	BIC Scores for Gaussian Models (of GDP per capita relative to the EU average) of different number of components with Equal (E) and Unequal (V) variance	75
3.9	Different Gaussian mixture models for GDP per capita relative to the EU average, with their corresponding BIC scores	76
3.10	Model-Based Clustering for Economic Indicators	79
3.11	Laplace Distribution for the Rate of Gross Value Added	80
3.12	Power Law Distribution for Gross Value Added and Compensation	81
3.13	Maps of Household Disposable Income and Unemployment Rate	89
3.14	Maps of Rate of GVA, Labor Productivity, and Gross Fixed Capital Formation	90
3.15	Correlation between Economic Indicators	91
3.16	Multi-Variate Classification	92
3.17	Multi-Variate Classification	93
3.18	Model-Based Clustering for Sectoral Indicators	94
3.19	Cluster Classification for Sectoral Labor Shares	95
3.20	Cluster Classification for Sectoral Labor Productivity	96
3.21	European Distributions of Regional Household Disposable Income	97
3.22	European Maps of Regional Household Disposable Income, Unemployment Rate, and Gross Fixed Capital Formation	98

List of Tables

1.1	Parameters and Initial Conditions of the Model	9
2.1	2003 US Direct Requirements Matrix	35
2.2	OLS for adjustment parameters $\delta_{p,i}$, $\delta_{x,i}$, and γ	38
2.3	OLS for aggregate adjustment coefficients	44
3.1	List of Socio-Economic Indicators for Italy	64
3.2	List of Socio-Economic Indicators for Italy (Continued)	65
3.3	Table of Indicators with 1 as Optimal Cluster Number	70
3.4	BIC Score Differences between Optimal and 1 Number of Clusters	70
3.5	Key Indicators on the Italian Economic Geography	78

Chapter 1

Economic Life: An Evolutionary Approach to Classical Competition

This paper proposes a simple model of firm growth along classical, computational and evolutionary lines that attempts to capture Anwar Shaikh's classical theory of real competition within a single industry. Equilibrium in prices is not presumed, but dynamically attained by the systematic, decentralized operation of economic competition over time. Economic competition is viewed as a statistical process that results in the dynamic equalization of prices over time. The model is disaggregated by profit-seeking, price-setting firms under an evolving heterogeneous technology operating under alternative micro-foundations, without any recourse to utility or profit maximization, but instead based on reproduction as fundamental closure. In order to relate macro-level patterns to micro-level observations over time, the theoretical framework re-conceptualizes equilibrium as a stationary property of statistical distributions of micro-economic variables, which allows a more realistic empirical description of the economy beyond the conventional notion of equilibrium as a single-point, rest state. The model generates a variety of stylized facts as macro-level emergent outcomes of the competitive process: price dispersion around the competitive value, a spectrum of profitability differentials reflecting cost differences, evolutionary selection of the lowest-cost firm, tent-shaped distributions of profit and growth rates that are consistent with the empirical evidence, endogenous cost-plus markup pricing, and business cycles in prices and inventories.

1.1 Introduction

The question of economic equilibrium is fundamental in economic theory. Equilibrium, understood as the delicate balance of supply and demand, is often accepted as a matter of fact, an achieved end result, or an *ad hoc* assumption, leading to frame the problem in terms of comparative statics. An alternative is to investigate it as the actual, dynamic outcome of the market forces of economic competition over time in an out-of-equilibrium situation. In particular, the question of the dynamic convergence of a competitive economy to equilibrium, either allowing trade outside of it or not, is still debated more than a century after Walras

set the problem. Tâtonnement processes replicating a centrally planned economy were eventually found to be generally unstable except under very restrictive conditions [Scarf, 1989], while non-tâtonnement processes of the Edgeworth or Hahn form reached limited success [Fisher, 1989b; Gintis and Mandel, 2012; Fisher, 2013]. In terms of computational complexity, at least good approximations of equilibrium cannot often be reached in reasonable (that is, polynomial) time [Cheung et al., 2019]. In general, equilibrium is not computable [Velupillai, 2005]. For Franklin Fisher, the problem of the stability of equilibrium constituted an important *lacuna* at the center of microeconomic theory, casting doubt on the adequacy of the utility theory of value [Fisher, 1989a, 2013]. Critics ponder whether general equilibrium theory is still relevant for economic analysis [Kirman, 1989, 1992; Rizvi et al., 2006; Colander, 2018], including dynamical stochastic general equilibrium [Caballero, 2010; Blanchard, 2016; Romer, 2016].

This paper provides a dynamical model of competitive equilibrium that is based instead on the central notion of reproduction, not optimization, under alternative micro-foundations defined along classical, statistical and evolutionary lines. Following Anwar Shaikh’s classical theory of real competition, this paper sets to explore how the dynamic, decentralized process of economic competition within a single industry unfolds over time in order to observe then how a notion of equilibrium can emerge in that context. Competition is understood not as an achieved end result but as a disorderly dynamic process that still manages to produce stable macro-level emergent outcomes in the aggregate. Competition between capitals does not bring a smooth process of adjustment and convergence towards equilibrium prices and quantities, but disequilibria and long-lasting deviations from the center of gravity [Semmler, 1981]. Production is like life in the sense that the economy is conceived as an ecosystem in the form of a self-reproducing set of interdependent processes of production or firms [Sraffa, 1960]. In order to survive in the marketplace over time, competitive firms require realized above-average profits at the expense of each other. In the classical view, competition is the direct result of the self-expansion of capital [Semmler, 1981]. From an evolutionary-classical viewpoint, profitability is what drives the stochastic, interdependent growth of firms, which in the model depends on the probabilistic choices of all firms over price (firms are price-setting) and industry (firms are profit-seeking) under competitive conditions.

The issue of differences in profit rates across firms and industries is “a, perhaps the, fundamental issue in industrial organization” [Megna and Mueller, 1991, p.632]. While persistent profit differences are the norm, attempts to explain them in terms of market power, collusion, barriers to entry, differences in efficiency, and even alternate measures of profit and capital including R&D, have generally been unsuccessful [Shaikh, 2016, p.290]. Instead of perfect homogeneity in profitability across firms and industries, the model assumes the profit rates of individual firms dynamically follow a well-defined classical structure as articulated by Anwar Shaikh’s theory of real competition [Shaikh, 2016, ch.7]. The real-competition framework features two levels of abstraction that structure firm profitability through the systematic operation of competition over time: within a single industry –the focus of this chapter–, cost-cutting price competition among firms leads to the equalization of prices around their competitive value, the existing spectrum in costs produces intra-industrial

profit rate differentials, and thus differential growth leads to the evolutionary selection of the lowest-cost technology over time. Between industries, the dynamical process of the free mobility of profit-seeking capital across lines of production equalizes the profit rates of their lowest-cost capitals over time. Instead of perfect equality of supply and demand at any point of time, supply and demand are roughly (i.e. “turbulently”) equalized over time in ceaseless fluctuations due to the constant inflow and outflow of capital between industries. Business cycles result from the dynamic gravitation of the actual prices and quantities of firms and industries over time around their competitive values as defined by technology and demand.

The model articulates Shaikh’s theory of real competition within a single industry by imposing on individual firm profitability an intra-industry price adjustment at the firm level (with respect to the average industry price) in conversation with the classical and neoclassical literature on dynamical adjustments [Morishima, 1981; Mas-Colell, 1986; Flaschel and Semmler, 1987; Fisher, 1989a; Flaschel, 1990; Duménil and Lévy, 1993]. Unlike the latter, an important contribution of our model is that it is not disaggregated by industrial sectors, but by the microeconomic behavior of individual heterogeneous firms, which requires to reconceptualize equilibrium under a statistical view, following the literature on statistical equilibrium in economics and econophysics [Steindl, 1965; Farjoun and Machover, 1983; Mirowski, 1989; Foley, 1994; Drăgulescu and Yakovenko, 2001; Cockshott et al., 2009; Scharfenaker and Foley, 2017; Shaikh, 2020]. Under the alternative approach, decentralized, probabilistic behavior of minimal-rationality, satisficing firms at the micro-level produces emergent statistical regularities in the aggregate under the systematic operation of competition over time. In this context, economic equilibrium is no longer conceived as a single-point, rest state, but as a statistical distribution of micro-economic variables that becomes stationary due to the systematic operation of the competitive process.

Under alternative micro-foundations, competitive firms are seen as profit-seeking, price-setting, cost-cutting, and generating variations in technology, instead of anomalies or market frictions as in the context of imperfect competition [Shaikh, 2016, p.19]. In the competitive struggle for survival within a single industry, price-setting, profit-seeking firms only choose their price (their output is determined by their capital stock and average costs), which respectively have an interdependent impact on the intra-industry adjustment on profitability and thus growth. Firms set disequilibrium prices in a trial-and-error, decentralized, interdependent fashion that mimics groping or tâtonnement on the basis of minimal rationality for survival, on the lines of Herbert Simon’s “satisficing” principle of bounded rationality [1956]. The realized profitability of one firm (and thus growth) depend on the probabilistic choices in prices that all firms make following a minimal-rationality for survival that is fundamentally driven by profit. This introduces a systematic difference between what firms produce and what they manage to sell (that is, between expected and actual outcomes) that constitutes the source of uncertainty in the model.

Without any recourse to optimization, this model of real competition, which employs agent-based simulations at the base level, reproduces a variety of stylized facts as macro-level emergent outcomes of micro-economic behavior: price dispersion and tent-shaped distributions for the rates of profit and growth in agreement with the empirical evidence [Bottazzi and

Secchi, 2006; Coad, 2009; Bottazzi et al., 2010; Alfarano et al., 2012; Kaplan and Menzio, 2015; Williams et al., 2015], endogenous cost-plus markup pricing (where the plus is related to the residual, profit, which can also be negative), downward-sloping demand curves, a kinked demand curve that is reminiscent of oligopolistic competition, and the dynamic equalization of prices within a single industry. As expected within a single industry, a shock in technology as the reduction in production costs reduces the competitive price; a positive shock in demand increases the competitive price. In the model, the competitive price is an emergent property of single-industry competition.

1.2 On Theories of Competition

In the conventional story of perfect competition, the balance of supply and demand is achieved at instantaneous speed as the product of the decisions of utility-maximizing consumers and an infinite set of identical price-taking, profit-maximizing firms of identical technology at the competitive price, computed by the Walrasian auctioneer as an index of scarcity, under instantaneous tâtonnement. In this scenario, all firms within an industry would feature the same profit rates and the same price –the “law of one price”– and competition between industries would equalize profit rates due to the free mobility of capital. However, the empirical evidence shows systematic violations of the law of one price in the form of persistent price dispersion, as well as frequent price changes [Engel and Rogers, 1996; Haskel and Wolf, 2001; Hopkins et al., 2006; Crucini and Shintani, 2008; Kaplan and Menzio, 2015]. Further, the distributions of prices are more peaked than Gaussian [Kaplan and Menzio, 2015] and the distributions of price change sizes tend to be quite irregular and have no clear shape [Cavallo and Rigobon, 2011]. Firm profitability also shows persistent dispersion, but its distribution is very stable over time and shows a clear double-exponential (tent) shape [Bottazzi and Secchi, 2006; Coad, 2009; Bottazzi et al., 2010; Alfarano et al., 2012; Williams et al., 2015; Scharfenaker and Foley, 2017]. Attempts have been made to characterize micro-level firm behavior that induces the observed tent-shaped distribution of growth rates, for instance Gibrat’s law that considers growth to be independent of firm size, which generates a log-normal distribution [Gibrat, 1931; Kalecki, 1945; Ijiri and Simon, 1967], but they have been generally unsuccessful.

In reality, firms are not price-takers as the theory of perfect competition establishes, but in fact set prices following a cost-plus rule (i.e. sum of average costs plus some conventional net profit margin as the markup) [Hall and Hitch, 1939; Lee, 1999], show increasing returns to scale [Sraffa, 1926; Andrews, 1964] and are finite in number. Rather than reflecting market power in line with post-Keynesian approaches, in the classical tradition this markup or residual over costs is in fact related to the profit rate and is endogenous to the operation of economic competition: profits are not fixed over time, they are not certainly guaranteed but subject to fundamental uncertainty, and may even be negative. The first systematic examination of actual business studies by the Oxford Economists Research Group (in 1933) revealed to what extent common firm behavior departs from the predictions of perfect-competition theory, which came as a shock to many academic economists [Harrod, 1972, p.ix].

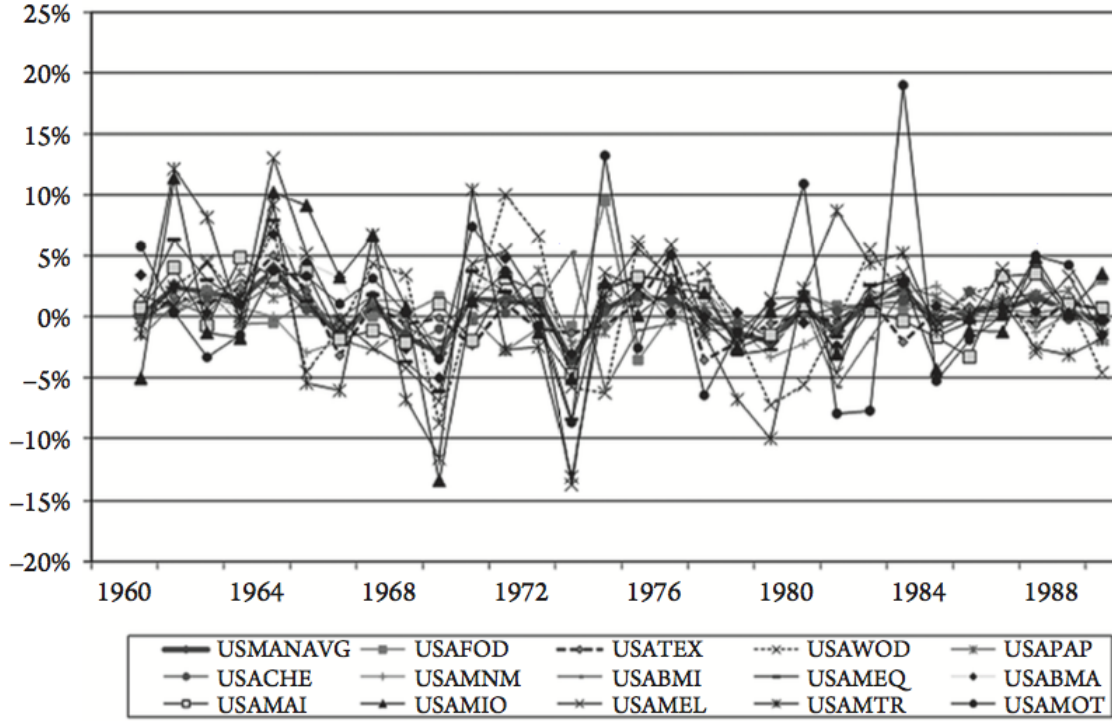


Figure 1.1: **Profit Rate Equalization: US Manufacturing Incremental Rates of Profit, 1960-1989** [Shaikh, 2016, p.304]

Within an industry, individual firms vary in technology and thus have different operating costs (and hence different markups) [Salter, 1969], but are forced by competition to keep their prices in line with those of the price leader, which will be the lowest-cost firm. Hence, the net margins set of price-followers are simply those which kept their own prices in line [Hall and Hitch, 1939]. This generates a spectrum of profitability based on cost differentials [Shaikh, 2016, p.268]. For Andrews and Brunner, the OERG findings described the actual behavior of competitive, profit-driven firms, which see themselves in a constant competitive struggle that forces them to equalize prices within an industry [Andrews and Brunner, 1962; Andrews, 1964; Andrews and Brunner, 1975]. Firms constantly jockey for room by cutting prices to enhance their market shares, but since these maneuvers are limited by their production costs, they constantly seek to create and adopt lower-cost technologies [Shaikh, 2016, p.274]. As observed in figure 1.2, this implies that changes in unit labor costs drive changes in selling prices [Salter, 1969].

Such anomalies are often taken as empirical evidence supporting the more realistic theory of monopolistic competition [Chamberlin, 1949; Robinson, 1969] in the form of frictions from the ideal theory of perfect competition as benchmark. For instance, price dispersion is explained in terms of search costs by consumers under imperfect information and to menu costs by firms that cannot adjust prices instantaneously [Barro, 1972; Weitzman, 1979; Hopkins et al., 2006]. Under the classical view of economic competition, price-setting,

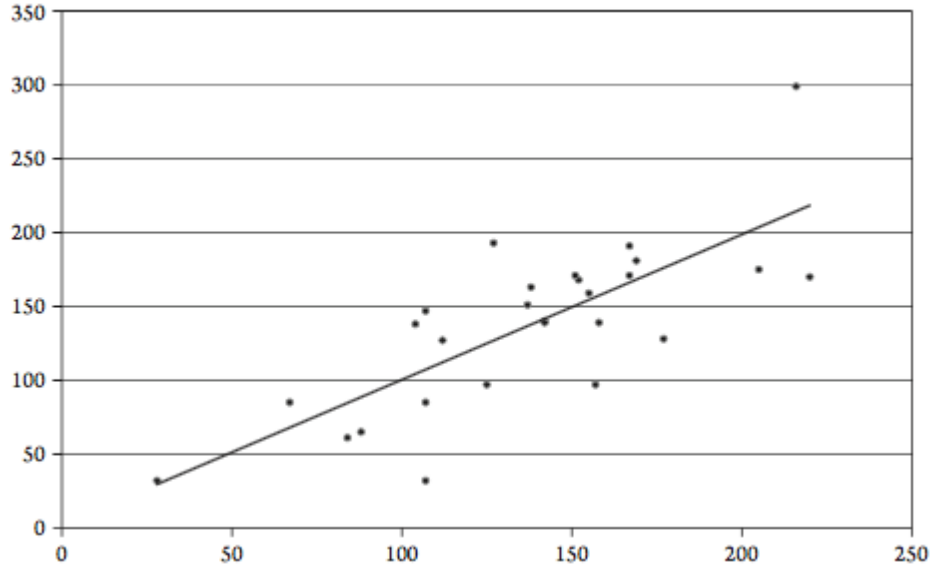


Figure 1.2: **Change in Selling Price versus Change in Unit Labor Cost, US 1923-1950 (Ratio of Each Variable in 1950 to its 1923 Value)** [Shaikh, 2016, p.287] extracted from [Salter, 1969, p.197]

cost-cutting behavior and technology variations are instead seen as intrinsic to competition, not as anomalies or imperfections far from an ideal benchmark [Shaikh, 2016, p.19]: price dispersion is seen as the outcome of the process of price competition within a single industry itself. Constant efforts of competitive firms to undercut each others' prices induces a "law of correlated prices" (as opposed to the neoclassical law of one price), where the lowest-cost firm is often the price leader and higher-cost firms become the price followers [Shaikh, 2016, p.262].

In the classical context, competition is understood not as an achieved end result but as a disorderly dynamic process that still manages to produce stable macro-level emergent outcomes in the aggregate. Competition between capitals does not bring a smooth process of adjustment and convergence towards equilibrium prices and quantities, but disequilibria and long-lasting deviations from the center of gravity [Semmler, 1981]. Actual, market prices systematically diverge over time from their competitive or equilibrium values [Shaikh, 2016, pp.16-17], which are determined by the production structure and distribution under Sraffa's classical theory of price [Sraffa, 1960]. Firm and industry prices and quantities ceaselessly gravitate around their competitive values as centers of gravity in the dynamic and turbulent process of economic competition, fraught by over- and undershooting, subject to fundamental uncertainty over the systematic divergence between expected and realized outcomes [Harris, 1988]. Following the classical conception of the long-period method [Foley, 2011], a notion of equilibrium emerges then as a central statistical time-averaging tendency of micro-economic variables, an emergent regular outcome at the macro-level that is produced by the generalized operation of economic competition at the micro-economic level. Between

industries, profit rates are never perfectly uniform at any particular point in time. Instead, inter-sectoral profit rates roughly equalize over time due to the free mobility of capital between lines of production in constant search of higher profitability [figure 1.1]. However, the dynamical process of capital mobility is fraught by constant over- and undershooting (i.e. it is “turbulent”) so that perfect equality is never realized as an achieved end result.

1.3 Firm Growth as an Evolutionary Process

In agreement with the literature in evolutionary economics, the classical theory of economic competition does not assume that firms maximize profits over the whole technology space [Nelson and Winter, 1974, 1978, 2009]. For Herbert Simon, firms ‘satisfice’ instead by using simple rules of thumb to search for available alternatives until a threshold is met, because their rationality is in fact bounded by information-gathering costs [Simon, 1956]. Firms are not hyperrational in the sense that they know all production plans beforehand and choose the one that maximizes their profits; instead, they are subject to fundamental uncertainty, so realized outcomes diverge from their expected outcomes: profits are not guaranteed. For Alchian, ‘profit maximization’ in the context of fundamental uncertainty is in fact “meaningless as a guide to specifiable action” [Alchian, 1950]. In an early paper, Gary Becker argued that it is the market that behaves rationally, while individual behavior only needs to be subject to budgetary constraints even if it is irrational [Becker, 1962].

While attempting to justify the supposedly profit-maximizing behavior of firms, neoclassical authors do not appeal to empirical evidence or general equilibrium, but employ (and actually misunderstand) an extraneous argument to conventional economic theory: Darwin’s natural selection. For instance, in his *Theory of Industrial Organization*, Jean Tirole lays out the selectionist argument supporting profit maximization in the words of Scherer [Tirole, 1988]:

When forced into the trenches on the question of whether firms maximize profits, economists resort to the ultimate weapon in their arsenal: a variant of Darwin’s natural selection theory. Over the long pull, there is one simple criterion for the survival of a business enterprise: Profits must be nonnegative. No matter how strongly managers prefer to pursue other objectives, and no matter how difficult it is to find profit-maximizing strategies in a world of uncertainty and high information costs, failure to satisfy this criterion means ultimately that a firm will disappear from the economic scene. [Scherer, 1980, p.38]

The analogy with natural selection identifies biological progeny with economic profits. “Nonnegative profits” are understood in the neoclassical sense, that is, as above-normal profits where the normal profit rate is identified as the interest rate. Sustained failure to satisfy the criterion of above-normal profits will indeed induce the long-term disappearance of the firm, just like the sustained production of below-average progeny implies the long-run extinction of any genetic population. However, this does not imply necessarily that such profits (or progeny) will be maximal given all possible production plans (or all genetic mutations),

or that all firms in the market (or species) employ the same profit-maximizing production plan (or feature the same genetic profile). In fact, natural selection fundamentally relies on the *differential* survival and reproduction of individuals; natural selection in the marketplace critically depends on the heterogeneity of firms and their growth rate differentials. Biological organisms do not maximize their progeny, but instead only require that their progeny is *greater* than their biological competition over time – a *satisficing*, not optimizing, criterion.

In this sense, the selectionist argument that is commonly employed to justify the profit maximization hypothesis requires that, for natural selection to occur, at a particular moment in time many firms will show suboptimal behavior by exhibiting below-normal or even negative profitability, just like many maladaptive organisms die without progeny: firms will simply disappear eventually or go back to above-normal profitability if it manages to correct its production plans. In this sense, this paper supports Alchian’s view that the requirement for firm survival is only excess profits to be “positive” in the neoclassical sense (i.e. above-normal in the classical sense), not necessarily maximal [Alchian, 1950]. In order to maintain its market share at each time period, firms require that their revenues cover their production costs plus the average profit margin in their industry, which plays the role of an endogenous markup. Profits are fed back in the form of investment in the next time period in a sequential cumulative process that regulates firm growth, because capturing a larger share of profits allows the firm to grow larger in the next time period at the expense of its competitors, which conversely grow smaller. Profitability drives the growth of firms; sustained, above-normal profitability ensures their reproduction over time. Hence, profitability can be considered as the economic equivalent of evolutionary fitness in biology.

Firms will dominate the marketplace if they have a cost advantage in the form of lower production costs or a differentiation advantage [Porter, 1998] and attempt to maintain its market share, even resorting to anti-competitive monopolistic practices. The brief, temporary existence of firms dominating particular industries does not imply that the conventional process of economic competition no longer applies; it just implies they possess the competitive advantage at the current moment over their competition, regardless of scale or market concentration. As soon as new firms enter the market with lower production costs or better quality, the older firms, no matter their size, will start losing market share as soon as they lose their competitive advantage. The history of capitalism is full of examples of formidable firms that once cornered particular markets and today no longer exist, just like the history of biological life is full of extinct fossils.

Such an understanding of economic competition is fundamentally evolutionary and biological, similarly in line with the insights of evolutionary economics [Nelson and Winter, 2009]. For P.W.S. Andrews and authors in the post-Keynesian tradition such as Lee, the firm is an organic entity, “living” in historical time, whose ultimate objectives are survival and, if possible, growth [Andrews, 1964; Lee, 1999]. In his 1960 seminal analysis, Sraffa also emphasized the interdependence of processes of production as a fundamental requirement for their self-reproduction as a whole over time [Sraffa, 1960]. In sum, it is reproduction – in the satisficing sense of survival over time among competitive, yet interdependent, entities, and not optimization – that constitutes the regulating principle in economic competition,

Parameters		
Number of Firms	N	
Constant Aggregate Mass of Capital Stock	M	
Aggregate or Average Profitability	$\bar{r}$	$\langle r \rangle = \sum_i r_i \frac{M_i}{M} = \bar{r}$ holds
Intra-Industry Competition Intensity	α	
Price Step	θ	
Normal Inventory/Sales Ratio	u_{eq}	
Inventory/Sales Half-Band	u_d	
Initial Conditions		
Vector of Capital Stocks	$M_i \sim N(1/N, 0.1/N) \times M$	$\sum_i M_i = M$ holds
Vector of Average Costs	ac_i	
Vector of Prices	$p_i = (1 + \bar{r})ac_i N(1, 0.01)$	
Vector of Inventories	$u_{eq} \frac{M_i}{ac_i}$	

Table 1.1: **Parameters and Initial Conditions of the Model**

producing emergent statistical regularities of micro-economic variables at the macro-level over time.

1.4 A Classical-Evolutionary Model of Economic Competition

1.4.1 Compact Description

The realized mass of revenue is also constant, $M' = (1 + \bar{r})M$, as the mass of profit $M' - M = \bar{r}M$.

The initial vector of capital stocks is extracted from a Gaussian distribution centered around M/N and rescaled so their sum equals total mass of capital stock M .

1. Profitability

$$r_i = \bar{r} + \alpha \varepsilon_i; \quad \sum_i \varepsilon_i M_i = 0 \quad \text{holds} \quad (1.1)$$

2. Intra-Industry Deviation

$$\varepsilon_i = 1 - \frac{p_i}{\langle p \rangle} \quad \begin{cases} \varepsilon_i < 0 & \text{if } p_i > \langle p \rangle \\ \varepsilon_i > 0 & \text{if } p_i < \langle p \rangle \end{cases} \quad (1.2)$$

3. Growth

$$g_i = \frac{\varepsilon_i}{1 + \bar{r}} = \frac{r_i - \bar{r}}{1 + \bar{r}}; \quad \langle g \rangle = 0 \quad \text{holds} \quad (1.3)$$

4. Output Produced

$$X_i = \frac{M_i}{ac_i} \quad (1.4)$$

5. Output Sold

$$S_i = (1 + r_i) \frac{M_i}{p_i} \quad (1.5)$$

6. Inventory Change

$$inv_i(t) = inv_i(t-1) + X_i - S_i \quad (1.6)$$

7. Price Rule

$$p_i(t+1) = \begin{cases} p_i(t) & \text{if } (u_{eq} - u_d) < \frac{inv_i}{S_i} < (u_{eq} + u_d) \\ p_i(t) - \theta \times N(1, 0.025) & \text{if } \frac{inv_i}{S_i} > (u_{eq} + u_d) \\ p_i(t) + \theta \times N(1, 0.025) & \text{if } \frac{inv_i}{S_i} < (u_{eq} - u_d) \end{cases} \quad (1.7)$$

In the single-industry model, the only variable of choice is price, which firms adjust with respect to their individual inventory/sales ratio. In the aggregate, this micro-level behavior generates a double-exponential distribution for prices, profit rates, and growth rates.

Definition 1. The economy is described by $i = 1, \dots, N$ firms of capital stock M_i . The micro-level characteristics of the firms, which are treated as random real variables, are capital stock M_i , average costs under constant returns to scale ac_i (capital/output ratio), price p_i , output produced X_i , output sold S_i , inventory inv_i , profitability r_i , and growth g_i . All capital is circulating and hence depreciation $\delta = 1$ (corn model).

The sum of capital stock is assumed to be constant at M : average firm growth is thus assumed to be zero, which imposes firms to grow at the expense of each other under economic competition. In the classical view, competition is the direct result of the self-expansion of capital [Semmler, 1981, p.41]. The mass of profit $\bar{r}M$ is also constant over time, as determined by the average profit rate $\bar{r}$, which is an exogenous parameter as well as the average profitability weighted by firm capital shares

$$\bar{r} \sum_i^N M_i = \sum_i r_i M_i$$

Profitability, which drives the growth of competitive firms at the micro-level, is the linchpin of the model and is understood in a statistical sense, following the literature in econophysics. In order to survive in the marketplace (the reproduction condition to be satisfied), competitive firms only require above-average, not maximal, profitability $r_i > \bar{r}$, which regulates their growth g_i over time. If firm profitability is above-average (below-average), the firm will expand (contract). The profitability of each firm $r_i(p)$ is defined as a statistical deviation from average profitability in the form of an intra-industry firm-specific price adjustment

and an inter-industry sector-specific quantity adjustment. These adjustments depend on the minimal-rationality probabilistic choices of each firm following a search process over price and line of production in relation to the choices of all other firms. They attempt to capture the classical profitability structure of Shaikh’s theory of real competition:

- within a single industry, the process of price competition by price-cutting firms to attract demand at the expense of each other equalizes prices in the form of a statistical dispersion around the competitive price (the classical “law of correlated prices” as opposed to the empirically elusive neoclassical law of one price [Engel and Rogers, 1996; Haskel and Wolf, 2001; Hopkins et al., 2006; Crucini and Shintani, 2008; Kaplan and Menzio, 2015]), a spectrum in profitability and thus growth arises due to cost differentials, and the lowest-cost (or “regulating”) capital is selected over time.
- between industries, the free mobility of profit-seeking capital induces the equalization of profit rates across the lowest-cost capitals of each industry over time. Inter-industry profit rates are not uniform at any particular time, but roughly equalize over time. This results for each industry in the dynamic gravitation of market prices around their competitive values and industry supply around their demand (equalization of supply and demand over time) following inflows and outflows of capital – in the form of business cycles.

The intra-industry profitability adjustment attempts to capture price competition within a single industry. If the own firm’s price is lower (higher) than the industry average, it will attract larger demand from its competition so that its deviation from the industry average profitability will be positive. For the sake of simplicity, the model does not capture the potential ambiguity on the sign of the profitability deviation that can arise in price-cutting behavior between a price reduction and an increase in quantity sold over profit. Instead, the quantity sold and quantity produced are determined by accounting identities: the former is a function of firm profitability, capital stock, and price, and the latter is a function of capital stock and the capital/output ratio; their difference is added or subtracted to the inventory. Price-setting firms vary their prices in random small steps in relation to their own observed demand, which is given by their inventory/sales ratio: they decrease (increase) their price if their inventory/sales ratio is too high (low). Firms will thus grow if they can reduce their price enough with respect to the average industry price, but there is an ultimate lower limit to their price reductions, which is to obtain enough revenue to cover their production costs. Hence, the firm with the lowest-cost technology (the regulating capital) will be the price leader. The competitive industry price around which firm prices turbulently gravitate through over- and undershooting will be the production costs of such firm times the normal profit rate as a markup.

At each time period, firms produce an output X_i under a technology structure of constant returns to scale, which they attempt to sell as sales S_i . In the model, realized output (output sold) S_i randomly diverges from output planned (produced) X_i . The stochastic nature of competitive firm behavior creates a statistical dispersion on profitability r_i and thus growth g_i that is consistent with the empirical evidence in following a tent-shaped distribution

[Bottazzi and Secchi, 2006; Coad, 2009; Bottazzi et al., 2010; Alfarano et al., 2012; Williams et al., 2015]. Price dispersion also follows a tent shape, in agreement with the empirical evidence [Kaplan and Menzio, 2015].

1.4.2 Basic Dynamic Setup

Definition 2. Formally, the circuit of capital can be written as:

$$M_i(t) - \underbrace{C}_{\text{input}} \dots \underbrace{P}_{\text{Production}} \dots \underbrace{X_i}_{\text{output produced}} \rightarrow \underbrace{S_i}_{\text{output sold}} - \underbrace{(1 + r_i)M_i}_{\text{revenue}} \rightarrow M_i(t + 1) \quad (1.8)$$

Firm Profitability and Growth

The model assumes that the sum of capital stocks M_i has constant value M (an exogenous parameter) and realizes a constant mass of profit $\bar{r}M$ at each time period, with the average profit rate $\bar{r}$ as another exogenous parameter. Hence:

Assumption 1. *Average firm growth $g_i(t)$ is zero.*

$$\langle g_i(t) \rangle = 0 \quad (1.9)$$

Assumption 2. *The average firm profit rate $r_i(t)$ is $\bar{r}$.*

$$\sum_i^N (1 + r_i(t))M_i(t) = (1 + \bar{r})M \quad (1.10)$$

Definition 3. Firm Growth By accounting definition, the capital stock at the next time period $M_i(t + 1)$ is defined by the growth rate $g_i(t)$ and capital stock $M_i(t)$:

$$M_i(t + 1) = M_i(t)(1 + g_i(t)) \quad (1.11)$$

Assumption 3. Reproduction Condition

Profitability $r_i(t)$ drives growth $g_i(t)$:

$$g_i = g_i(r_i(t)) \quad (1.12)$$

Survival requires above-normal profits over time. *Growth $g_i(t)$ depends on the difference on the profit rate from the reference rate $\bar{r}$, which is the average rate of profit:*

$$g_i(t) = f(r_i(t) - \bar{r}) \quad (1.13)$$

The function f is further assumed to be increasing and $f(0) = 0$: if the firm profit rate $r_i(t)$ is above (below) average profitability $\bar{r}$, the firm will grow (decline) at the expense of its competitors in the next time period.

Firms with above-normal (below-normal) profitability expand (contract) in the next time period. The growth rate may have the following functional form for f :

$$g_i(t) = \frac{r_i(t) - \bar{r}}{1 + \bar{r}} \quad (1.14)$$

1.4.3 Firm Profitability

In this model, an intra-industry price adjustment is imposed on profitability r_i following the classical dynamical structure of economic competition [Flaschel and Semmler, 1987]. Within a single industry, price-cutting competition among price-setting firms induces a “law of correlated prices” in the form of a price dispersion around the competitive value as a central tendency.

Assumption 4. Price Equalization *Within a single industry, the price adjustment on firm profitability at the firm level will be positive (negative) if the firm’s price p_i is below (above) the industry average price $\langle p \rangle$.*

In other words, the price adjustment reflects that a firm will gain above-normal profits within a single industry if it under-cuts the average price and thus its competition in that line of production.

Definition 4. Firm Profitability Firm profitability r_i thus is defined by the following regulating function, where an intra-industry price adjustment is imposed at the firm level with respect to the average industry price:

$$r_i = \bar{r} + \alpha \left[1 - \frac{p_i}{\langle p \rangle} \right] \quad (1.15)$$

where α is an adjustment coefficient that captures the relative intensity of intra-industry competition.

Firm profitability will be equal to the average profit rate $\bar{r}$ and growth will be zero if the firm’s price p_i equals industry average price $\langle p \rangle$.

1.4.4 Firm Behavior: Pricing

Firms set prices entirely by trial and error in blind attempts to undercut its competitors while preserving its continued reproduction in the marketplace – that is, actual *tâtonnement*: “to search or attempt to find something in the dark, or, as a blind person, by feeling; to move about hesitatingly, as in darkness or obscurity; to feel one’s way, as with the hands, when one can not see”¹. Empirical research in the business literature lists up to 19 different pricing strategies, with most firms employing 3 or 4 of them [Rao, 2009, pp.18-19]. It is rather more cavalier to assume minimal rationality (in the satisficing sense) on firm behavior

¹Wiktionary definition.

by making price-setting probabilistic under simple rules of thumb instead of imposing very specific behavior in their pricing strategies – in a way that it mimics a price-setting iterative process of step-wise groping.

In the model, firms vary prices by striving to maintain their own inventory/sales ratio (which is taken as a proxy of individual demand) within a particular a band in order to ensure its survival over time. If the firm's inventory/sales is too high, this signals to the firm that its price is too high with respect to its competition so the firm decreases its price to attract more demand. If the firm's inventory/sales ratio is too low, this signals to the firm that its price is too low in order to cover its production expenses, so it increases its price to ensure its survival.

Assumption 5. Price Rule *Firms increase and decrease prices a step of length θ with a Gaussian random error, attempting to maintain the inventory/sales ratio around a normal value u_{eq} within a band of length $2u_d$.²*

$$p_i(t+1) = \begin{cases} p_i(t) & \text{if } (u_{eq} - u_d) < \frac{inv_i}{S_i} < (u_{eq} + u_d) \\ p_i(t) - \theta \times N(1, 0.025) & \text{if } \frac{inv_i}{S_i} > (u_{eq} + u_d) \\ p_i(t) + \theta \times N(1, 0.025) & \text{if } \frac{inv_i}{S_i} < (u_{eq} - u_d) \end{cases} \quad (1.16)$$

In reality, the empirical distribution of price change sizes [Cavallo and Rigobon, 2011] is very irregular and not double-peaked as we impose, although size step θ can be replaced by a uniform distribution to approximate it better. The rationale of the modeling choice is to characterize tâtonnement by price-setting firms in a decentralized way, which is also amenable to the empirical description of actual price behavior in firms by marketing studies [Rao, 2009]. Asymmetry in the price response function can be further added by including a price decrease if the capital share decreases $M_i(t-1) > M_i(t)$ and/or setting different threshold parameters for the price increase and decrease (e.g. making firms less responsive to high demand than low demand, i.e. they are more probable to decrease than to increase prices).

Accounting Identities

Simple accounting identities define output produced X_i , output sold S_i , and inventory inv_i ³:

Definition 5. At each time period t , firm M_i produces output X_i at average cost ac_i under constant returns to scale (so that ac_i is also the capital/output ratio):

$$X_i = \frac{M_i}{ac_i} \quad (1.17)$$

Definition 6. The source of uncertainty in the model is that output sold S_i systematically diverges from output produced X_i , as it is determined by the firm's profit rate r_i , depend-

² $N(\mu, \sigma^2)$ represents a number drawn from the Gaussian distribution of mean μ and variance σ^2 .

³We drop time indices except when it is strictly necessary.

ing on the probabilistic choices of all price-setting, profit-seeking firms over prices p and industries k :

$$S_i = [1 + r_i(p)] \frac{M_i}{p_i} \quad (1.18)$$

Definition 7. The difference between output sold $S_i(t)$ and $X_i(t)$ goes to the inventory:

$$inv_i(t) = inv_i(t-1) + X_i(t) - S_i(t) \quad (1.19)$$

The inventory/sales ratio the firm observes for itself regulates its price-setting behavior, as it describes the demand conditions it faces at the micro-level.

1.4.5 Competitive Values for Prices and Equilibrium

The model follows a statistical conception of economic equilibrium such as in econophysics, which allows a more fine-scaled statistical characterization of the economy that can be contrasted with empirical evidence that can be observed and measured [Steindl, 1965; Farjoun and Machover, 1983; Mirowski, 1989; Foley, 1994; Drăgulescu and Yakovenko, 2001; Cockshott et al., 2009; Scharfenaker and Foley, 2017]. The micro-economic variables are treated as random, but still yield particular stationary distributions over time through the operations taken over them:

Definition 8. Equilibrium Under such a view, equilibrium is the stationary property of statistical distributions of micro-economic variables that are treated as stochastic.

In the model, equilibrium in prices and quantities is not presumed, but firms dynamically attain it by the systematic, decentralized operation of economic competition over time. Competition is fundamentally thought as a disorderly process that still manages to produce macro-level emergent outcomes as statistical regularities in the micro-economic variables. Equilibrium is not considered as a mechanical single-point rest state that is the static outcome of the decisions of optimizing agents. In the conventional story of perfect competition, the balance of supply and demand is achieved at instantaneous speed as the product of the decisions of utility-maximizing consumers and an infinite set of identical price-taking, profit-maximizing firms at the competitive price, computed by the Walrasian auctioneer as an index of scarcity under instantaneous tâtonnement. Under the classical view, firm and industry prices and quantities ceaselessly gravitate around their competitive values in a turbulent process, fraught by over- and undershooting, subject to fundamental uncertainty over the systematic divergence between expected and realized outcomes. Equilibrium emerges then as a central statistical time-averaging tendency of micro-economic variables, conceived then as an emergent regular outcome at the macro-level produced by the generalized operation of economic competition at the micro-economic level, following the classical conception of the long-period method [Foley, 2011].

Definition 9. Competitive Equilibrium Competitive equilibrium within a single industry is defined on statistical lines when the average over firm prices over time Δt equals the

competitive value under the market forces of economic competition:

$$\langle p_i(t) \rangle_{\Delta t} = p^*; \quad (1.20)$$

In contrast to the neoclassical and classical literature in dynamical adjustment processes [Morishima, 1981; Mas-Colell, 1986; Flaschel and Semmler, 1987; Fisher, 1989a; Flaschel, 1990; Duménil and Lévy, 1993], a contribution of our model is the re-conceptualization of equilibrium along statistical lines.

Definition 10. The technology structure is exogenously defined by a vector of constant average costs ac_i of size N . The lowest-cost technology determines the industry competitive price p^* as a central tendency or equilibrium value:

$$p^* = (1 + \bar{r}) \min_i \{ac_i\} \quad (1.21)$$

The vector of firm technology ac_i can be taken as homogeneous across firms within a single industry in order to observe better the oscillatory dynamics that the model produces under constant technology. The cost structure can also be taken as evolving under repetitive or one-off productivity shocks in order to observe how firm prices dynamically adjust to the new competitive price.

1.5 Simulation Results

In the single-industry scenario of price competition within a single industry, I explore how price competition stochastically unfolds over time (under technology that can be constant or evolving and homogeneous or heterogeneous across firms) in a decentralized set of price-setting, profit-seeking firms that grow in an interdependent way in constant competition over a mass of profit. Firms grow over time in their capital shares in what it seems a random walk, which is in fact regulated by the profits obtained by under-cutting its competition through the price adjustment. If a firm is able to consistently undercut its competitors' prices over time, it will certainly be able to grow with respect to the others and eventually take over. A firm cannot set prices too low either because it will not obtain enough profits to cover for its production expenses.

For constant-technology dynamics, costs are homogeneous across firms and the simulations just show price equalization around its competitive value, so that the industry price cycles in the form of constant price wars with no victor [figure 1.5]. For heterogeneous-technology dynamics, the lowest-cost firm is able to sustain a lower price than its competition over time. Cost differences constrain price choices in firms so that cost differentials induce a spectrum in firm profitability and thus growth. The higher profitability of the lowest-cost firm leads to the evolutionary selection of its technology over time [figure 1.4].

For price competition within a single industry, one can show that equation (1.15) for the

profitability regulating function is equivalent to:

$$r_i = \bar{r} + \frac{\sum_{j=1}^N x_j(p_j - p_i)}{\langle p \rangle} \quad (1.22)$$

where x_j is the firm's share in total capital stock M_j/M . This version reflects better the under-cutting dynamics of price competition between dyads of competitive firms: if $p_i > p_j$ ($p_i < p_j$) a firm i loses (attracts) demand to firm j in proportion to their capital share x_j . In this situation, the average $\langle p \rangle$ is now *weighted* by capital share x_j . In order to show how price competition for higher profits operates among firms, we plot in figure 1.8 the potential variation in profits with respect to potential variations in their prices *ceteris paribus* at the prevailing set of firm prices: all firms gain profits if they reduced their price with respect to the others, which they will do in case of too high inventory/sales ratio or loss in market share. However, any price change in a particular firm will alter the profitability schedule of all firms, which is thus highly unpredictable. The firms' linear relations of potential profits with respect to their own price intersect at the industry average price $\langle p \rangle$, which is different than the competitive price p^* in a way that resembles the classic kinked demand curve described by Hall and Hitch [figure 1.8].

Assumption 6. Initial Conditions *Total capital stock is $M = 1000$, normal profit rate $\bar{r} = 0.2$, normal inventory/sales ratio $u_{eq} = 1$, inventory/sales half-band $u_d = 0.25$, and price step $\theta = 0.1$.*

For constant-technology dynamics, initial costs are homogeneous across firms over time: $ac_i = 5$ for $i = 1, \dots, N$.

For heterogeneous-technology dynamics, initial costs are obtained from a uniform distribution between 5 and 5.5.

Initial firm prices p_i and capital shares x_i are obtained from the Gaussian distributions (capital shares are re-scaled to one):

$$p_i(0) = (1 + \bar{r})ac_i \times N(1, 0.01) \quad (1.23)$$

$$x_i(0) = \frac{1}{N} \times N(1, 0.01) \quad (1.24)$$

Capital shares are re-scaled to one and $M_i(0) = x_i(0)M$. Finally, the initial inventory is

$$inv(0) = u_{eq} \frac{M_i(0)}{ac_i} \quad (1.25)$$

The visualization of the computer simulations of the dynamic evolution of the statistical ensemble of firm observations under the systematic operation of price competition as an iterative process over time can take many forms. In figures 1.5 and 1.6, we plot two simulation runs of the evolution over time of the following micro-economic variables for hundreds of firms in price competition within a single industry, under constant technology dynamics: capital stock (M_i), price (p_i), profit rate (r_i), inventory (inv_i), output sold/demand (S_i),

and output produced/supply (X_i); as well as their respective averages over firms. Figure 1.7 shows the distributions of these micro-economic variables for a particular simulation that are aggregated over time.

Firms grow following a random walk with growth g_i in their capital shares x_i ; the average capital and average output produced are always constant, across firms and over time:

$$\langle M_i \rangle = \frac{1}{N}; \quad \langle X_i \rangle = \frac{M}{acN} \quad (1.26)$$

Figure 1.5 shows price-setting competition following the groping step-wise process as defined by the price rule (1.16), where firms set prices with a step θ with a Gaussian random error, yielding a double-peaked distribution of price changes. Pronounced price oscillations emerge in tandem with inventories, so that the industry average price gravitates around the competitive value in a way that average supply $\langle X_i \rangle$ equalizes average demand $\langle S_i \rangle$ over time:

$$\langle X_i \rangle_{\Delta t} = \langle S_i \rangle_{\Delta t} \quad (1.27)$$

Figure 1.5 shows price-setting competition following the decentralized groping step-wise process as defined by the price rule (1.16), where firms set prices with a step θ with a Gaussian random error, yielding a double-peaked distribution of price changes. Instead, figure 1.5 shows a pricing rule where the step θ is taken to be from a uniform distribution, more in line with the empirical evidence that shows high irregularity [Cavallo and Rigobon, 2011]. The differences are remarkable. When price changes take a double-peaked form, pronounced oscillations emerge in prices in tandem with inventories, with a dispersion of firm prices around the industry average, which conspicuously gravitates around the competitive value [figure 1.11]. When price changes take a uniform distribution instead, firms can adjust to the competitive price much faster so that the price-inventory oscillations are much less pronounced and only feature short random deviations from the competitive value. Further, the amplitude of price oscillations depends on the number of firms N : the greater the number, the random deviations around the normal profit rate become zero.

Figure 1.7 shows the distributions of these micro-economic variables for a particular simulation that are aggregated over time. Price dispersion is observed around its competitive value $p^* = (1 + 0.2) \times 5 = 6$. Firm profitability and growth show a statistical dispersion around their respective values $\bar{r}$ and 0 in the form of a tent-shaped distribution that is consistent with the empirical evidence [Bottazzi and Secchi, 2006; Coad, 2009; Bottazzi et al., 2010; Alfarano et al., 2012; Williams et al., 2015]. The remaining distributions appear as Gaussian, with a tilt due to asymmetric price competition. Figures 1.9 and 1.10 shows further evidence of the tent shape on a log-linear plot.

Under heterogeneous technology that is constant in time [figure 1.4], initial costs ac_i are obtained from a uniform distribution between 5 and 5.5 and initial prices are set as $(1 + \bar{r})ac_i$, whose average lies above the competitive price p^* , which is still 6. Under price-setting competition, lower-cost firms possess a competitive advantage to undercut its competition's prices and still service its production costs. Price competition will thus lower the industry price to the competitive value, which induces a spectrum in profitability following cost differen-

tials leading to differential firm growth. Price competition thus functions as an evolutionary mechanism of selection of the lowest-cost technology, which eventually dominates the market over time. Because firm prices roughly equalize around a particular value due to competition, profitability will be roughly linear in the cost inverse over time [figure 1.4]:

$$\langle r_i \rangle_{\Delta t} = \frac{p^*}{ac_i} - 1 \quad (1.28)$$

In order to show the stability of the decentralized pricing process, a robustness check is run with 100 simulations where the initial prices are drawn from a Gaussian distribution centered between 5.5 and 6.5. The trajectories of the industry average price are plotted in figure 1.12. In spite of such variations in initial prices and the minimal rationality of firm behavior, competitive firms still manage, by decentralized groping, to adjust prices under conditions of reproduction so that their average price gravitates around the competitive value over time, first adopting more deterministic oscillations until the randomness of the pricing process sets in. Hence this can be taken as evidence of the stability of this notion of competitive equilibrium in price and quantity within a single industry developed along classical-evolutionary lines.

In the model, cost-plus markup pricing thus arises as an emergent macro-level outcome of the decentralized, ceaseless groping by firms under price competition, rather than being imposed as an *ad hoc* assumption. In the classical case, the aggregate markup is not determined by market power, but by the average profit rate $\bar{r}$. At the firm level, the markup is not necessarily positive, since profitability can also be negative.

1.6 Conclusions

This paper takes seriously Franklin Fisher’s concerns over the lack of stability of competitive equilibrium and the questions it raises on the validity of the theory of value. Such issues have eventually led some critics to complain about an over-mathematization of economics under a “pretence-of-knowledge” syndrome [Caballero, 2010; Romer, 2016]. For Eatwell and Milgate, “instead of providing the starting point for an empirical analysis of the workings of a market economy, the concept of equilibrium has been reduced to a name for the solution of any set of equations, divorced from the systematic workings of a competitive economy” [Eatwell and Milgate, 2005, p.82]. This paper sets to explore how the dynamic, decentralized process of economic competition unfolds over time in order to observe then how a notion of equilibrium can emerge in that context. The view of this paper follows von Neumann’s observation that the problem in economic analysis does not lie in its use of mathematics, but its conceptual categories [Mirowski, 1992] – specifically the concept of optimization.

The purpose of this paper is to provide an alternative formal description of the dynamic process of economic competition that is based on classical, evolutionary, and statistical lines, which requires to re-define the notion of equilibrium. In this context, equilibrium in prices and quantities is not presumed as the static result of the optimizing decisions of economic agents, but dynamically attained by decentralized, minimal-rationality firms under

the systematic operation of economic competition. Competition is envisioned as a dynamic disorderly process that operates on firms at the micro-level over time, but still yields macro-level emergent outcomes as statistical regularities in the aggregate.

The model shows that no optimization is actually necessary as an assumption in order to reproduce many stylized facts of economic competition: price dispersion around the competitive value, a spectrum in profitability that reflects cost differentials, the selection of the lowest-cost technology, tent-shaped distributions for firm profitability and growth, cost-plus markup pricing, downward-sloping demand curves, a kinked demand curve that is reminiscent of oligopolistic competition, business cycles in prices and inventories, and the dynamic equalization of prices within a single industry, profit rates between industries, and industry supply and demand.

Instead of optimization, reproduction is the fundamental closure in the model. The competitive firm requires not maximal, but above-average profits over time for its long-lasting survival in the marketplace, following a view that is both classical and evolutionary. Firm profitability is what drives the growth of firms, which is made stochastic and interdependent through their probabilistic, minimal-rationality choices. Economic production is like life in the sense that the economy is conceived as an ecosystem in the form of a self-reproducing set of interdependent processes of production [Sraffa, 1960]. Such a view emphasizes reproducibility over optimization [Leontief, 1928], production over exchange [Pasinetti, 1993], or structural formation (that is, how an economy emerges in the first place, and grows and changes structurally over time) over allocation [Arthur, 2014]. This view compares the classical conception of the economy as a circular flow of competitive, yet interdependent, evolving firms to complex biological and ecological systems. This approach is particularly amenable to the domain of complexity science and in particular the literature on auto-catalytic sets and chemical reaction networks, which originated in evolutionary theory in order to explain the emergence of self-reproducing structures regarding the origin of biological life [Smith, 1968; Eigen and Schuster, 1979; Wolfram, 1983; Farmer et al., 1986; Jain and Krishna, 2001; Kauffman, 2011; Smith and Krishnamurthy, 2017] and is now slowly being applied to economics [Padgett and Powell, 2012; Hordijk, 2013].

In order to complete the analogy of the economy as a circular flow, an immediate extension of the model is the modeling of the process of profit rate equalization in a multi-sector scenario, where the additional action variable is on the line of production. In this case, the effects of relative prices that are not competitive on aggregate profit must be taken into account, subject to the fundamental constraint of a constant labor surplus, of which profit is its monetary form. Further, the introduction of input-output linkages in production would allow to fully characterize self-reproduction – at the moment, the model only assumes exogenous demand for final goods.

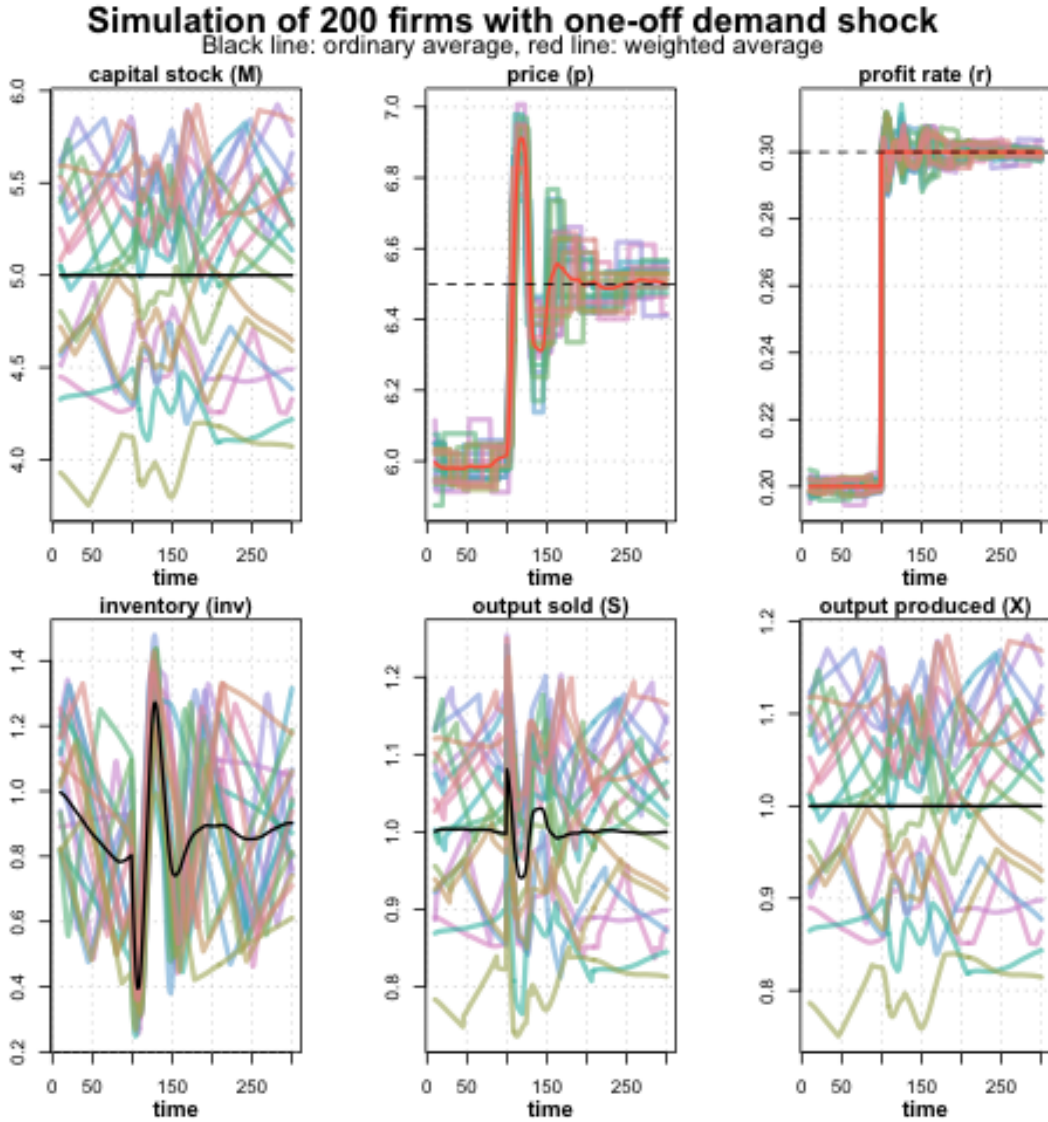


Figure 1.3: **Simulation with Positive Demand Shock at $t = 100$: Average Price Increases**
 A sample of only 20 firms is shown for clarity (with colored lines). Average profitability parameter is changed from 0.2 to 0.3; firms re-adjust their prices so that the average gravitates from the old competitive value (6) to the new one (6.5)

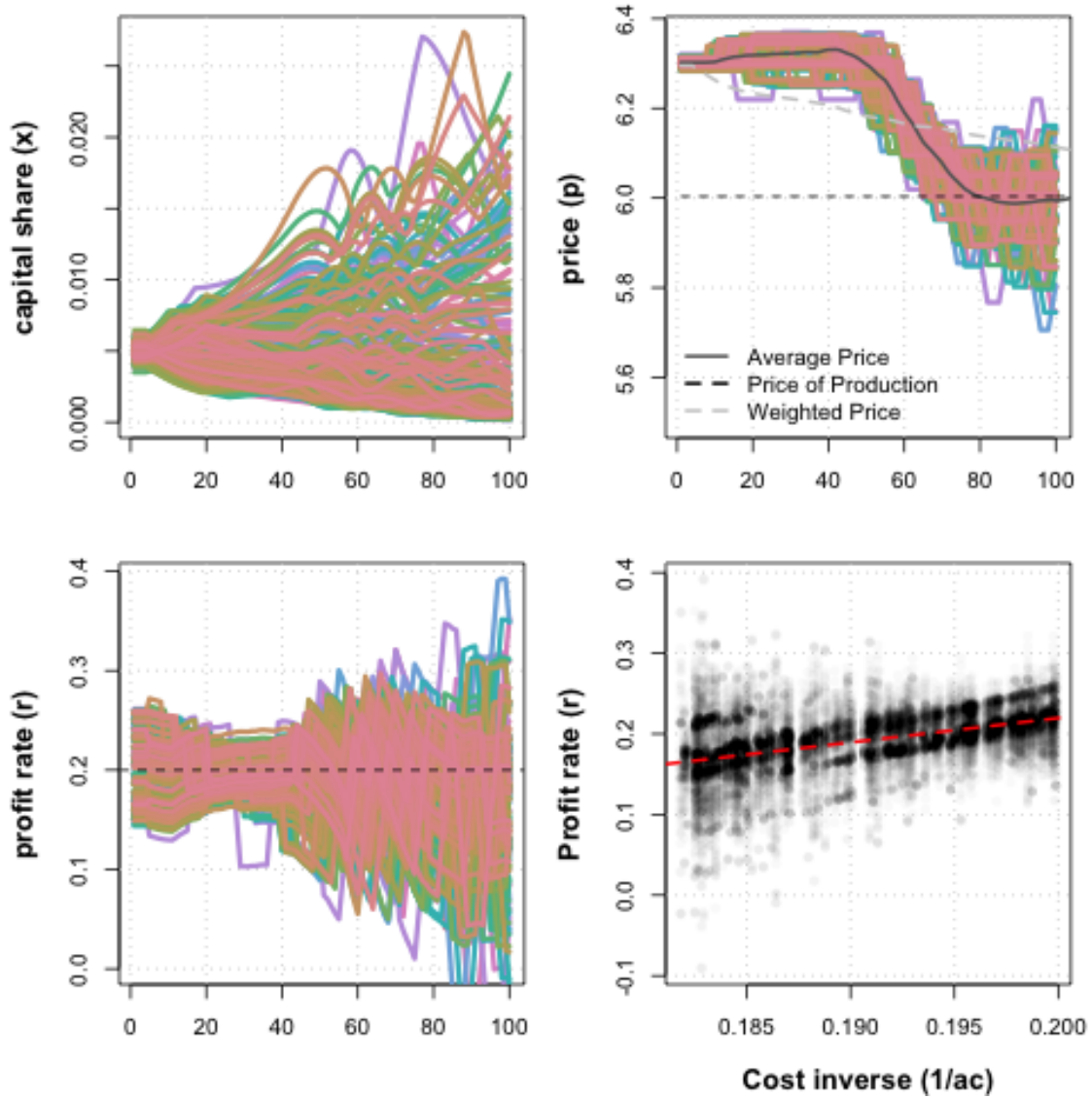


Figure 1.4: **Spectrum in profit rates following cost differences within a single industry** Simulation with heterogenous costs from $t = 1$ to $t = 100$ for $N = 200$ firms with costs obtained from a uniform distribution between 5 and 5.5 units. In the bottom-right panel, one observes the expected spectrum of profitability, which follows a linear relation between the profit rate and the inverse of costs [equation 1.28].

Simulation of 300 starting firms

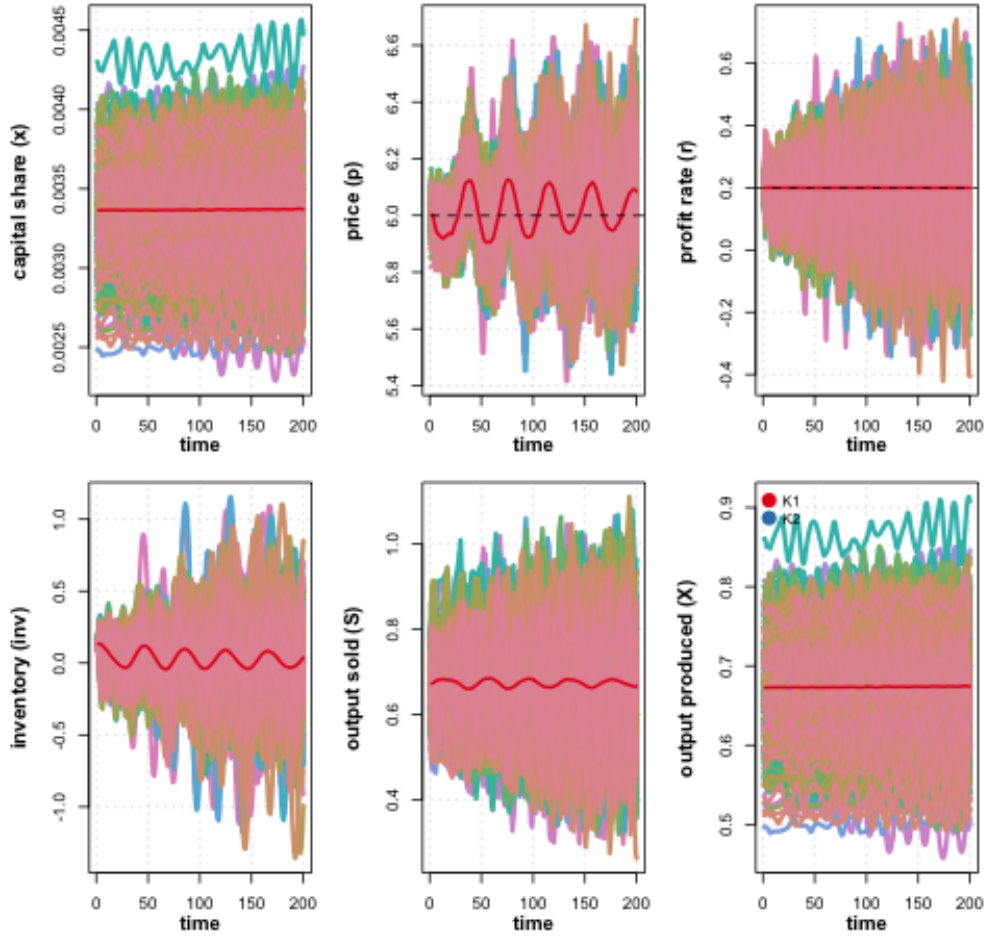


Figure 1.5: **Simulation under Constant Technology** trajectories (in color) of 200 timesteps and 300 firms of the following variables: capital stock (M), price (p), profit rate (r), inventory (inv), output sold/demand (S), and output produced/supply (X). The average is shown in bright red; oscillations in average price, inventory, and demand.

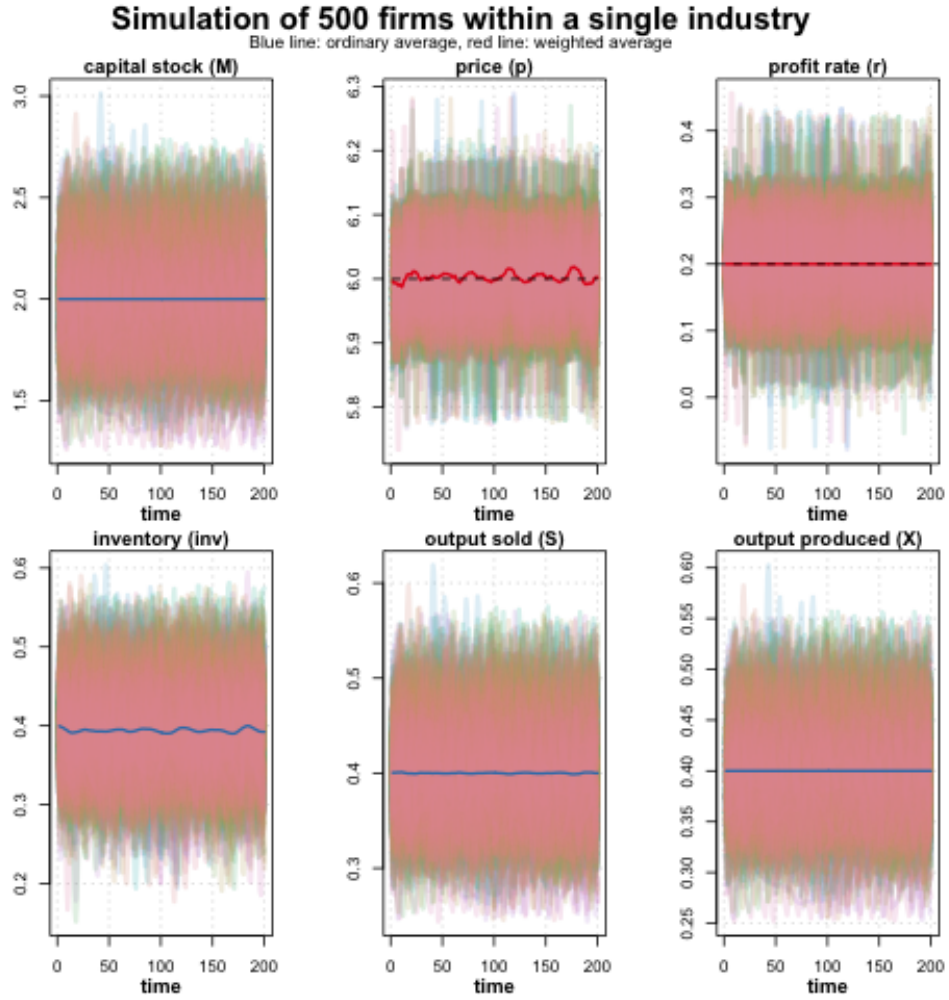


Figure 1.6: **Simulation of 200 timesteps under constant technology** Each colored line represents the trajectory of the corresponding microeconomic variable for one particular firm. Here the pricing rule involves steps that follow a uniform distribution, in agreement with the empirical evidence [Cavallo and Rigobon, 2011].

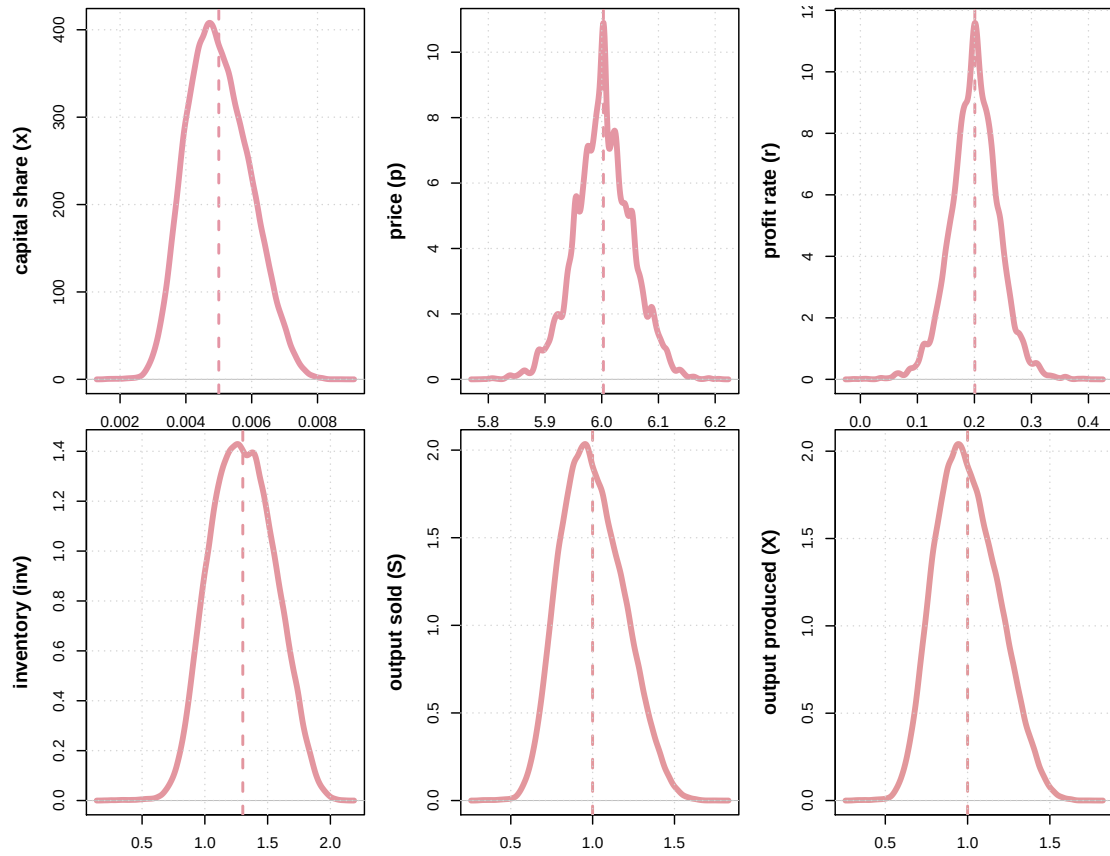


Figure 1.7: **Distributions of Micro-Economic Variables** for a simulation of 200 firms and 200 timesteps. The profit rate distribution follows the typical double-exponential shape [Coad, 2009]

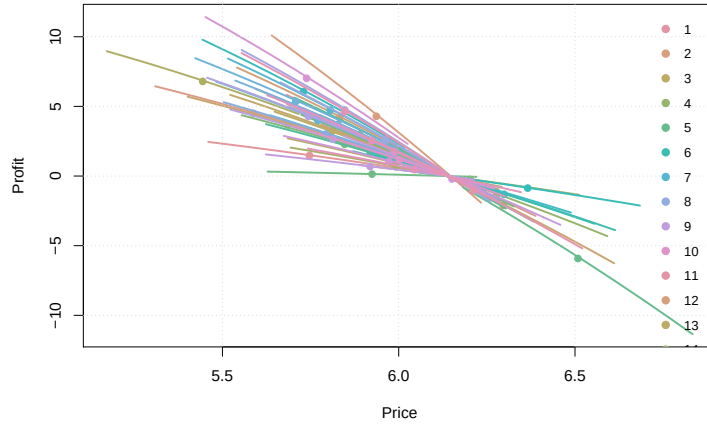


Figure 1.8: **Kinked Demand Curve: Static price decisions of firms and their impact on own profits at a particular timestep** *ceteris paribus* all firms gain profits if they reduce their price with respect to the others, but interestingly enough we still retrieve something akin the classic kinked demand curve of Hall and Hitch. The intersection point is the price average, the dots represent the actual prices set by firms and the straight lines the potential profit they can obtain by changing their price keeping the others' fixed.

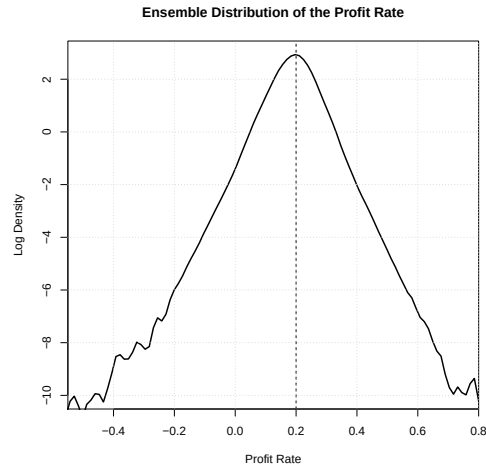


Figure 1.9: **Log-Linear Plot of the Ensemble Distribution of Profitability, 100 simulations aggregated** The tent-shaped distribution for the profit rate (and thus growth) emerges clearly when running many iterations in the model and aggregating the observations in the form of a statistical ensemble.

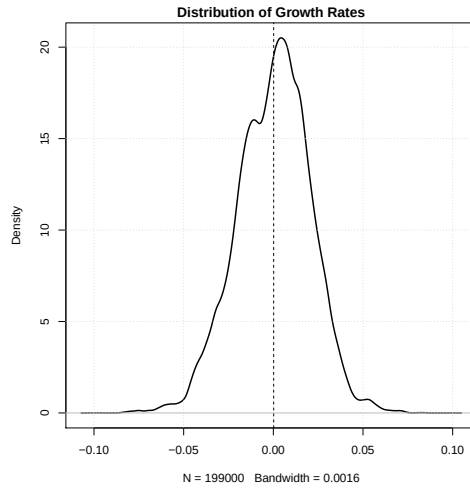


Figure 1.10: **Distribution of Growth Rates** for a particular simulation with 200 firms and 1000 timesteps, which follows the double-exponential shape of the empirical evidence [Coad, 2009] (linear relation in log-linear plot)

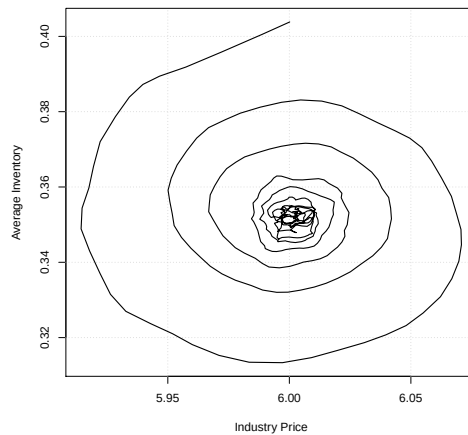


Figure 1.11: **Oscillations and limit cycle in the industry price-inventory plane for a particular simulation** Excess inventory (i.e. low demand) makes firms reduce their price below the competitive value, raising demand again (reducing the inventory stock) and subsequently allowing firms to raise their price. At the limit, irregular oscillations emerge.

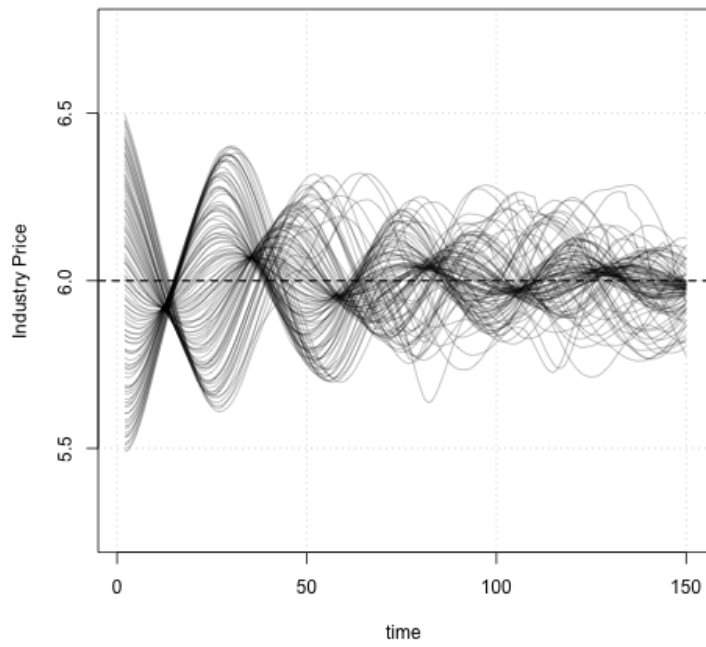


Figure 1.12: **Stability of the Pricing Process: Average Price Trajectories for a Range of Initial Values** 100 simulations are run, where initial values for the price vector are now drawn from Gaussian distribution centered between 5.5 and 6.5.

Chapter 2

The Green Transition: Directed Technical Change Towards Decarbonization

This chapter employs the Flaschel-Semmler dynamical model of multi-sector growth to study the technical substitution of carbon sectors by green sectors under directed technical change. The Flaschel-Semmler model of linear production is based on the dynamic cross-dual linear adjustment between prices and quantities in the form of a law of excess demand and a law of excess profitability, which produces a complex pattern of oscillations around their equilibrium values. The linear adjustment coefficients of the model are empirically calibrated for six countries using a simple OLS model on EU KLEMS and WIOD data. The paper concludes by evaluating analytically and computationally the tax rates that allow decarbonization to meet the targets of the UN Intergovernmental Panel on Climate Change. Directed technical change is enforced by a revenue-neutral, pro-active fiscal policy of a tax-subsidy form, which has the effect to greatly accelerate the phase-out of the carbon sector, in particular at its earliest stages. Without fiscal policy, it is impossible for any economy to reach the IPCC targets.

2.1 Introduction

The climate crisis is one of the defining issues of our time. According to a recent landmark report by the UN Intergovernmental Panel on Climate Change (IPCC), 16 years are left for a rapid structural change of the economy towards decarbonization in order to avoid climate catastrophe by keeping world temperatures below the 1.5-Celsius-degree target. Consensus among economists views climate change as a negative externality that can be corrected by a Pigouvian tax on carbon internalized by polluting firms, which may be further used to subsidize and direct structural technical change towards green technology. In this context, most economic analysis on climate change policies operates in the tradition of computational general equilibrium and new growth theory, which endogenizes innovation [Goulder

and Schneider, 1999; Nordhaus and Boyer, 2000; Nordhaus, 2002; Jaffe et al., 2002; Goulder and Parry, 2008; Gillingham et al., 2009; Acemoglu et al., 2012; Gans, 2012; Golosov et al., 2014; Acemoglu et al., 2016]. These contributions envision optimal taxation based on the fundamental neoclassical trade-off between consumption today and consumption in the future, leading to the design of damage functions that relate climate change with losses in economic output [Keen, 2020]. More recent studies study the role of public finance and green bonds in promoting the green transition [Heine et al., 2019; Deleidi et al., 2020].

In particular, Acemoglu et al. [2012; 2016] posit two pressing questions: (1) how much time the structural transition from carbon to renewable energy may take, and (2) to what extent fiscal policy of the tax-subsidy kind can accelerate the process under directed technical change. This paper addresses these two questions from a different theoretical perspective that is particularly well-suited to parsimoniously study dynamical behavior with well-grounded empirics: an extension of the Flaschel-Semmler dynamical model of multi-sector growth that incorporates technological dynamics in the form of one-off innovation and extinction of economic processes [Flaschel and Semmler, 1987, 1992]. In contrast to the contributions based on new growth theory, the multi-sector character of the model allows to explicitly explore the effect of input-output linkages in production in the context of the green transition. Within the broader literature on competitive dynamical adjustments in prices and quantities [Jorgenson, 1960; Hahn, 1970; Morishima, 1981; Mas-Colell, 1986; Duménil and Lévy, 1987; Flaschel and Semmler, 1987; Fisher, 1989a; Flaschel, 1990; Duménil and Lévy, 1993], the Flaschel-Semmler model theoretically relies on the laws of excess demand and excess profitability, which imply cross-dual linear adjustments in prices and quantities triggered by supply-demand and revenue-cost imbalances, respectively (i.e. deviations between supply and demand and from normal profitability).

Using WIOD and EU KLEMS data, the Flaschel-Semmler model is calibrated by estimating empirically the adjustment coefficients for six developed economies (Germany, France, Japan, Italy, Netherlands, and the US), which are used to compute simulations of the dynamic substitution of two carbon-based sectors of economic production by their green or carbon-free equivalents: in particular, “Electricity, gas, steam and air conditioning supply” (D) and “Manufacture of coke and refined petroleum products” (C19). The theoretical model can be conceived as a machine learning algorithm that uses the WIOD and EU KLEMS datasets as training data. The impact on the speed of adjustment is evaluated analytically and computationally with respect to the efficiency in production of the green sector relative to the carbon sector (in terms of capital intensity), the magnitude of the tax-subsidy mix, and the initial output ratio between both sectors. A relevant next step in the research is to compare the effects of an exclusively carbon-tax policy and one that adds subsidies on the speed of technical substitution.

In absence of fiscal policy, efficiency in production, which currently lies between 0.8 and 1.1 [see figure 2.6], is, as expected, the critical parameter that regulates the speed of the green transition, by imposing sustained profit and growth rate differentials between the carbon and the green sector. In line with Acemoglu et al. [2012; 2016], lower costs in production alone induce too slow of an adjustment speed in the phase-in of the green sector to fall within

IPCC targets or any reasonable time horizon, even when starting many magnitudes above its current levels of output relative to the carbon sector. The findings of the paper show to what extent fiscal policy that taxes carbon output to subsidize green output is indispensable to expand the profit and growth rate differentials enough to meet the IPCC targets for decarbonization. Without fiscal policy, it is impossible for any economy to reach the IPCC targets. Furthermore, revenue-neutral tax-subsidy policy shows its greatest effect at the earliest stages of the green transition, in agreement with Acemoglu et al. [2012; 2016]. The paper finally suggests appropriate tax rates to accelerate the green transition within IPCC targets.

2.2 A multi-sector growth model of the green transition

2.2.1 Constant-Technology Dynamics

The Flaschel-Semmler model of multi-sector growth with circulating capital and constant technology describes price and quantity oscillations over time around the equilibrium values of N prices p (a row vector) and N quantities x (a column vector) of the Sraffa-von Neumann system [Von Neumann, 1945; Sraffa, 1960]. It follows a dynamical cross-dual adjustment in a linear model of production with matrix A as inputs and matrix B as outputs¹, which take real positive values. By the Perron-Frobenius theorem, unique positive equilibrium values p^* and x^* solve for the positive gross rate of return or expansion rate $R > 1$, in matricial notation:

$$Bx^* = RAx^* \quad \text{supply equals demand} \quad (2.1)$$

$$p^*B = Rp^*A \quad \text{profit rates are uniform across sectors} \quad (2.2)$$

where p^*B is the equilibrium unit revenue and Rp^*A is the equilibrium unit cost in relation to the expansion rate R . In scalar notation,

$$\sum_j b_{ij}x_j^* = R \sum_j a_{ij}x_j^* \quad i = 1, \dots, N \quad (2.3)$$

$$\sum_i p_i^*b_{ij} = R \sum_i p_i^*a_{ij} \quad j = 1, \dots, N \quad (2.4)$$

The expansion rate R is the inverse of the unique largest positive real eigenvalue of the input-output matrix A/B , equilibrium prices p^* are its associated positive row eigenvector, and equilibrium output x^* its the associated positive column eigenvector. The second largest eigenvalue of the input-output matrix A/B determines the speed of convergence to equilibrium [Brody, 1997]. R thus can be considered the “maximum expansion rate” [Shaikh, 2016],

¹These matrices are “augmented” in the sense that they also incorporate labor supply and its price, the wage rate. This contribution emphasizes issues of multi-sector growth and technical change over distribution.

which is associated to the aggregate profit rate when wages are zero and there is no capitalist consumption, that is, all capitalist profit is re-invested:

$$1 + r = R = \frac{pBx}{pAx} \quad (2.5)$$

When the system is in maximum expanded reproduction, the demand for final goods will be proportional to the total output vector [Shaikh, 2016]. The empirical statistical analysis of the input-output matrix for the US between 1967 and 2007 shows a remarkably persistent exponential distribution for the distribution of its (moduli) eigenvalues, which cluster around zero in the complex plane [Torres-González and Yang, 2019]. From a dynamical-systems perspective, this suggests a very complex pattern of oscillations operating at different speeds of adjustment, which the Flaschel-Semmler model attempts to capture by appealing to two abstract laws: the law of excess demand and the law of excess profitability.

Models of dual and cross-dual adjustment in prices and quantities have a long history in economics [Morishima, 1981; Mas-Colell, 1986; Duménil and Lévy, 1987; Flaschel and Semmler, 1987; Fisher, 1989a; Flaschel, 1990; Flaschel and Semmler, 1992; Duménil and Lévy, 1993]. In the Flaschel-Semmler model [1987], a deviation of quantities from equilibrium will trigger a response in prices, while a deviation of unit profits from equilibrium will induce a response in quantities. Cross-duality in its simplest form gives rise to stability, but not to asymptotic stability for the equilibrium assumed, but rather ceaseless over- and undershooting of prices, quantities, and profit rates around their natural values as centers of gravity [Flaschel and Semmler, 1987; Shaikh, 2016], see figure 2.1.

Following the **law of excess demand**, market prices p will decline (rise) if supply Bx is greater (smaller) than demand:

$$\left(\frac{\dot{p}}{p}\right)^T = -\delta_p[B - RA]x = \delta_p[\underbrace{RAx}_{\text{demand}} - \underbrace{Bx}_{\text{supply}}] \quad (2.6)$$

Following the **law of excess profitability**, quantity x_i will rise (decline) if unit revenues pB are greater (smaller) than unit costs times R , RpA , since capital will flow out of the sectors with below-normal profitability into the sectors with above-normal profitability:

$$\frac{\dot{x}}{x} = +\delta_x[B - RA]^T p^T = \delta_x[\underbrace{B^T p^T}_{\text{revenue}} - \underbrace{RA^T p^T}_{\text{cost}}] \quad (2.7)$$

where $\frac{\dot{x}}{x}$ is the column vector of the growth rates in relative quantities, $\frac{\dot{p}}{p}$ is the row vector of the growth rates in relative prices, and δ_p and δ_x are diagonal matrices with N positive adjustment coefficients (so they can also be understood as vectors).

In discrete-time, scalar form,

$$\frac{p_{t+1}^i}{p_t^i} - 1 = -\delta_p^i \sum_j (b_{ij} - Ra_{ij})x_t^j \quad i = 1, \dots, N \quad (2.8)$$

$$\frac{x_{t+1}^i}{x_t^i} - 1 = +\delta_x^i \sum_j p_t^j (b_{ji} - Ra_{ji}) \quad i = 1, \dots, N \quad (2.9)$$

In the Flaschel-Semmler model, firms are not profit-maximizing in the sense that they maximize profits given all available techniques, but profit-seeking instead. The dynamic process of the free mobility of profit-seeking capital among sectors of production induces fluctuations in outputs: if an industry earns higher-than-average profits, capital will move there raising output. Since market prices react positively to excess demand and negatively to excess supply, the increase in supply caused by capital inflows will eventually drive prices down, reducing industry profitability and eventually forcing capital to flow out to other sectors of production with higher profit rates, and in the process reducing sector output. Market prices gravitate around their equilibrium values, which are ultimately determined by the technological structure A and B . This is the basic dynamic structure of the business cycle in the theoretical model.

In order to retrieve dynamic convergence to equilibrium, Flaschel and Semmler add an adjustment, with parameter $\gamma > 0$, where capitalists also take account of the sign of change of extra profits (or losses) when moving their capitals between sectors, so the law of excess profitability is modified [Flaschel and Semmler, 1987]:

$$\frac{\dot{x}}{x} = +\delta_x [B - RA]^T (p^T + \gamma \dot{p}^T) \quad (2.10)$$

This modified cross-dual adjustment process is proven to be globally asymptotically stable as a special case of what Hahn and Fisher call quasi-global stability [Flaschel and Semmler, 1987, p.26]².

2.2.2 Technical change with process innovation and extinction

In a subsequent contribution [Flaschel and Semmler, 1992], Flaschel and Semmler propose a generalization based on their classical competitive process of dynamical adjustment of the model of technical change presented in Silverberg [1984], which is based on the Goodwin model of class struggle and capital accumulation [1982]. The Goodwin model assumes neutral, exponential, disembodied technical progress, under fixed coefficients production, with fluctuating unemployment regulating changes in the level of real wages. Instead of disembodied technical progress, the contribution by Silverberg presents an economy with a fixed production process and then proceeds by examining the stability of the resulting equilibrium state when a second production process embodied in a new capital good with different technical coefficients is introduced.

The Flaschel-Semmler model of technical change allows the input-output A, B to be $K \times N$ rectangular and evolve over time, with $i = 1, \dots, K$ commodities (corresponding to rows) and $j = 1, \dots, N$ processes (corresponding columns). Hence, the j th column of A corresponding to the j th process indicates the input requirements of commodities $i = 1, \dots, N$.

²[Hahn 1982 pp 749, Fisher 1983]

Now the quantity vector is of dimension K for each process (as adjustment parameter vector δ_x), while the price vector has dimension N for each commodity (as adjustment parameter vector δ_p). The output matrix B has the same dimensions than the input matrix A . If the system is single-product, the $K \times N$ output matrix is composed exclusively by 0 and 1s, that is, b_{ij} is 1 if the j th process produces commodity i and 0 if not. For joint-product systems, b_{ij} can take any real number between 0 and 1.

In particular, the model considers material (or wage) saving innovations (i.e. capital- or labor-saving, which are indistinguishable in the framework), substitution effects where a more efficient process competes to take over a less efficient process, and innovation in a joint-product system. If a new process is introduced, a square $K \times K$ A matrix at time t is replaced at time $t+1$ by a rectangular $K \times N$ A matrix where now $N = K+1$. A newer, more efficient process j' emerges to compete with an older, less efficient process j : formally, $a_{ij'} \leq a_{ij} \quad \forall i = 1, \dots, K$ commodities. In their contribution, Flaschel and Semmler examine the three scenarios separately (material/wage-saving innovations, substitution effects, and joint-product innovation), but the three cases can be simulated altogether to model innovation in more general terms.

In discrete time, the simulations for a $K \times N$ rectangular system proceed in the following way, where commodity 1 is produced by two competing sectors [Flaschel and Semmler, 1992]:

$$\begin{pmatrix} x_1^1 \\ x_1^2 \\ x_2 \end{pmatrix}_{t+1} = \begin{pmatrix} x_1^1 \\ x_1^2 \\ x_2 \end{pmatrix}_t + \underbrace{\begin{pmatrix} d_{x_1^1} & 0 & 0 \\ 0 & d_{x_1^2} & 0 \\ 0 & 0 & d_{x_2} \end{pmatrix}}_{\langle d_x \rangle} \underbrace{\begin{pmatrix} x_1^1 & 0 & 0 \\ 0 & x_1^2 & 0 \\ 0 & 0 & x_2 \end{pmatrix}}_{\langle x \rangle} [B - RA]^T \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}_t \quad (2.11)$$

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix}_{t+1} = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}_t - \underbrace{\begin{pmatrix} d_{p_1} & 0 \\ 0 & d_{p_2} \end{pmatrix}}_{\langle d_p \rangle} \underbrace{\begin{pmatrix} p_1 & 0 \\ 0 & p_2 \end{pmatrix}}_{\langle p \rangle} [B - RA] \begin{pmatrix} x_1^1 \\ x_1^2 \\ x_2 \end{pmatrix}_t \quad (2.12)$$

where in a single-product scenario

$$B - RA = \underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_B - R \underbrace{\begin{pmatrix} a_{11}^1 & a_{11}^2 & a_{12} \\ a_{21}^1 & a_{21}^2 & a_{22} \end{pmatrix}}_A \quad (2.13)$$

As in the square $N \times N$ case with constant technology, an additional investment criterion of firms is included via the $N \times N$ term $S(p_t)$:

$$\frac{S(p_t)}{p_t} = -(B - RA)^T \delta_p (B - RA) \quad (2.14)$$

so that the new discrete system becomes:

$$\frac{x_{t+1}}{x_t} = 1 + \delta_x [(B - RA)^T p_t + \gamma S(p_t)] \quad (2.15)$$

	1	2	3	4	5	6	7
1 Agriculture	0.2008	0.0000	0.0011	0.0338	0.0001	0.0018	0.0009
2 Mining	0.0010	0.0658	0.0035	0.0219	0.0151	0.0001	0.0026
3 Construction	0.0034	0.0002	0.0012	0.0021	0.0035	0.0071	0.0214
4 Manufacturing	0.1247	0.0684	0.1801	0.2319	0.0339	0.0414	0.0726
5 TTU	0.0855	0.0529	0.0914	0.0952	0.0645	0.0315	0.0528
6 Services	0.0897	0.1668	0.1332	0.1255	0.1647	0.2712	0.1873
7 Other	0.0093	0.0129	0.0095	0.0197	0.0190	0.0184	0.0228

^a TTU: Trade, Transportation, and Utilities

Table 2.1: **2003 US Direct Requirements Matrix**

$$\frac{p_{t+1}}{p_t} = 1 - \delta_p(B - RA)x_t \quad (2.16)$$

In their contribution, material-saving innovation is explored with an input matrix A that evolves over time featuring 2 commodities and 3 processes with the following coefficients:

$$A(t) = \underbrace{\begin{pmatrix} 0.4 & 0.6 \\ 0.3 & 0.5 \end{pmatrix}}_{\text{before innovation}} \rightarrow \underbrace{\begin{pmatrix} 0.4 & 0.2 & 0.6 \\ 0.3 & 0.15 & 0.5 \end{pmatrix}}_{\text{during innovation}} \rightarrow \underbrace{\begin{pmatrix} 0.2 & 0.6 \\ 0.15 & 0.5 \end{pmatrix}}_{\text{after innovation}} \quad (2.17)$$

Initially, there are two processes producing commodity 1 and 2 with input coefficients $(a_{11}, a_{21}) = (0.4, 0.3)$ and $(a_{21}, a_{22}) = (0.6, 0.5)$. A material-saving innovation takes place with the introduction of a newer, more efficient process producing commodity 1 $(a'_{11}, a'_{21}) = (0.2, 0.15)$ which are half of the older process. Eventually, the more efficient process drives out the older process, yielding another square matrix with smaller coefficients.

Substitution effects are computed by the following evolving matrix A :

$$A(t) = \underbrace{\begin{pmatrix} 0.4 & 0.2 & 0.6 \\ 0.3 & 0.15 & 0.5 \end{pmatrix}}_{\text{before innovation}} \rightarrow \underbrace{\begin{pmatrix} 0.15 & 0.2 & 0.6 \\ 0.1 & 0.15 & 0.5 \end{pmatrix}}_{\text{after innovation}} \quad (2.18)$$

Initially, the more efficient process is $(0.2, 0.15)$, which has an absolute-cost advantage over $(0.4, 0.3)$: the former process produces the same output twice more efficiently than the latter, that is, it requires half the circulating capital to produce one unit of output. The one-off innovation turns the tables by making the latter more efficient, with coefficients $(0.15, 0.1)$.

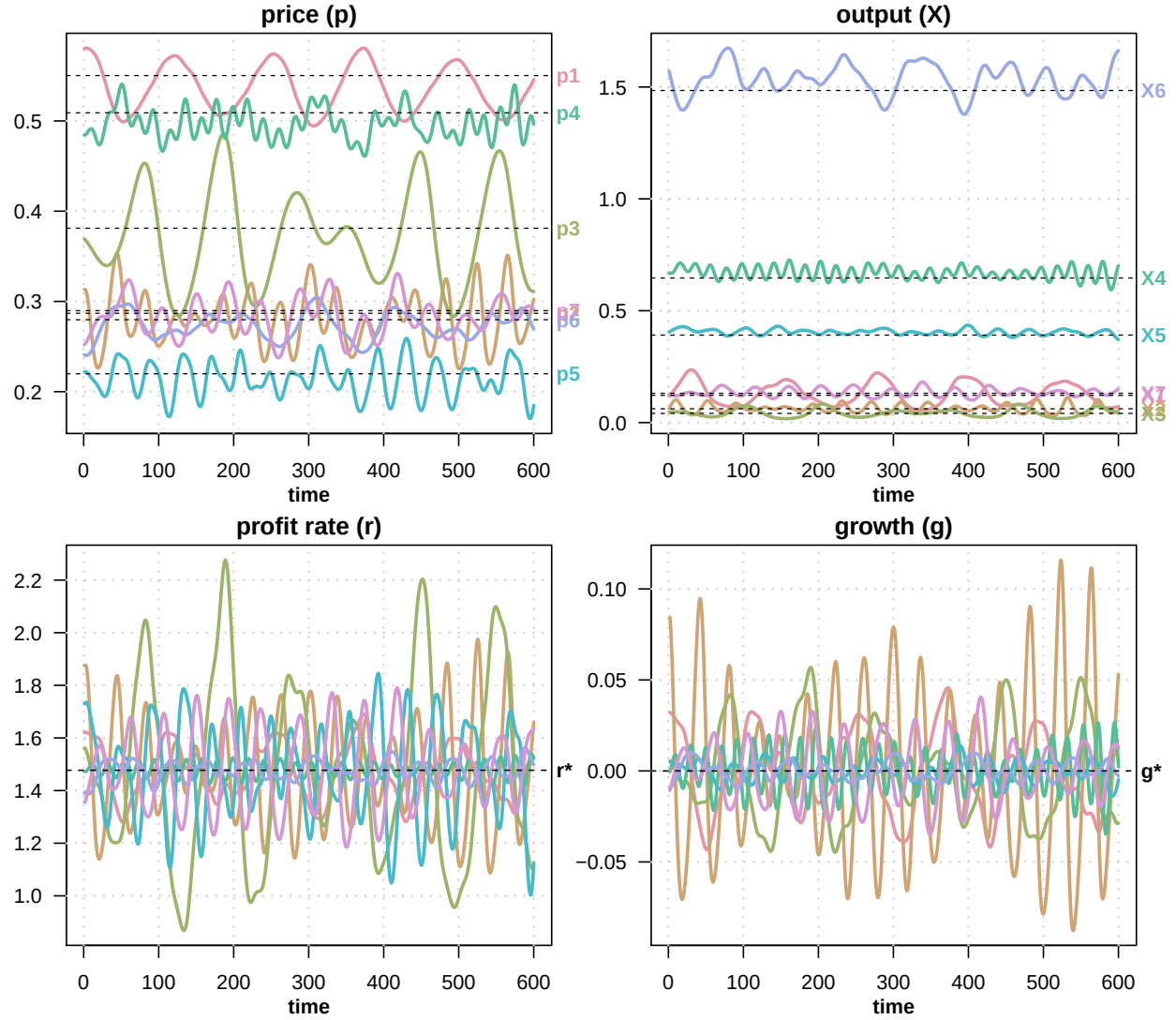


Figure 2.1: **Flaschel-Semmler Constant-Technology Dynamics for the 2003 US Direct Requirements Matrix** Dashed horizontal lines indicate equilibrium values for the profit rate r^* , prices p^* , quantities x^* , and aggregate growth $g^* = 0$.

2.3 Model Calibration

2.3.1 Synthetic Data

As a first example, constant-technology dynamics are simulated for the 2003 US direct requirements matrix, disaggregated into 7 industries (table 2.1) [Miller and Blair, 2009, p.29] in figure (2.1). The equilibrium values for prices, quantities, and the expansion rate are:

$$\begin{aligned} p^* &= (0.550, 0.287, 0.381, 0.509, 0.220, 0.280, 0.290) \\ x^* &= (0.0727, 0.0367, 0.0244, 0.3854, 0.2331, 0.8853, 0.0780) \\ R &= 2.477343 \end{aligned}$$

The resulting simulations produce a complex pattern of deterministic coupled cyclic oscillations in the price and quantity vector around its equilibrium values. Oscillations vary in amplitude and frequency depending on the adjustment parameters chosen, in this case:

$$\begin{aligned} \delta_p &= (0.1, 1, 0.5, 0.5, 1, 0.1, 0.5) \\ \delta_x &= (1, 2, 0.5, 2, 0.25, 1, 1) \\ \gamma &= 1 \end{aligned}$$

The N adjustment parameters for prices $\delta_{p,i}$ [equation 2.8] can be estimated from the simulation data with an ordinary linear regression without intercept:

$$y_{i,t} = \alpha_i x_{i,t} + \epsilon_{i,t} \quad i = 1, \dots, N \quad (2.19)$$

where the dependent variable is the growth rate of prices, the independent variable is excess demand, and the linear slopes α_i correspond to the adjustment parameters $\delta_{p,i}$. Likewise, the N adjustment parameters for quantities $\delta_{x,i}$ [equation 2.9] can be estimated with a linear regression:

$$y_{i,t} = \alpha_{1i} x_{i,t} + \alpha_{2i} x_{2,i,t} + \epsilon_{i,t} \quad i = 1, \dots, N \quad (2.20)$$

where the dependent variable is the growth rate of quantities, the independent variable is the excess unit profit, and the linear slopes α_{1i} are the adjustment parameters $\delta_{x,i}$, while $\alpha_{2i} = \gamma \delta_{x,i}$.

The OLS regression suffices to find the parameters with p -values of 0, which is expected since it is synthetic data. Maximum-entropy linear regression obtained the same results. For the estimations using real data, the profit sign $x_{2,i,t}$ is dropped and excess unit profit is used as single regressor as in the first linear regression.

In fact, it seems much more convenient to apply a Generalized Linear Model on the whole dataset using country and sector as dummy variables, instead of performing an OLS regression per country and sector.

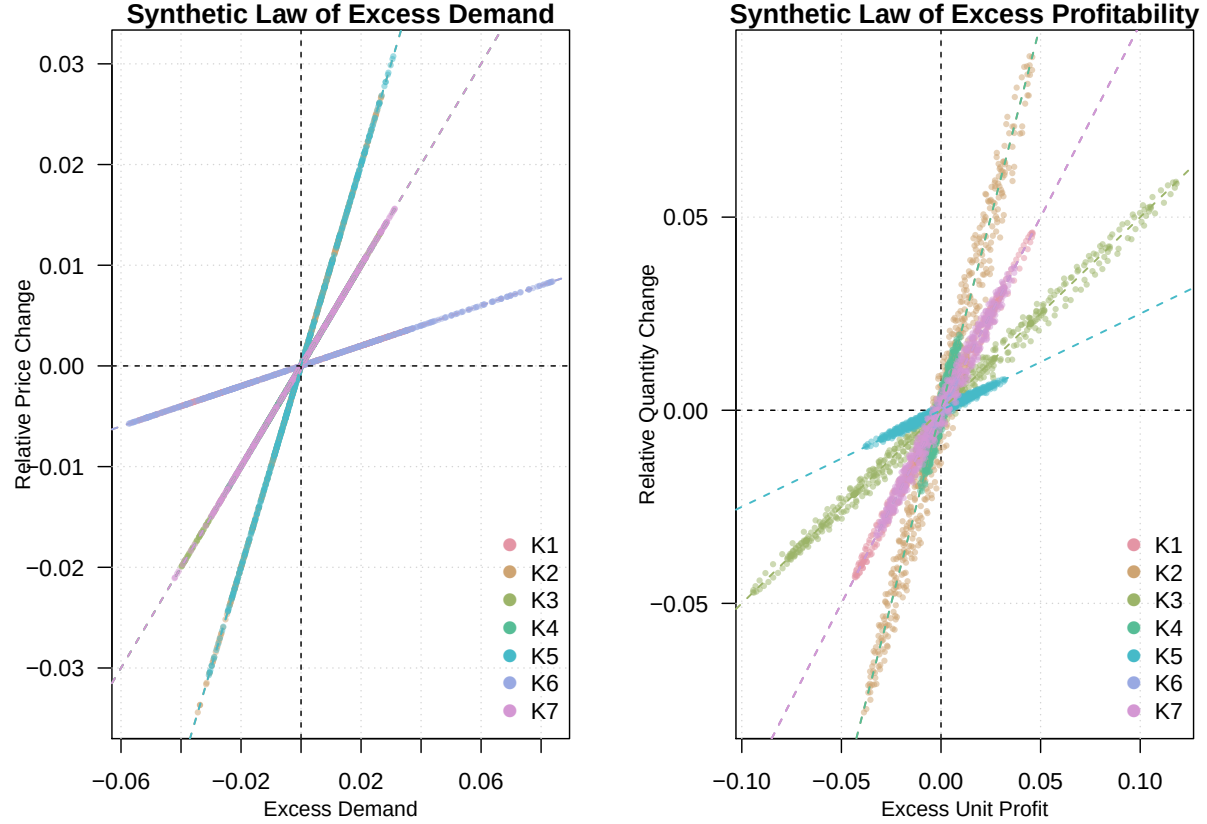


Figure 2.2: **Relative price and quantity changes with respect to excess demand and excess unit profit** The linear slopes correspond to the adjustment parameters δ_p and δ_x . The synthetic law of excess profitability shows a slight departure from strict linearity due to the stability adjustment γ . Dashed horizontal lines indicate equilibrium values r , x^* , p^* , and $g^* = 0$. Dashed color lines indicate the linear regressions for each sector.

sector	delta.p	p.value	delta.x	delta.x.gamma	p.value1	p.value2
1	0.1	0	1.00	1.00	0	0
2	1.0	0	2.00	2.00	0	0
3	0.5	0	0.50	0.50	0	0
4	0.5	0	2.00	2.00	0	0
5	1.0	0	0.25	0.25	0	0
6	0.1	0	1.00	1.00	0	0
7	0.5	0	1.00	1.00	0	0

Table 2.2: OLS for adjustment parameters $\delta_{p,i}$, $\delta_{x,i}$, and γ

2.3.2 Estimation using real training data

At the current moment of the research, the empirical estimation of the linear adjustment parameters is unsatisfactory: the theoretical requirement imposed by the laws of excess demand and excess profitability is that they are all positive, which is not observed at the sector level, but only at the aggregate level for each country. As a first approximation for the simulations of the green transition, uniform adjustment parameters across sectors are assumed as a first approximation, which is unrealistic.

This raises three potential issues: first, the assumption of only circulating capital and no fixed capital is too stringent. In fact, Flaschel and Semmler already provide an additional extension of their theoretical model where depreciation is included, following Bródy's analysis [Bródy, 1974; Flaschel and Semmler, 1986]. Second, multiple OLS regressions are employed per each sector $i = 1, \dots, N$ and country $k = 1, \dots, K$. Instead, a panel-data linear model with country and sector as dummy variables may be more convenient to shed light on the hierarchical structure of the dynamic adjustments, of the form:

$$y_{a,it} = \beta_0 + \beta_1 x_{a,it} + \gamma \text{GEO}_{it} + \delta \text{SECTOR}_{it} \quad a = p, x \quad i = 1, \dots, N \times K; \quad t = 2000, \dots, 2014$$

Third, the European datasets used may not be appropriate for the calculations due to their short timespan and it would be better to rely on US datasets.

Definition of Variables

Two datasets are used as training data for the Flaschel-Semmler algorithm: EU KLEMS (for information on prices and quantity variations) and WIOD (for information on input-output relations and final demand). EU KLEMS and WIOD have different sector disaggregations (WIOD is more disaggregated) so the elements of WIOD objects are summed to match the EU KLEMS sector disaggregation. All values detailed here are vectors indexed by country, industrial sector, and year, except the expansion rate that is aggregate for all sectors (only indexed by country and year). Years span between 2000 and 2014.

The growth rates of prices and quantities can be directly computed from the time series offered by the EU KLEMS database on “Gross output, price indices, 2010=100” (to compute $y_p \equiv \frac{\dot{p}}{p}$) and “Gross output, volume 2010 ref. prices, NAC mn” (to compute $y_x \equiv \frac{\dot{x}}{x}$). EU KLEMS only offers these indices for six countries (Germany, France, Italy, Japan, Netherlands, and USA) so we can compute the adjustment coefficients only for these. Since the Flaschel-Semmler model employs relative activity levels and relative prices, the indices suffice for our computations.

For the law of excess demand, quantity supply (Bx in the model) is the sum of the gross output, volume 2010 ref. prices (from EU KLEMS as before) plus imports (from WIOD), while demand (RAx in the model) is computed as the sum of intermediates ($A \times x$) and final demand (consumption, investment, and exports) extracted from the WIOD. The WIOD offers exports, imports, and the different components of final demand in monetary value, which are linearly transformed into volumes by dividing by its sector gross outputs and

multiplying by its gross outputs in volume (from EU KLEMS):

$$\text{supply} = \text{GO (quantity, KLEMS)} + \frac{\text{imports (value, WIOD)}}{\text{GO (value, WIOD)}} \times \text{GO (quantity, KLEMS)} \quad (2.21)$$

$$\text{demand} = \frac{\text{intermediates (A, WIOD)} + \text{final demand (WIOD)}}{\text{GO (value, WIOD)}} \times \text{GO (quantity, KLEMS)} \quad (2.22)$$

An alternative, much simpler approach following Shaikh [?] employs the input-output category of unintended excess inventories as a proxy for excess supply.

For the **law of excess profitability**:

$$\text{unit revenue} = \frac{\text{GO (value, KLEMS)}}{\text{GO (quantity, KLEMS)}} \quad (2.23)$$

$$\text{unit cost} \times \text{expansion rate} = R \times \frac{\text{intermediates} + \text{labor compensation (value, KLEMS)}}{\text{GO (quantity, KLEMS)}} \quad (2.24)$$

$$R = 1 + \frac{\text{capital compensation (KLEMS)}}{\text{intermediates} + \text{labor compensation (KLEMS)}} \quad (2.25)$$

2.3.3 Regressions

First, standard OLS regressions without intercept are performed for all countries and sectors. The expectation from the laws of excess demand and profitability is that the regression slopes should be positive. However, of all 216 linear regressions for the law of excess demand, 146 return positive slopes, while 70 return negative slopes. Out of the 146 positive slopes, only 79 have a p -value smaller than 0.05. Of all 218 linear regressions for the law of excess profitability, 137 return positive slopes and 81 return negative slopes. Out of the 137 positive slopes, only 42 have a p -value smaller than 0.05. This implies violations by specific sectors of the laws of excess demand and profitability, which may be expected to happen in reality, but makes the estimation of the adjustment parameters nonsensical. Similar results arise if data is aggregated by country to find sector adjustment parameters. As an example, linear regressions for Germany and the law of excess demand are plotted in the following figure, with the dashed colored lines representing the linear fits by sector [figure 2.3].

Hence as a first approximation, adjustment parameters are assumed to be uniform across sectors so standard OLS regressions without intercept are performed (i.e. $y_t = \alpha x_t$) for each country aggregating by sector. This returns reasonable fits, especially for the law of excess profitability: all of the p -values are highly significant (below 8×10^{-3}) except for the USA, which is 0.26.

For the law of excess demand [figure 2.4], there are two sectors that are consistently outliers for all countries: construction (F) and professional, scientific, technical, administrative and support service activities (M-N). The former always has excess supply and the latter always has excess demand. If those sectors are excluded, the OLS regressions returns

Sectoral Law of Excess Demand, Germany

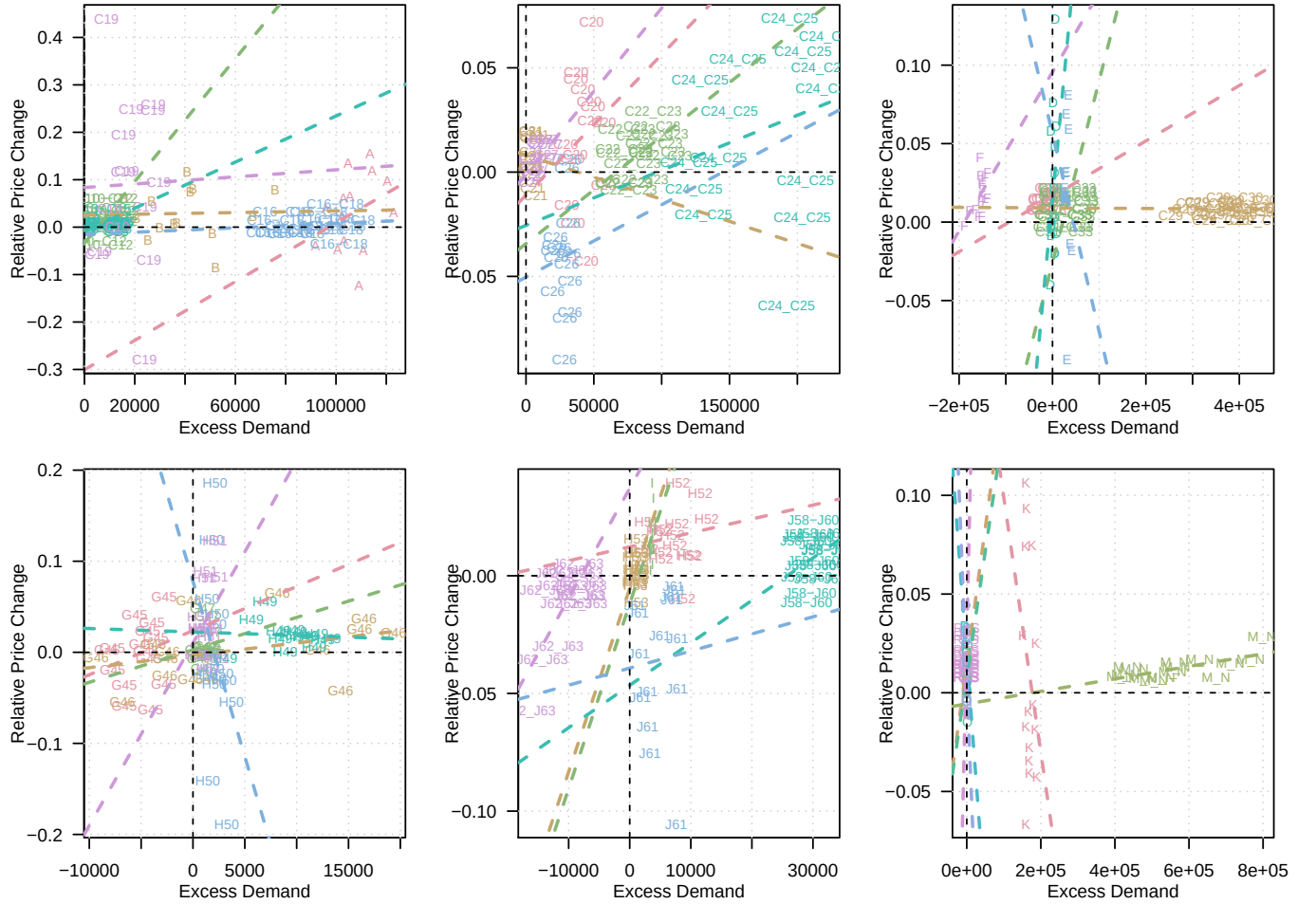


Figure 2.3: **Sector Law of Excess Demand for 37 sectors, Germany** The linear regressions by sector are plotted as colored dashed lines.

highly significant p -values, all below 0.01, and positive linear slopes. In the figure for excess demand, two vertical gray dashed lines are plotted within which the values that are included in the regression. The colored dashed lines indicate regressions with and without intercept, and including and excluding outlier values. The most significant p -values are for the linear regressions without intercept and with excluded outlier values.

2.4 The Green Transition

In this section, the effects of a tax-subsidy policy mix on the profit rate and growth differentials between a carbon sector and a green sector are evaluated analytically and computationally.

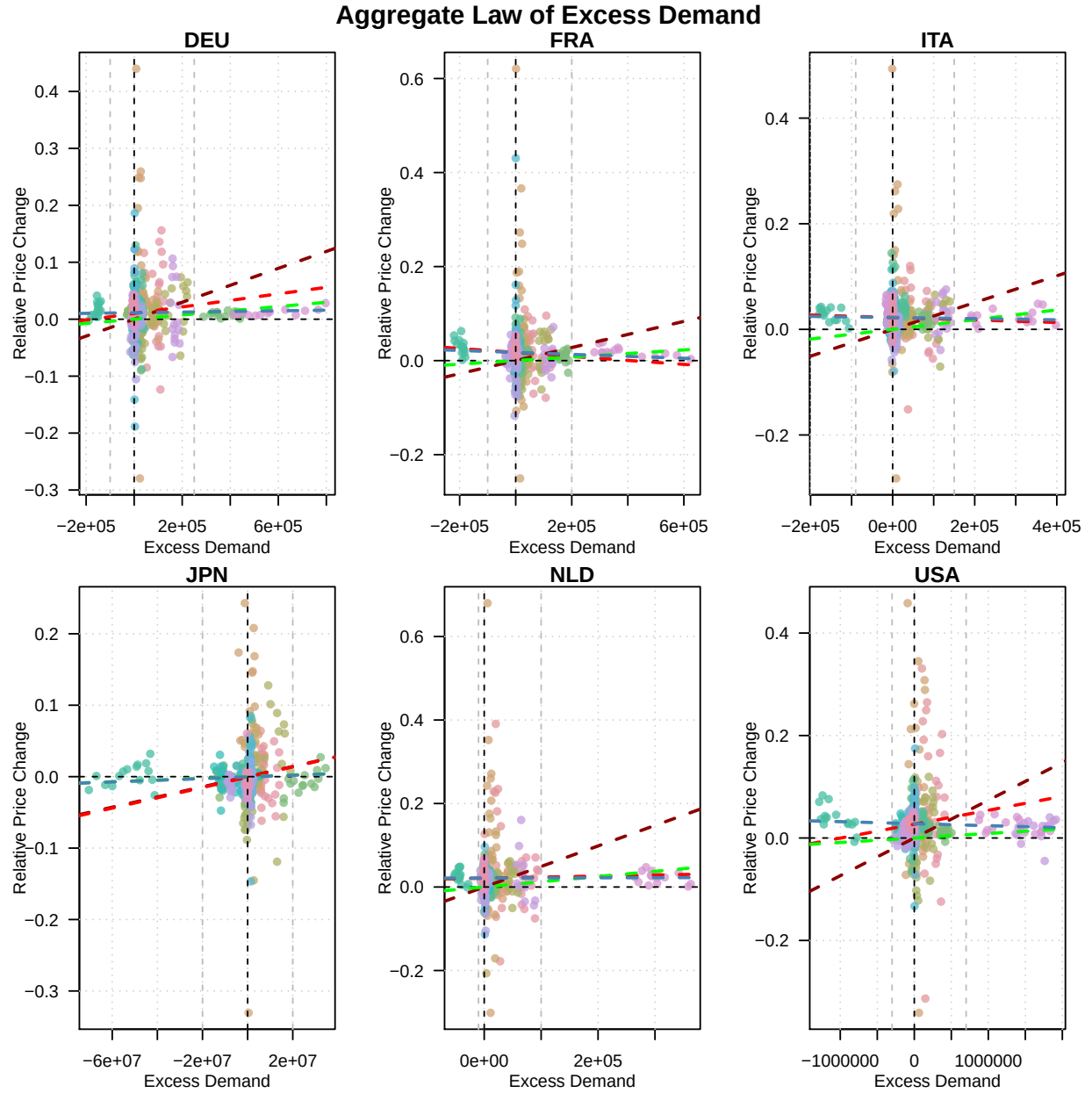


Figure 2.4: **Aggregate Law of Excess Demand for 6 countries** The color scale identifies different sectors. The aggregate linear regressions are plotted as colored dashed lines: dark red (without intercept, without outliers), bright red (with intercept, without outliers), green (without intercept, with outliers), and blue (with intercept, with outliers).

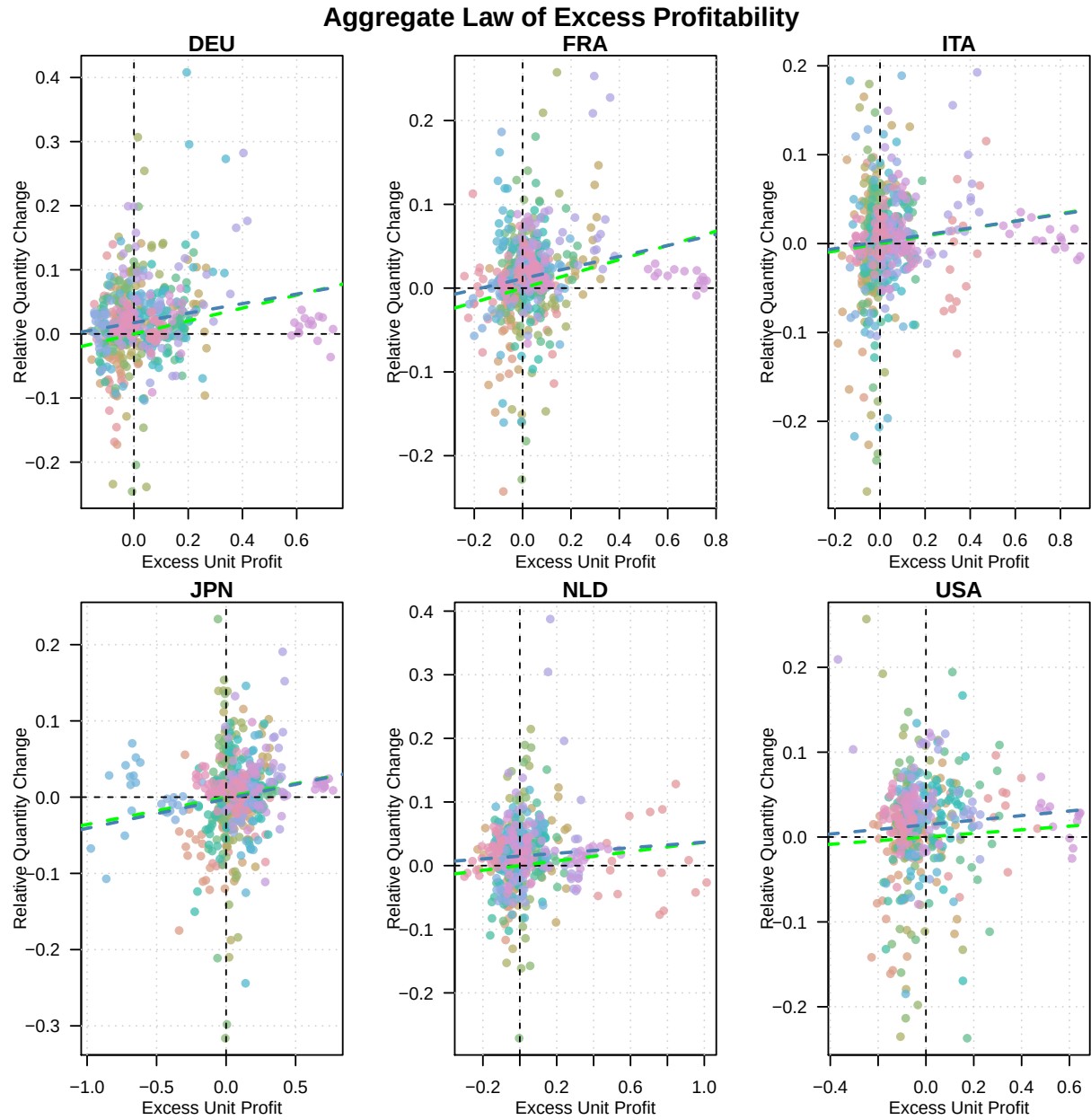


Figure 2.5: **Aggregate Law of Excess Profitability for 6 countries** The color scale identifies different sectors. The aggregate linear regressions are plotted as colored dashed lines: green (without intercept) and blue (with intercept).

	delta.p	p.value	delta.x	p.value.1
DEU	1e-07	0.0002289	0.1008606	0.0000001
FRA	1e-07	0.0100843	0.0848255	0.0000000
ITA	3e-07	0.0000051	0.0421259	0.0052491
JPN	7e-10	0.0376675	0.0358134	0.0009043
NLD	5e-07	0.0000507	0.0367410	0.0082892
USA	1e-07	0.0015982	0.0214045	0.2689342

Table 2.3: **OLS for aggregate adjustment coefficients**

2.4.1 Comparative Statics of Tax-Subsidy Policy on sector Growth

For simplicity, input-output coefficients of the green sector g are defined as constant over time and proportional to the carbon sector c ,

$$\theta = \frac{a_{ig}}{a_{ic}} \quad \forall i = 1, \dots, N$$

so their unit costs are also proportional over time (labeled with superindices c and g),

$$\theta = \frac{\kappa_t^g}{\kappa_t^c} = \frac{\sum_i p_t^i a_{ig}}{\sum_i p_t^i a_{ic}} = \frac{\sum_i p_t^i \theta a_{ic}}{\sum_i p_t^i a_{ic}} \quad (2.26)$$

Profitability and Growth Rate Differentials without Policy

In the Flaschel-Semmler model, the green and carbon sectors produce the same output so that they share a common price $p_t^c = p_t^g$. Hence, relative efficiency in production θ , the proportion of the capital/output ratio of the green sector with respect to the carbon sector, is the only parameter regulating their profit rate differentials,

$$1 + r_t^c = \frac{p_t^c x_t^c}{\kappa_t^c x_t^c} = \frac{p_t^c}{\kappa_t^c} \quad (2.27)$$

$$1 + r_t^g = \frac{p_t^c}{\theta \kappa_t^c} = \frac{1 + r_t^c}{\theta} \quad (2.28)$$

The growth rates for the carbon and green sectors can be then calculated re-writing the laws of excess demand and profitability in discrete-time, scalar form:

$$\frac{p_{t+1}^i}{p_t^i} = 1 - \delta_p^i \sum_j (b_{ij} - R a_{ij}) x_t^j \quad (2.29)$$

$$\frac{x_{t+1}^i}{x_t^i} = 1 + \delta_x^i \sum_j p_t^j (b_{ji} - R a_{ji}) \quad (2.30)$$

in order to compute³

$$1 + g_t^c = \frac{\kappa_{t+1}^c}{\kappa_t^c} \frac{x_{t+1}^c}{x_t^c} = \left[1 - \frac{\sum_i p_t^i a_{ic} \delta_p^i \sum_j (b_{ij} - Ra_{ij}) x_j}{\sum_i p_t^i a_{ic}} \right] [1 + \delta_x^c (p_t^c - R\kappa_t^c)] \quad (2.32)$$

$$1 + g_t^g = \frac{\theta \kappa_{t+1}^c}{\theta \kappa_t^c} \frac{x_{t+1}^g}{x_t^g} = \left[1 - \frac{\sum_i p_t^i a_{ic} \delta_p^i \sum_j (b_{ij} - Ra_{ij}) x_j}{\sum_i p_t^i a_{ic}} \right] [1 + \delta_x^g (p_t^c - R\theta \kappa_t^c)] \quad (2.33)$$

so that the growth rate differential between the carbon and green sector is once again dependent on parameter θ , as well as the adjustment parameters δ_x^g , δ_x^c that relate the change in quantities with deviations from the equilibrium unit profit:

$$\frac{1 + g_t^c}{1 + g_t^g} = \frac{1 + \delta_x^c (p_t^c - R\kappa_t^c)}{1 + \delta_x^g (p_t^c - R\theta \kappa_t^c)} \quad (2.34)$$

Lower-cost green technology ($\theta < 1$) will thus ensure a greater profit rate ($r_t^g > r_t^c$) and growth rate ($g_t^g > g_t^c$).

In order to keep track of the phase-in dynamics of the green sector with respect to the carbon sector, it is convenient to define the output ratio

$$\sigma_t = \frac{x_t^g}{x_t^c} \quad (2.35)$$

which has evolution rule

$$\sigma_{t+1} = \frac{x_{t+1}^g}{x_{t+1}^c} = \sigma_t \frac{1 + \delta_x^c (p_t^c - R\kappa_t^c)}{1 + \delta_x^g (p_t^c - R\theta \kappa_t^c)} = \sigma_t \frac{1 + g_t^g}{1 + g_t^c} \quad (2.36)$$

In this paper, the Green Transition is considered to have been “successful” when the output of the green sector is equal or larger than the output of the carbon sector, so that the output ratio σ_t is equal or larger to 1. The success of the Green Transition without fiscal policy thus depends on the growth rate differential regulated by the relative efficiency parameter θ , as well as the initial output ratio σ_0 . There is a minimum time period $t^* > 0$, that is, the duration of the Green Transition, when carbon and green outputs are equal:

$$x_{t^*}^g = x_{t^*}^c \rightarrow x_0^g \prod_{t=1}^{t^*} (1 + g_t^g) = x_0^c \prod_{t=1}^{t^*} (1 + g_t^c) \quad (2.37)$$

³The time rule for unit costs is:

$$\kappa_{t+1}^i = \sum_j p_{t+1}^j a_{ji} = \sum_j p_t^j a_{ji} [1 - \delta_p^j \sum_k (b_{jk} - Ra_{jk}) x_k^j] = \kappa_t^i - \sum_j p_t^j a_{ji} \delta_p^j \sum_k (b_{jk} - Ra_{jk}) x_k^j \quad (2.31)$$

so that t^* solves:

$$\sigma_0 = \frac{x_0^g}{x_0^c} = \frac{\prod_{t=1}^{t^*}(1 + g_t^c)}{\prod_{t=1}^{t^*}(1 + g_t^g)} = \prod_{t=1}^{t^*} \frac{1 + g_t^c}{1 + g_t^g} = \prod_{t=1}^{t^*} \frac{1 + \delta_x^c(p_t^c - R\kappa_t^c)}{1 + \delta_x^g(p_t^c - R\theta\kappa_t^c)} \quad (2.38)$$

Profitability and Growth Rate Differentials with Policy

The welfare problem that the central planner faces employs action variable tax τ to maximize the ratio of green output over carbon output σ_t over time, which now depends on a tax-subsidy mix τ :

$$\max_{\tau} \sum_t \beta^t \sigma_t \quad (2.39)$$

Since discount factor $\beta < 1$, the central planner wants to accelerate directed technical change towards decarbonization as much as possible. In order to increase the green growth rate differential and thus output proportion σ_t , the central planner introduces a tax $0 < \tau < 1$ on carbon output τx_c after production, which is used to finance a subsidy of the same value to the green sector. The outputs using the tax-subsidy mix (where the hat notation distinguishes the variable with and without policy) thus become:

$$\hat{x}_t^c = x_t^c - \tau x_t^c = x_t^c(1 - \tau)$$

$$\hat{x}_t^g = x_t^g + \tau x_t^c = x_t^c(\sigma_t + \tau)$$

Output proportion with policy becomes:

$$\hat{\sigma}_t = \frac{\hat{x}_t^g}{\hat{x}_t^c} = \frac{\sigma_t + \tau}{1 - \tau} \quad (2.40)$$

The profit rates for the carbon and green sectors internalize the tax-subsidy policy:

$$1 + \hat{r}_t^c = \frac{p_t^c x_t^c(1 - \tau)}{\kappa_t^c x_t^c} = (1 + r_t^c)(1 - \tau) \quad (2.41)$$

$$1 + \hat{r}_t^g = \frac{p_t^c x_t^c(\sigma_t + \tau)}{\theta \kappa_t^c x_t^c \sigma_t} = (1 + r_t^c) \frac{1 + \frac{\tau}{\sigma_t}}{\theta} \quad (2.42)$$

While the negative contribution of tax τ is linear on the carbon sector, its positive effects on green profitability depend on the fraction $\frac{\tau}{\sigma_t}$, i.e. they are the largest when carbon output is much larger than green output ($\sigma_t \sim 0$), that is, at the beginning of the introduction of the policy, and they are multiplied by capital efficiency θ . The profitability differential once the policy is introduced can be then computed in terms of capital efficiency θ , output proportion σ_t , and tax rate τ :

$$\frac{1 + \hat{r}_t^c}{1 + \hat{r}_t^g} = \frac{\sigma_t \theta (1 - \tau)}{\sigma_t + \tau} = \frac{\theta (1 - \tau)}{1 + \frac{\tau}{\sigma_t}} \quad (2.43)$$

which shows how a tax-subsidy policy can reinforce cost differentials (when $\theta < 1$) or even offset them (when $\theta > 1$). Higher policy-induced green profitability will make capital flow out of the carbon sector to the green sector faster than without policy. The growth rates with and without policy can be computed for the carbon and green sectors:

$$1 + \hat{g}_t^c = \frac{\kappa_{t+1}^c x_{t+1}^c (1 - \tau)}{\kappa_t^c x_t^c} = (1 + g_t^c)(1 - \tau) \quad (2.44)$$

$$1 + \hat{g}_t^g = \frac{\theta \kappa_{t+1}^c x_{t+1}^g + \tau x_{t+1}^c}{\theta \kappa_t^c x_t^g} = (1 + g_t^c) \frac{\sigma_{t+1} + \tau}{\sigma_t} = (1 + g_t^c) \left[\frac{1 + \delta_x(p_t^c - R\theta \sum_i p_t^i a_{ic})}{1 + \delta_x(p_t^c - R \sum_i p_t^i a_{ic})} + \frac{\tau}{\sigma_t} \right] \quad (2.45)$$

so the growth rate differential with policy can be compared with the growth rate differential without policy:

$$\frac{1 + \hat{g}_t^g}{1 + \hat{g}_t^c} = \left[\frac{1 + \delta_x(p_t^c - R\theta \sum_i p_t^i a_{ic})}{1 + \delta_x(p_t^c - R \sum_i p_t^i a_{ic})} + \frac{\tau}{\sigma_t} \right] \frac{1}{1 - \tau} = \left[\frac{1 + g_t^g}{1 + g_t^c} + \frac{\tau}{\sigma_t} \right] \frac{1}{1 - \tau} \quad (2.46)$$

Once again, the additive presence of the ratio $\frac{\tau}{\sigma_t}$ shows that the policy to direct technical change towards decarbonization is the most effective at the earliest stages of the phase-in (when $\sigma_t \sim 0$).

2.4.2 Simulations

The simulations target two of the economic sectors with highest carbon content, “Electricity, gas, steam and air conditioning supply” (D) and “Manufacture of coke and refined petroleum products” (C19), and study their phase-out by time t^* by equivalent green sectors with relative efficiency in production θ under a tax-subsidy rate τ and initial output ratio σ_0 .

The IPCC imposes many duration targets for the Green Transition [Hausfather, 2018]:

- 16 years for a 66% chance of avoiding a temperature increase of 1.5 degrees Celsius,
- 23 years for a 50% chance of avoiding a temperature increase of 1.5 degrees Celsius,
- 51 years for a 66% chance of avoiding a temperature increase of 2 degrees Celsius, and
- 65 years for a 50% chance of avoiding a temperature increase of 2 degrees Celsius

At the current moment, green energy is increasingly outcompeting carbon energy: the cost of green energy can be considered to be between 0.7 and 1.1 times the cost of carbon energy [figure 2.6]. Figure 2.7 plots many indicators that characterize the time evolution of the relevance of sector C19 within a variety of national economies, as computed from their corresponding input-output matrices. For OECD economies, the current share of final energy consumption in renewable sources over carbon sources is around 5% [Upadhyaya, 2010], which is the benchmark value that is taken for the output ratio σ_0 in the simulations.

Figures 2.8 and 2.9 show the simulations of the Green Transition without and with fiscal policy for the C19 sector and the economy of Germany, which show to what extent a small

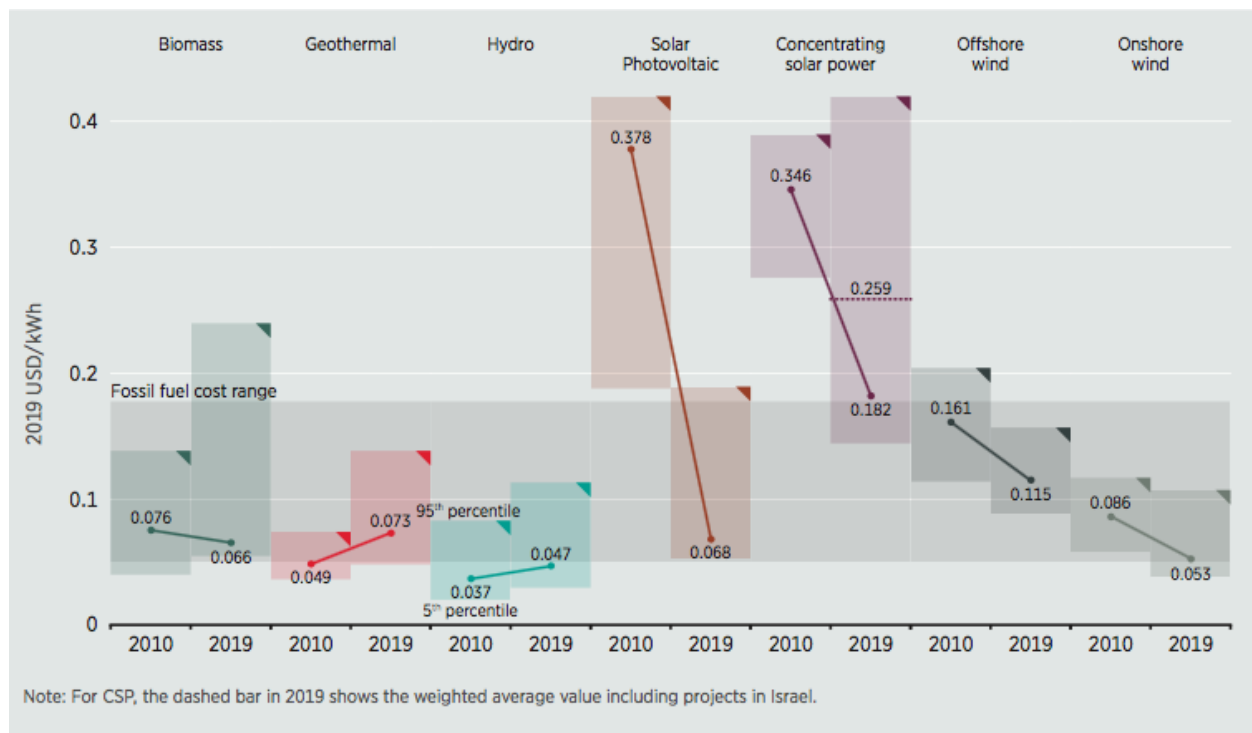


Figure 2.6: **Global weighted average levelised cost of electricity from utility-scale renewable power generation technologies, 2010 and 2019** Fossil fuel cost range is depicted in gray [IRENA, 2020].

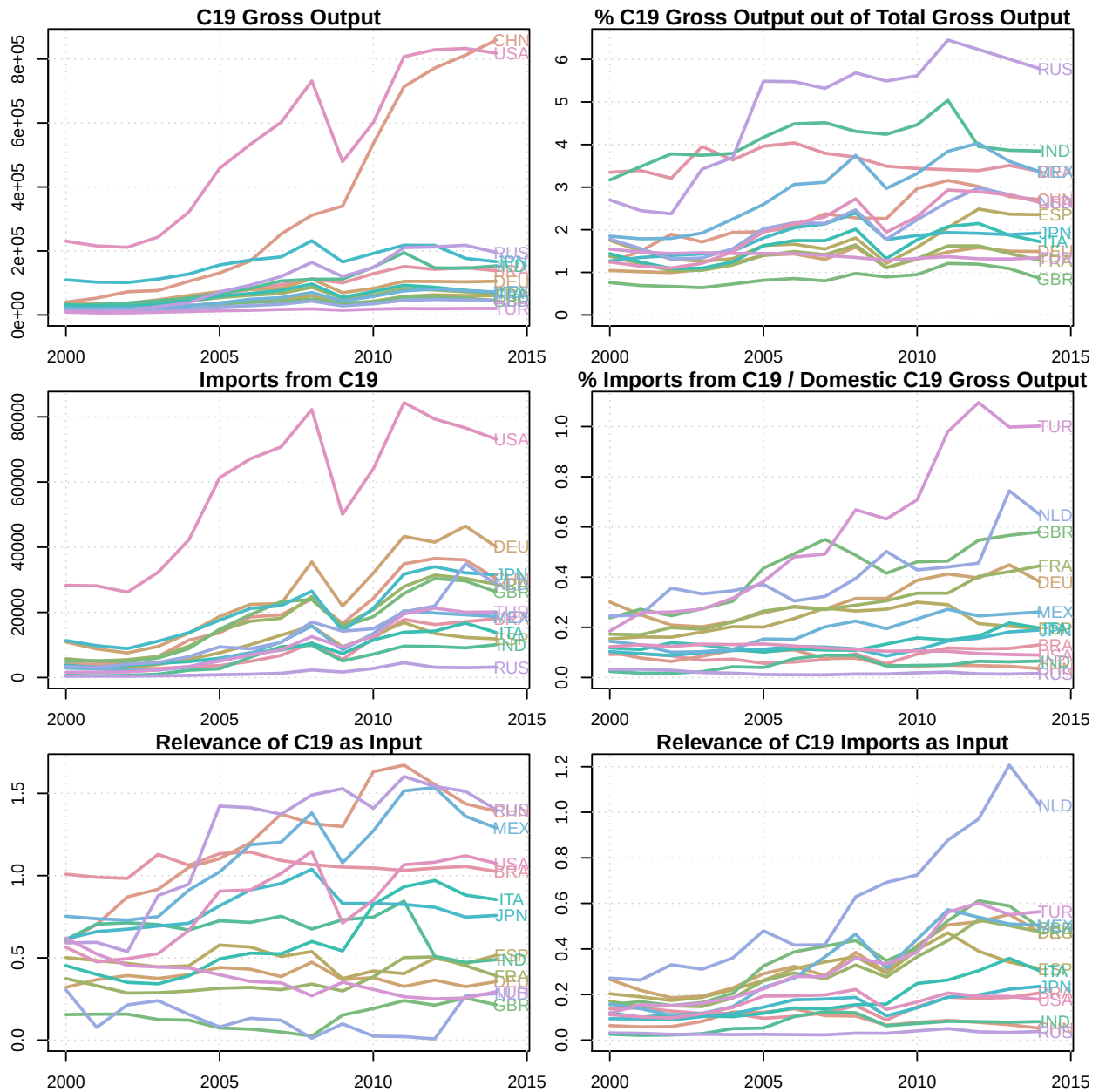


Figure 2.7: **Time evolution of the many indicators for sector C19 within a variety of national economies** The relevance of C19 as input and as import input is computed as the row-sum of the input-output matrix and import matrix, respectively.

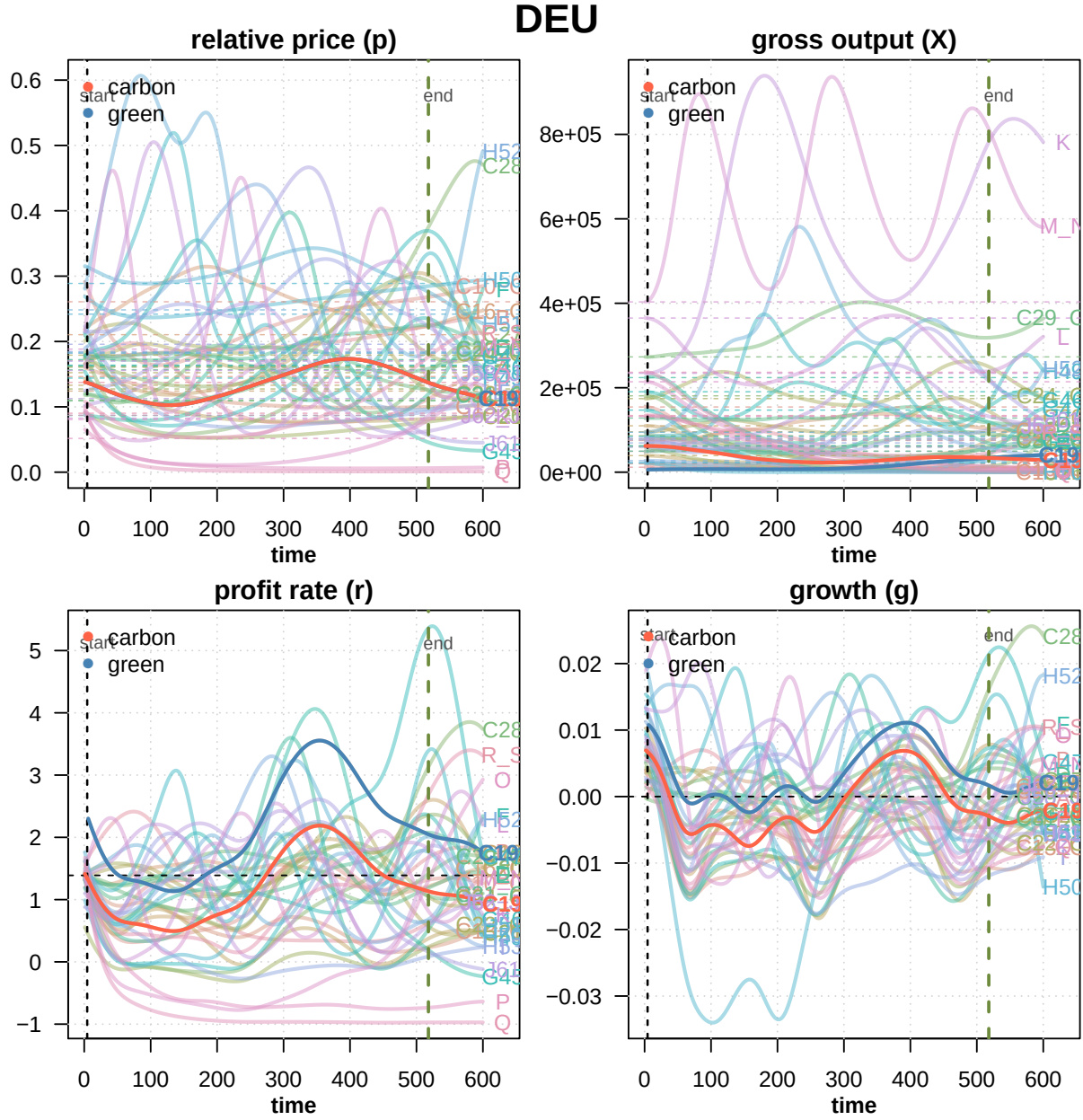


Figure 2.8: **Simulation of the Green Transition of sector C19 for Germany, with $\theta = 0.7$, $\sigma_0 = 0.1$, and no fiscal policy ($\tau = 0$)** Without fiscal policy, it takes more than $t^* \sim 500$ timesteps for the green sector to take over the carbon sector.

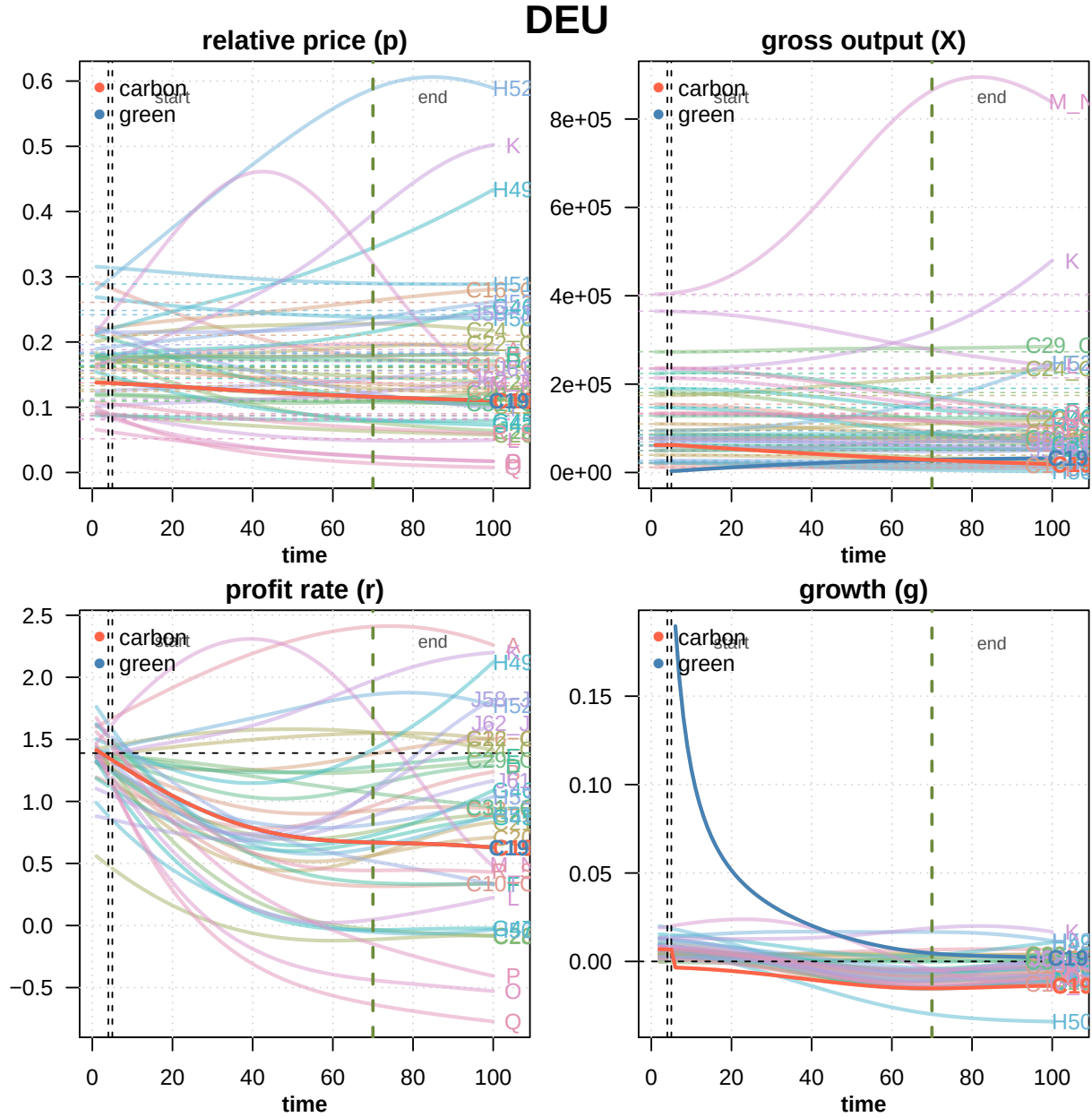


Figure 2.9: **Simulation of the Green Transition of sector C19 for Germany, with $\theta = 1$, $\sigma_0 = 0.055$, with fiscal policy ($\tau = 0.01$)** Despite the carbon sector being as cost efficient as the green sector, a tax rate of 1% greatly reduces the duration of the Green Transition to $t^* \sim 70$ timesteps.

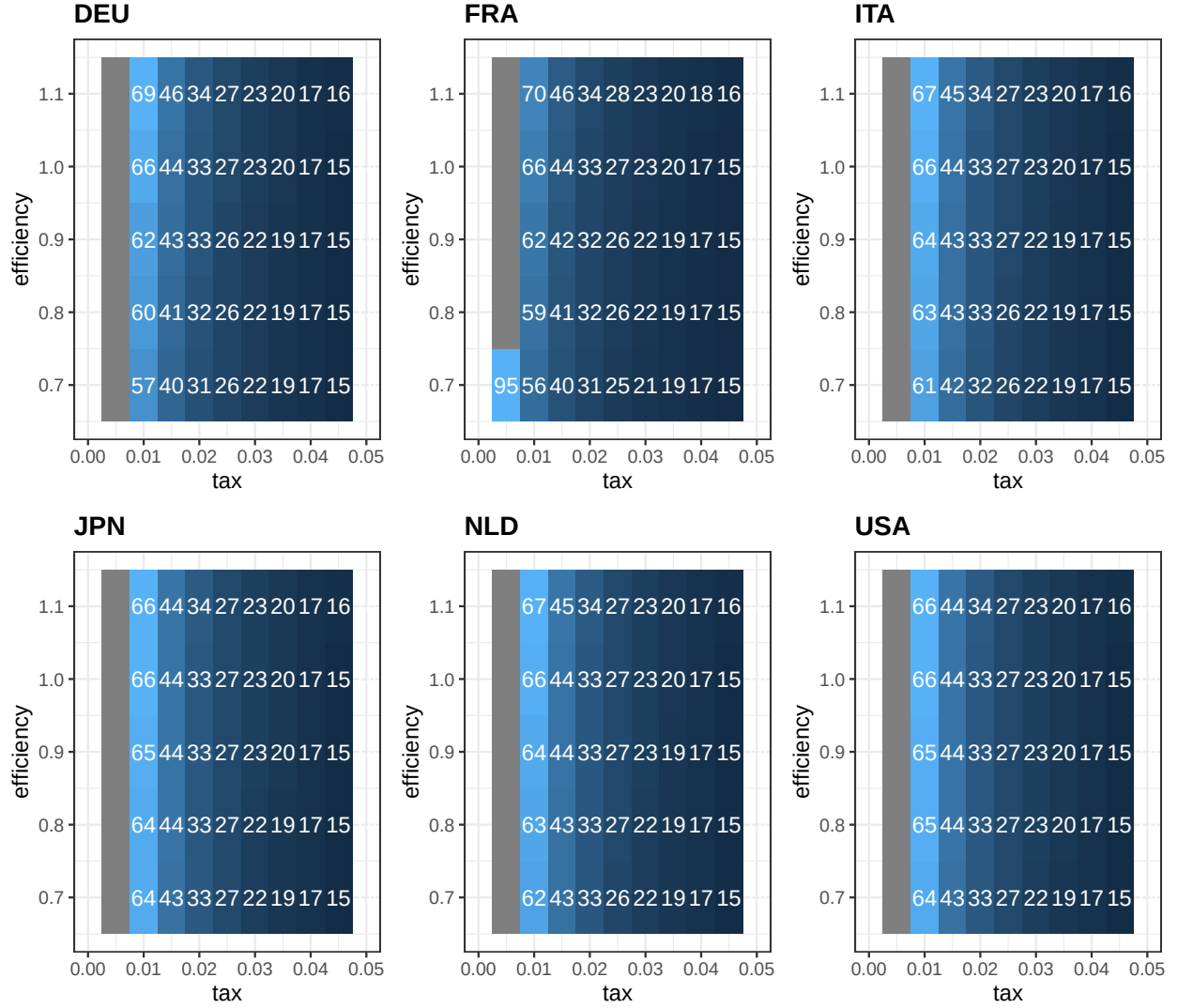


Figure 2.10: Duration of the Green Transition t^* of sector D for selected countries, with respect to the tax rate τ and relative efficiency θ , and initial output ratio $\sigma_0 = 0.05$. The tax rate appears to accelerate the Green Transition exponentially, while relative efficiency increases its speed linearly. Gray squares indicate a duration that is greater than 100 timesteps.

tax rate can greatly accelerate the Green Transition (even for green sectors that are less cost efficient). Each timestep can be considered as one year, given the time dimension included in the adjustment coefficients, which were computed with yearly data. Since the green sector is more cost efficient, the acceleration of the Green Transition also implies a faster reduction in the relative price of the targeted sector and thus a general increase in economic efficiency. Figure 2.10 shows the duration of the Green Transition for the selected countries with respect to cost efficiency θ and the tax rate τ : the tax rate appears to accelerate the Green Transition exponentially, while relative efficiency increases its speed linearly. A tax rate of 1% appears to be the value that brings the duration of the Green Transition within the least ambitious IPCC targets, while the most ambitious targets would require tax rates of 5%.

2.5 Conclusions

The Flaschel-Semmler dynamical model of multi-sector growth is employed as a machine-learning algorithm to study the speed of substitution of a carbon-based energy sector by a renewable-energy sector under directed technical change, that is, the Green Transition. This theoretical model of linear production features a complex pattern of oscillations of prices and outputs around equilibrium values as determined by the technology structure, augmented to include the wage rate. It relies on a linear form of cross-dual dynamic adjustment as established by two abstract laws, the law of excess demand and the law of excess profitability, which are evaluated empirically: using WIOD and EU KLEMS datasets, their linear adjustment coefficients for six countries (Germany, France, Italy, Netherlands, Japan, and the US) are computed in order to empirically calibrate the model. At the current moment, conventional ordinary least squares regressions are implemented per country and sector. This retrieves highly significant values when adjustment coefficients are assumed to be uniform across sectors, while results are mixed when disaggregating by sector.

The next step in the research is to obtain more precise datasets with a longer timespan (from national statistics bureaus rather than international tables) where to regress using country and sector as dummy variables that can better retrieve the hierarchical structure of the data, potentially employing Bayesian and maximum-entropy methods. In particular, the assumption of circulating capital is very stringent and the existence of fixed capital and depreciation may distort the calculations; Flaschel and Semmler address this very issue in a contribution of theirs that builds on Bródy [Bródy, 1974; Flaschel and Semmler, 1986]. Secondly, the method of calculating demand and supply is problematic: in national income accounts, the discrepancy between demand and supply is added to one side so that the two sides balance. This discrepancy can be captured by measuring unintended inventory change, as Shaikh shows [Shaikh, 2016, pp.120-128]. Furthermore, other functional forms of adjustment could be tested, for instance where the regressions could be logistic instead of linear, retrieving a logistic kind of dynamic adjustment process,

$$y_{i,t} = \frac{1}{1 + \exp(-\beta_i x_{i,t})} \quad (2.47)$$

which is very interesting to explore numerically stability-wise as an extension of the Flaschel-Semmler model. In this sense, this theoretical model works as a form of supervised machine learning using linear regressions on training data.

The speed of substitution of a carbon-based energy sector by a renewable-energy sector is evaluated analytically and computationally for the six selected countries with respect to the relative cost efficiency between the carbon and green sectors, the initial output ratio, and the tax rate. Directed technical change is enforced by a revenue-neutral, pro-active fiscal policy of a tax-subsidy form, which has the effect to greatly accelerate the phase-out of the carbon sector, in particular at its earliest stages, in line with the contributions of Acemoglu et al. [Acemoglu et al., 2012, 2016]. A comparison of the tax policy without subsidy (that is, a carbon tax) and with the subsidy seems very relevant in order to evaluate whether carbon taxation is enough to achieve effective decarbonization in time. Without fiscal policy, it is impossible for any economy to reach the IPCC targets for decarbonization.

Chapter 3

Uneven Development as the Outcome of Economic Integration: Analysis of the Statistical Structure of Italian Economic Geography

This paper empirically addresses the economic geography of uneven development in the context of the Italian regions during the European Monetary Union. It applies a machine-learning method, Gaussian mixture modeling, to identify clusters on a wide variety of socio-economic indicators of Eurostat data and classify Italian regions accordingly in a simple and parsimonious way. The analysis of the complex statistical structure of Italian regional inequality confirms that it is most importantly characterized by its marked dual polarization into North and South as core and periphery, but still shows the empirical relevance of Arnaldo Bagnasco's "Third Italy" hypothesis. Further analysis of sectoral wage shares of income corroborates the more complex bi-modal/tri-modal structure in the economy. Regional rates of gross value added follow a Laplace distribution, while the distributions of sectoral compensation and gross value added follow a power law. After 160 years of political and monetary unification, there is no evidence for any convergence of Italian regions in most of the data. In the broader context of European economic integration, the Italian economic geography is characterized by protracted peripheralization. The paper concludes with a political-economic argument on how the interplay of trade and capital-account imbalances leading to such peripheralization fuel the Euroskeptic populist backlash against EU-led globalization.

3.1 Introduction

Whether economic integration between countries or regions within the same country leads to income convergence or divergence (in the form of core-periphery dynamics of uneven development) is an important subject of debate among the different economic theories of free trade, especially within the current wave of globalization and European monetary unification.

When the Maastricht Treaty that founded the European Union was signed in 1992, the conventional expectation among economists was that poor countries would grow faster than rich countries so that they would eventually catch up [Solow, 1956; Barro and Sala-i Martin, 1995]. However, the rise in within-country income inequality, the robust persistence of bi-modal distributions of country incomes [Quah, 1996b], the existence of a middle-income trap [Felipe et al., 2012], and the most recent political backlashes against economic integration around the world have led many to review the supposed benefits of free trade [Stiglitz, 2002; Rodrik, 2011, 2020].

In fact, all economic theories (neoclassical, neo-Keynesian, structuralist, post-Keynesian, evolutionary, and classical) address the possibility of core-periphery dynamics and uneven development between countries and regions within the same country. Except the structuralist model that relies on the Prebisch-Singer hypothesis, all other theories agree in the fundamental role of increasing returns to scale and cumulative causation underlying the agglomeration of economies in the form of clusters of economic activity, which results in a geographical concentration and centralization of capital. However, since they disagree in the actual causal mechanisms at play, this makes difficult to empirically identify which particular countries and regions actually belong to the core-periphery dichotomy that most theoretical models presume. In other words, the fundamental and complex question of how ‘regional’ and ‘local’ economies can be meaningfully conceptualised, and how such conceptions can be translated into empirical terms, is not considered at all [Martin, 1999].

This paper empirically addresses the question of the proper characterization of clusters of economic activity by employing an atheoretical, ‘distribution’ approach, in line with Quah’s “twin-peaks” work [1996b] as well as the econophysics literature [Drăgulescu and Yakovenko, 2001; Scharfenaker and Foley, 2017; Shaikh, 2020], in the context of Italy, which is seen as a case study or ‘natural experiment’ of 160 years of political and economic integration. The ‘Southern Question’ on the origins and causes of the divergence between Northern Italy and the Mezzogiorno has been intensely debated in political and academic circles since Unification in 1871 [Graziani, 1978; DiGiorgi et al., 1980; Fenoaltea, 2003; Felice, 2012; Felice and Vecchi, 2015; de Oliveira and Guerriero, 2018; Federico et al., 2019]. At the current moment, there is disagreement whether the divergence started with unification or instead predated it [Missiaia, 2016; Ciccarelli and Fenoaltea, 2013]. The analysis of the statistical structure of a variety of Italian regional and socio-economic Eurostat indicators for the 2008-2017 period reveals a complex economic geography that confirms it to be, as expected, fundamentally characterized by the dual polarization between North and South, akin to the bi-modality found for the global distribution of country incomes. The clustering method allows to further evaluate empirical hypotheses for higher number of clusters, and the analysis shows qualified support for Arnaldo Bagnasco’s hypothesis of tri-modality for Italy.

In this direction, most of the empirical growth literature in neoclassical analysis emphasizes the role of group heterogeneity, differential club convergence, and regional variation in shaping economic growth [Quah, 1996a; López-Bazo et al., 1999; Boldrin and Canova, 2001; Magrini, 2004; Fischer and Stirböck, 2006; Geppert and Stephan, 2008; Franks et al., 2018; Haupt et al., 2018]. However, many of these studies employ the conventional cross-sectional

regression analyses of growth with respect to initial income and a variety of determinants as regressors and use the problematic Cobb-Douglas production function [Felipe and McCombie, 2014]. Such methods may suffer from the underspecification of regressors, as well as nonlinearities in the growth equation, which may obfuscate the actual statistical structure of the socio-economic variables and their interactions. This has led many to develop nonparametric models that can be time-varying [Phillips and Sul, 2007, 2009; Haupt and Petring, 2011; Haupt et al., 2018]. In a similar spirit, an unsupervised machine-learning method is employed for cluster identification, Gaussian mixture modeling, that presupposes the data as a superposition of Gaussian distributions and quantifies the probability of a particular region of belonging to a particular cluster. Rather than imposing a particular number of clusters on the data, the algorithm chooses the optimal number of components or clusters through the Bayesian information criterion (BIC).

I first examine the distribution of regional income per capita as a simple uni-variate case, which reveals, as expected, the markedly bi-modal polarization between North and South, although the potential existence of subclusters paints a much richer picture on its regional structure in conversation with the debates on the ‘Third Italy’ hypothesis [Bagnasco]. Clustering analysis is further applied on 260 socio-economic regional indicators in demographics, health, poverty, education, tourism, and transportation, which reveals a robust pattern of uneven development in most of them. Further, uni-variate and multi-variate cluster analysis is performed on more specific economic variables: gross fixed capital formation as % of GDP, the wage share of income (employee compensation as % of GDP), the unemployment rate, the rate of gross value added (GVA), household disposable income, and labor productivity (GDP over labor-hours). Income shows a very strong anti-correlation with unemployment and a weaker correlation with labor productivity. Considering that sectoral disaggregation is also available in Eurostat, I evaluate the complex geographical and sectoral structure of the wage share of income (as % of GVA) and labor productivity (as GVA over labor-hours). While the aggregate wage share of income (a proxy of unit labor costs) is one of the few dimensions for which the data is regionally homogeneous, the sectoral decomposition reveals a pattern of geographical differentiation in cost competitiveness between North and South. Further, maximum-likelihood estimation returns with high significance a characteristic tent-shaped Laplace distribution for the regional rates of gross value added, as well as power-law distributions for sectoral wage bill and gross value added.

3.2 Uneven development and theories of trade

In this section, I briefly review what the different economic theories of trade and growth can say with respect to Italy’s Southern Question.

In general, neoclassical theory expects income convergence between trading countries and regions. However, the neoclassical approach is widely based on the problematic Cobb-Douglas production function, which is just a mathematical approximation to the accounting rule that net output is divided into profits and wages [?]. The conventional Heckscher-Ohlin model of free trade [1991], which builds on the comparative-advantage theory of Ricardo

[1891], precisely assumes away what characterizes trade between core and a periphery: in the model, there are no differences in technology, no mobility of capital and labor, and no unemployment. On the basis of diminishing returns to capital, the Solow growth model predicts all closed economies to eventually reach a balanced-growth steady state that is determined as the ratio of the savings rate to the growth of the labor force, technological progress, and the depreciation rate. Poorer countries should catch up with richer ones in the case all these parameters are constant and homogeneous across countries; in fact, these parameters not only change over time [Barro and Sala-i Martin, 1995] but they are interrelated between countries [Quah, 1993; Pesaran and Smith, 1995]. Following the development of endogenous growth theory, which simply consisted in adding further determinants of income times an elasticity into a logarithmic specification of the Cobb-Douglas function, Barro and Sala-i-Martin also predicted unconditional convergence under constant returns to scale for both countries and regions [Barro and Sala-i Martin, 1995]. Such a linear functional specification became extremely convenient to perform cross-sectional regression panels and estimate the elasticities (that the model itself tautologically imposes on the data), triggering a wealth of literature in empirical growth. However, most of these studies found very low convergence rates [Martin, 1999] and instead emphasized the role of group heterogeneity, differential club convergence, and regional variation in shaping economic growth and found evidence for persistent bi-modal distributions of country income [Quah, 1996a; López-Bazo et al., 1999; Boldrin and Canova, 2001; Magrini, 2004; Fischer and Stirböck, 2006; Geppert and Stephan, 2008; Franks et al., 2018; Haupt et al., 2018]. Variations in regional convergence are thus significant and reflect complex, geographically-differentiated processes [Dunford, 1993; Vickerman and Armstrong, 1995].

In contrast, the New Economic Geography is premised on the argument that the observed patterns of trade and specialization are driven by increasing returns, economies of scale, cumulative causation, and imperfect competition; and that the market, technological and other externalities underpinning increasing returns arise through a process of regional or local economic agglomeration [Krugman, 1991, 1998; Arthur et al., 1994; Venables, 1996a,b; Ottaviano and Puga, 1998]. Local clustering in economic activity is driven by Marshallian externalities such as technological spillovers and supply-and-demand linkages in intermediates goods [Marshall, 1890; Mongiovi, 1996]. On a broader regional level, pecuniary externalities in the form of market-size effects are also important, leading to a large-scale core-periphery pattern of economic development within nations [Krugman, 1991]. Unfortunately, unemployment is once again assumed away. Transport costs and labor mobility are the key determinants to account for spatial agglomeration or dispersion: the more mobile labor is, the more agglomeration will prevail over dispersion. Bertola develops an endogenous growth model to show how capital and labor mobility lead, via localized increasing returns, to the increasing concentration of economic activity and growth in some locations at the expense from others; Bertola thus expects European economic integration to likely lead to greater regional concentration of production [Bertola, 1993]. In a critical review, Martin argues that the New Economic Geography is a mere re-invention, by applying highly abstract mathematical models that fail to encapsulate the empirical realities of economic

geography, of the long-standing German tradition of equilibrium location theory, as well as regional science [Martin, 1999].

Krugman's argument shares its emphasis on cumulative causation and path dependency with Kaldor's export-led growth model in the demand-led post-Keynesian tradition, which also features core-periphery divergence on the basis of increasing returns to scale [Kaldor, 1970; Setterfield, 1997]. In Dixon and Thirlwall's formalization of Kaldor's model (as well as Setterfield's), technical progress occurs endogenously through Verdoorn's law, which expresses productivity growth as an increasing function of output growth, e.g. "learning by doing" [Dixon and Thirlwall, 1975, 1978; Setterfield, 1997, 2013]. It can thus be interpreted as representing increasing returns akin to Adam Smith's notion of the extent of the market and thus constitutes the mechanism at the heart of cumulative causation. As well, exports are considered the key source of autonomous demand. Kaldor already presented a verbal model specifically to address interregional differences in growth, where core-periphery divergence then becomes locked in due to initial differences in regional income. Due to the conceptual similarities to New Economic Geography, Kaldor's model as formalized by Thirlwall has been argued to have anticipated it by twenty years [McGregor and Swales, 1986; Bhattacharjee, 2010].

In a similar direction, evolutionary economic geography extends the work on increasing returns to scale by the new economic geography and post-Keynesian theory to focus on the study of agglomeration economies by formalizing the existence of spillovers in technological innovation on the stock of knowledge [Silverberg and Verspagen, 1995; Maurseth and Verspagen, 2009; Boschma and Martin, 2010; Harris, 2011]. Caniels and Verspagen [2001] model core-periphery dynamics considering that the intensity of spillovers depends on geographical distance between regions. The removal of country borders, thought as barriers to technological diffusion, may exacerbate centrifugal polarization, which leads Caniels and Verspagen to caution against European economic integration.

Thirlwall further extended Kaldor's theory into the balance-of-payments constrained growth model [Thirlwall, 1979; McCombie and Thirlwall, 2016], which espouses the view that international trade can drive long run growth without creating external imbalances [Setterfield, 1997, 2011]. It thus assumes away the chronic trade deficits that peripheries characteristically suffer amidst global and regional competition, as well as changes in the terms of trade (which are assumed constant), and international financial flows can only influence short-run growth [Setterfield, 1997, 2011, 2013]. By relaxing these assumptions, Thirlwall and Hussain find significant and persistent fluctuations in terms of trade changes and capital inflows, which provides a better picture of the conditions of the periphery [Thirlwall and Hussain, 1982]. While the structuralist literature also emphasizes the role of cumulative causation in interregional divergence [Hirschman, 1958], the Prebisch-Singer hypothesis precisely argues contra the assumption of constant terms of trade, which implies non-price competition prevails in international trade, that the price of primary commodities –produced by the periphery– declines relative to the price of manufactured goods –produced by the core– over the long term, which causes the terms of trade of primary-product-based economies to deteriorate. Empirical evidence gives qualified support to the Prebisch-Singer hypothesis [Harvey

et al., 2010]. Dutt [2002] interestingly builds on Thirlwall's law to provide a core-periphery model that addresses explicitly changing terms of trade and the distinction between short-run and long-run dynamics, where a core with excess capacity follows demand-led Kaleckian growth while the periphery is based on a Marx-Lewis model with fixed real wage and unemployed labor. Dávila-Fernández and Sordi incorporate Goodwin's distributive cycle model to a balance-of-payments-constrained open economy in the direction of Pugno [1996; 1998] by allowing technical change to be endogenous to the cyclical dynamics of the system, which generates a Hopf bifurcation [Dávila-Fernández and Sordi, 2019]. Spinola [2020] addresses the transition dynamics incompatibility between the balance-of-payments constrained model and the Prebisch-Singer hypothesis. Spinola finds damped cycles towards convergence by extending Dutt's model by adding endogenous productivity gap dynamics, including a Phillips wage curve, and modelling Goodwin labor supply dynamics that considers the labor transfer issue between traditional and modern sectors.

In the context of Thirlwall's law, Setterfield argues that permanent net inflows of capital (in domestic currency) that cover persistent imbalances have actually no effect in the long run [Setterfield, 1997]. Moreno-Brid reaches the same result by assuming the growth of financial inflows is equal to the growth in output [Moreno-Brid, 1998]. The classical theory of trade [Shaikh, 1980] critically disputes this argument. In the classical theory of trade, trade imbalances follow absolute, not comparative, cost differentials due to differences in increasing-to-returns technology between countries. Economies of scale also play a fundamental role in classical competition, since they are major means of lowering costs and hence gaining competitive advantage – in addition to wage repression. The feedback effect from prices to costs as identified by Sraffa implies that comparative costs may not change at all in response to changes in the exchange rates and hence do not follow Ricardo's comparative-cost story of self-stabilizing trade imbalances [Shaikh, 2016, p.509].

As observed in reality, trade imbalances are not self-stabilizing by automatic real exchange rate adjustments on the basis of the quantity theory of money; instead, they are remarkably chronic and persistent. Persistent trade imbalances will create imbalances of payments in the capital account that will affect interest rates and induce short-term international capital flows from the core to the periphery until the interest-rate differentials are balanced. Economic integration will consolidate: countries with absolute cost advantage will recycle their trade surpluses by lending to the periphery, which will use foreign borrowing to cover their trade deficits and thus chronically burying it in debt due to interest payments. Between regions within the same country, the state will have to cover the trade imbalance via fiscal transfers – which is what happened in Italy with the advent of Fascism in the 1920s and in the postwar period of 1951-71 [Graziani, 1978; DiGiorgi et al., 1980]. A core-periphery pattern in interregional and international development will thus inevitably emerge under classical competition.

Under a regime of fixed exchange rates and in the absence of stabilizing automatic fiscal transfers, fiscal consolidation in the advent of the sovereign debt crisis will repress real wages and thus transfers income from low-saving workers to high-saving rent-seeking landowners, who had already been benefiting from the inflows of capital from core to periphery. In fact,

political and economic integration of the core and the periphery under the state always follows a social pact between the capitalists in the core and the landlords in the periphery, where the former sells their excess produce to the latter – one can think of the Italy, Spain, US, or the European Union as a whole. Landlords as owners of nontradable goods constitute an essential actor to understand core-periphery dynamics of political and economic integration that models of trade tend to omit [Shaikh, 2016, p.516].

The effects of competitive wage deflation –wage-seeking labor flows from core to periphery, heightened social tensions, and polarizing income inequality– are thus confronted by pro-austerity policy-making elites with increased domestic and border policing and militarization [Jayadev and Bowles, 2006]. This induces a protracted decline in the liberal-democratic features of the capitalist system [Lührmann et al., 2018] and destabilizing societal responses in the form of anti-establishment populisms vying for inflationary fiscal spending once again, focused on either revolution (socialism) or reaction (fascism) – a political business cycle of the kind described by Polanyi and Kalecki in their seminal works [Kalecki, 1943; Polanyi, 1944]. For Polanyi, it was the systematic devaluation of the wage, which resulted from the workings of the Gold Standard and laissez-faire trade, that regularly imposed a level of social dislocation that eventually led to the rise of political extremes on both sides of the political spectrum (e.g. fascism and communism). Hence, the core-periphery dynamic in its classical form eventually results in the interplay of growing authoritarianism and destabilizing political polarization, that is, “populist backlash” against economic integration – a “integration-breeds-disintegration” hypothesis in line with Rodrik [2020]. In particular, Arrighi arguably considered the peripheral status of Italy to be the main reason why it developed Fascism in the 1920s [Arrighi, 1985].

3.3 Italy’s Southern Question

The origins and the historical evolution of the gap in economic performance and living standards between Italy’s North and South remains an unsettled and controversial issue in Italian economics and politics since political unification in 1871 [Graziani, 1978; DiGiorgi et al., 1980; Fenoaltea, 2003; Felice, 2012; Felice and Vecchi, 2015; de Oliveira and Guerriero, 2018; Federico et al., 2019]. Since the early 2000s, economic historians have started to rely on data, but this ‘quantitative turn’ has not yet settled the issue. Trends in regional GDP per capita are fairly well established for the 20th century: in 160 of years of internal political and monetary union under free trade, regional incomes of North and South have constantly *diverged* starting from similar levels, except for the period of the *miracolo economico* between 1951 and 1971, during which much state investment was spent in the South [Felice, 2012; Federico et al., 2019]. In particular, the gap widened with the industrialization of the North in the three decades before World War 1 and peaked just after World War 2, reduced during the 1951-1971 period of the *miracolo economico*, and widened again after 1971 [Felice, 2012; Federico et al., 2019]. Hence, it appears that it was during the forty years after unification (1871-1911) that the South effectively de-industrialized while the North developed its manufacturing power [Ciccarelli and Fenoaltea, 2013]. Missiaia suggests home market access

as the prime driver of the North-South divergence in Italy, while Ciccarelli mentions initial productivity differentials – which would lend credence to an explanation based on increasing returns to scale and cumulative causation. However, Federico et al. [2019] contend that real wages already diverged before unification, so the gap must be located in the marked difference in literacy rates between North and South in shaping long-run economic growth.

For DiGiorgi [1980], the continuing underdevelopment of Southern Italy is the a result of deliberate state policy, rather than as a failed attempt at development. Before 1945, the State supported the Southern landowning oligarchies. Since then, it has supplanted their power with that of a new state bourgeoisie – privileged revenue recipients supported by public expenditure. This was done to respond to social tensions and to strengthen the Christian Democratic regime, as well as to create markets for Northern industry’s durable goods. In this direction, Graziani [1978] argues that economic developments in the Mezzogiorno can only be properly understood if they are considered in relation to the structural transformation of Italy as a whole. Graziani counters a common view that stresses that the implementation of development policy is distorted by ‘vested interests’ that prevent such measures from being successfully applied. Such ‘vested interests’ are treated somehow as exogenous to the political economy of core-periphery relations, but they are no-one else than the Southern landlord oligarchies that benefited the most from political and economic integration with the North and hence should be treated as endogenous. In this direction, Graziani [1978] finds a high correlation between Southern net imports and current transfers of external aid to the Mezzogiorno (including government contributions, subsidies, pensions and the remittances of emigrant workers), as well as a moderately good correlation for north-west Italy which, being the main source of finance for the Mezzogiorno.

In the literature on world systems, Immanuel Wallerstein [1974] identified “the southern tiers of Italy” as semi-peripheral economy, locked in between the core of advanced capitalist nations and the periphery in structural dependency, or Global South in more modern terms. For Giovanni Arrighi, Wallerstein’s definition of the semi-periphery was ambiguous, suggesting to consider it in terms of the international division of labor, following Prebisch’s core-periphery analysis for Argentina [Arrighi and Drangel, 1986]. Like Prebisch, Arrighi considered the semi-periphery to have a geographical character, to be a place, and identified it with southern Europe, with Calabria as a case study [Arrighi, 1985]. Following Arrighi, Massimo Paci denounced the particularly pronounced dualism afflicting the Italian economy in his compelling unified theory of Italian capitalism as an *economia periferica* in which Italy’s petty capitalist small firm economy not only served for Italian monopoly capitalist out-sourcing, but also undermined worker unrest by dividing the working class and its unions between core and periphery workers, and absorbed cheaply surplus laboring populations such as women, former rural migrants to the big cities, as well as young and old people expelled from mature industrial sector in times of economic crisis [Paci, 1973]. Based on the strength and the geographical distribution of small-firm economies, Arnaldo Bagnasco countered with the hypothesis that central and northeastern Italy constituted a social formation just as specific as that of the north and south: there were indeed not two but three clusters of related regions, each cluster with an empirically discernible way of life

and standard of living distinct from the other two. This “Third Italy” consisted of the regions of Trentino-Alto Adige, Veneto, Friuli-Venezia Giulia, Emilia Romagna, Toscana, Umbria, and Marche [Bagnasco, 1977]. In his formulation of the “Third Italy” hypothesis, Bagnasco searched for geographical clusters while analyzing a variety of empirical measures, such as criminality, divorce rates, homeownership, ratio of home space to persons, days lost due to labor-management conflicts, election results, non-food consumption, hospital beds per population, and family size.

3.4 A Gaussian Mixing Model of the Geographical Division of Labor

This paper aims for a methodological contribution in the form of an econometric application of machine-learning methods of cluster analysis in order to properly characterize the agglomerative nature of economic activity in the form of a geographical division of labor. Datasets are viewed as statistically interdependent distributions aggregated over time, instead of time series that require a convenient Cobb-Douglas functional specification that may suffer from underspecification and nonlinearity [Phillips and Sul, Haupt and Semmler]. Most economic indicators used for policy are generally provided in terms of statistical averages for whole countries in spite of potential regional dispersion, e.g. “the US unemployment rate is at 5%” or “Italian inflation is below 2%”, aggregate measures that may gloss over important social and geographical variation. If geographical variation was not relevant, the statistical average of any indicator would be sufficiently informative to characterize the whole of the regional distribution. In this situation, the statistical distribution $p(x)$ over geographical space and time of the values of any indicator, namely x , could be roughly approximated to a single Gaussian distribution of mean μ and variance σ^2 :

$$p(x|\mu, \sigma) = \frac{1}{\sqrt{4\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{4\sigma^2}} \quad (3.1)$$

where the variance is related to how the indicator varies over time, that is, it will be sharply peaked if the indicator is stable or flat and smooth when it evolves greatly over time. However, for many indicators a single Gaussian distribution with a corresponding single average for a whole country may give an imprecise description of its actual regional dispersion. In this situation, the regional values of a national distribution may be clustered, or agglomerated, or concentrated, around K particular groups, subsets, or clusters, with corresponding averages μ_k and σ_k that may be very different from the μ and σ from equation (3.1):

$$p(x|\mu_1, \dots, \mu_K; \sigma_1, \dots, \sigma_K) = \sum_{k=1}^K \frac{\theta_k}{\sqrt{4\pi\sigma_k^2}} e^{-\frac{(x-\mu_k)^2}{4\sigma_k^2}} \quad (3.2)$$

where $\theta_1, \dots, \theta_K$ are the weights of each k th Gaussian component, for which $\sum_k \theta_k = 1$ holds (normalization). The algorithm considers both the possibility of equal (E) and unequal

Table 3.1: List of Indicators

id	description
Demographics (10)	
population	Population
pop.change	Population change
births	Live births
deaths	Deaths
fertility	Total fertility rate
expectancy	Life expectancy at birth
Economic (18)	
gdp	Regional gross domestic product (million EUR)
gdp.pps	Regional gross domestic product (million PPS)
gdp.pps.hab	Regional gross domestic product (PPS per inhabitant)
gva	Real growth rate of regional gross value added (GVA)
income	Disposable income of private households
primary.income	Primary income of private households
EU.income	Regional gross domestic product relative to EU average
employment.1564	Employment rate of the age group 15-64
employment.5564	Employment rate of the age group 55-64
employment.2064	Employment rate of the age group 20-64
Industry (By Sector, 196)	
unemployment	Unemployment rate
gva.nama	Gross value added at basic prices
gfcf	Gross fixed capital formation
comp	Compensation of employees
emp.pop	Employment (thousand persons)
emp.h	Employment (thousand hours worked)
household.income	Income of households
profit.share	Profit Share
labor.share	Labor Share
productivity.pop	Productivity (GVA per worker)
productivity.h	Productivity (GVA per hour)
KL	Capital-to-Labor Ratio (gfcf to comp)
Health (8)	
ulc	Unit labor cost
death.causes	All causes of death
cancer	Death due to cancer
ischaemic	Death due to ischaemic heart diseases
accidents	Death due to accidents
transport.accidents	Death due to transport accidents
doctors	Physicians or doctors
dentists	Dentists
hospital.beds	Available beds in hospitals

Table 3.2: List of Indicators (Continued)

id	description
Education (9)	
tertiary	Tertiary educational attainment
HR	Human resources in science and technology (HRST)
hightech	Employment in high-tech sectors
patent	Patent applications to the European patent office (EPO)
hightech.patent	High-tech patent applications to the European patent office (EPO)
RD	Intramural R&D expenditure (GERD)
researchers	Researchers, all sectors
Poverty (5)	
longterm	Long-term unemployment rate (12 months and more)
atriskofpoverty.rate	At-risk-of-poverty rate
severedeprivation	Severe material deprivation rate
atriskofpoverty.pop	People at risk of poverty or social exclusion
lowworkintensity	People living in households with very low work intensity
Tourism (5)	
tourist.nights	Nights spent at tourist accommodation establishments
tourist.establishments	Number of establishments and bed-places
Transport (6)	
mar.passenger	Maritime transport of passengers
mar.freight	Maritime transport of freight
air.passenger	Air transport of passengers
air.freight	Air transport of freight
rail	Rail network
motorway	Motorways network

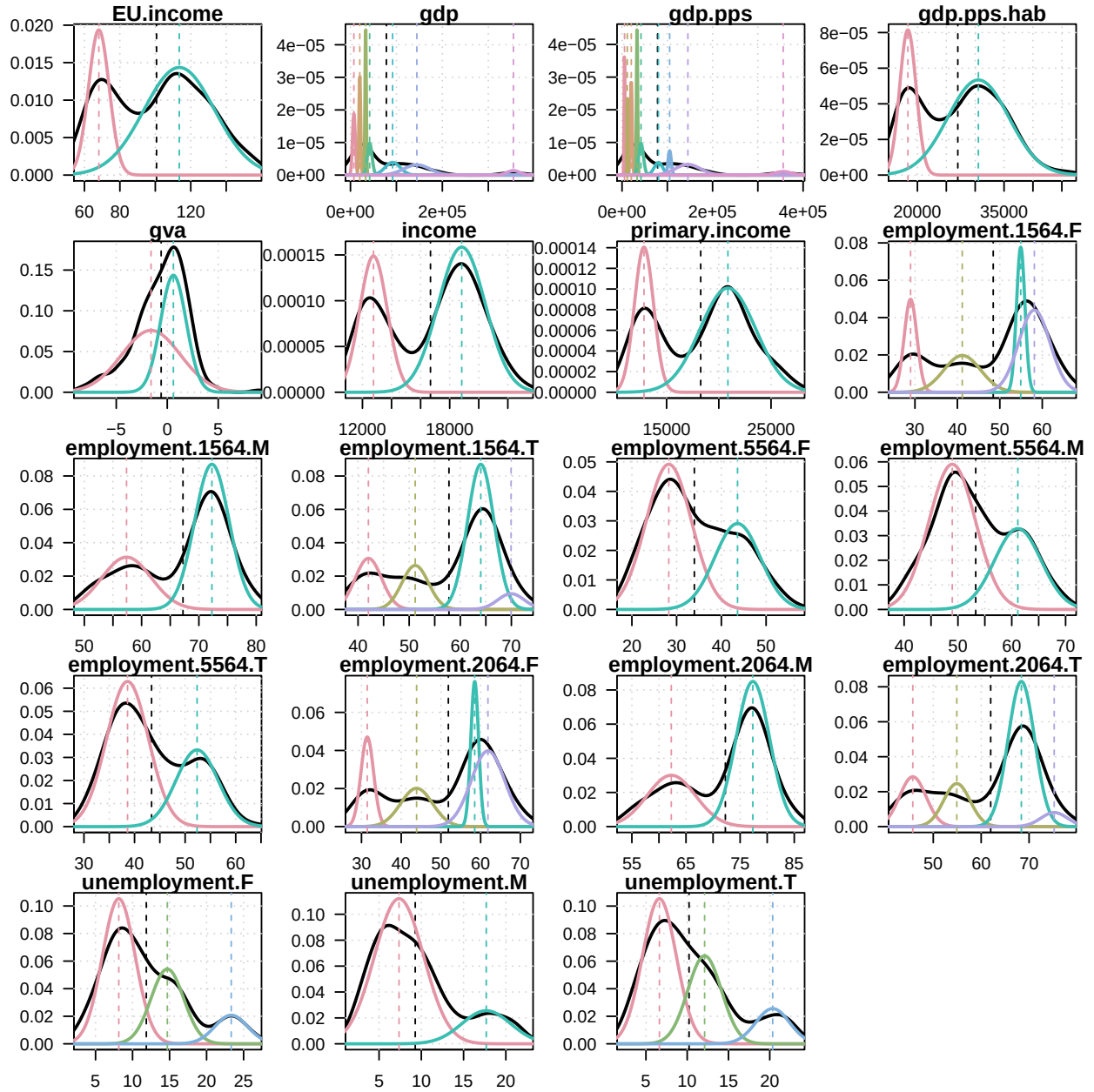


Figure 3.1: **Model-Based Clustering for Economic Indicators** Line in black depicts the distribution of regional values, colored lines depict the distributions of the distinct cluster components as found by the algorithm. Vertical lines indicate their corresponding averages.

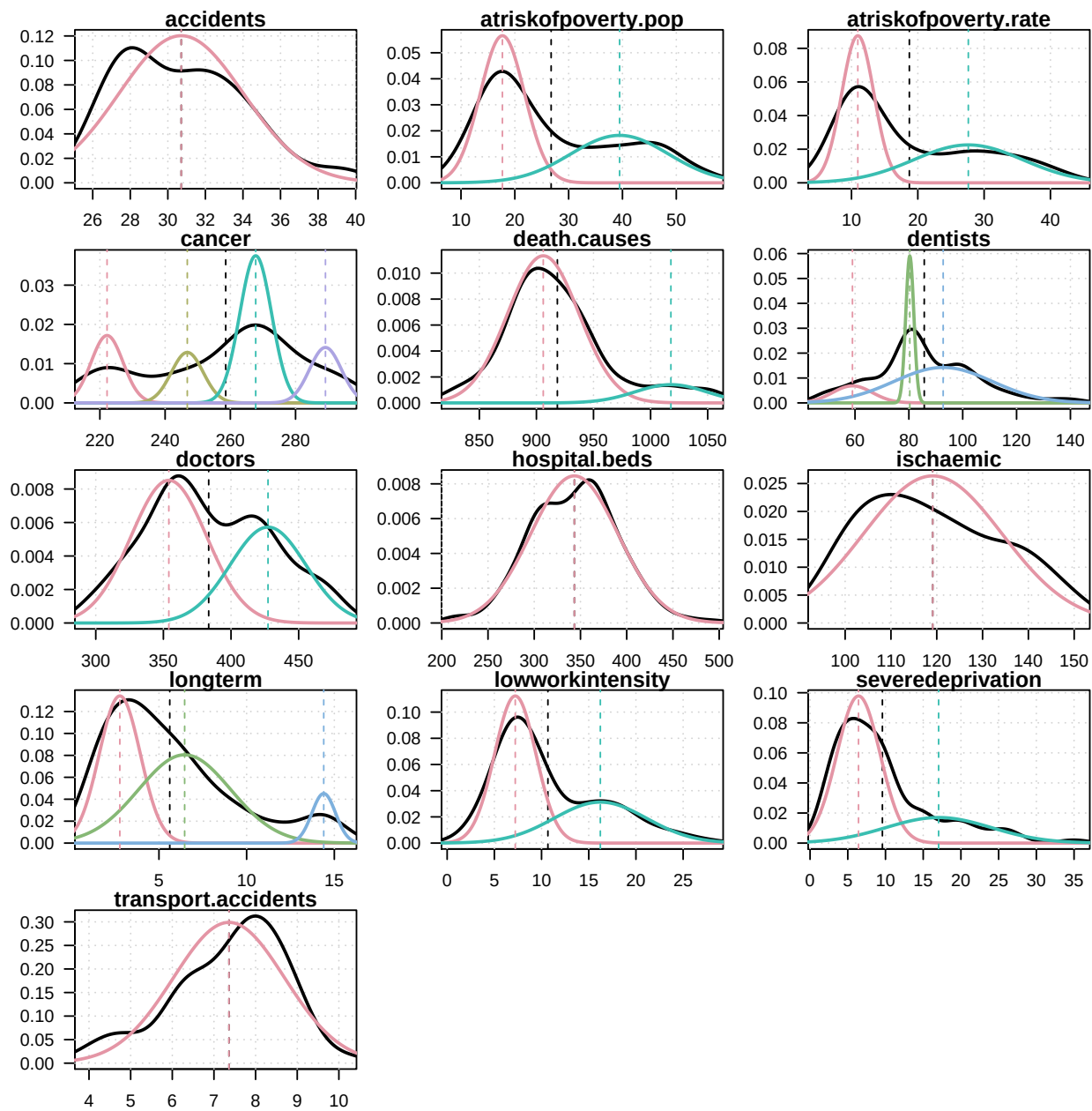


Figure 3.2: Model-Based Clustering for Health and Poverty Indicators

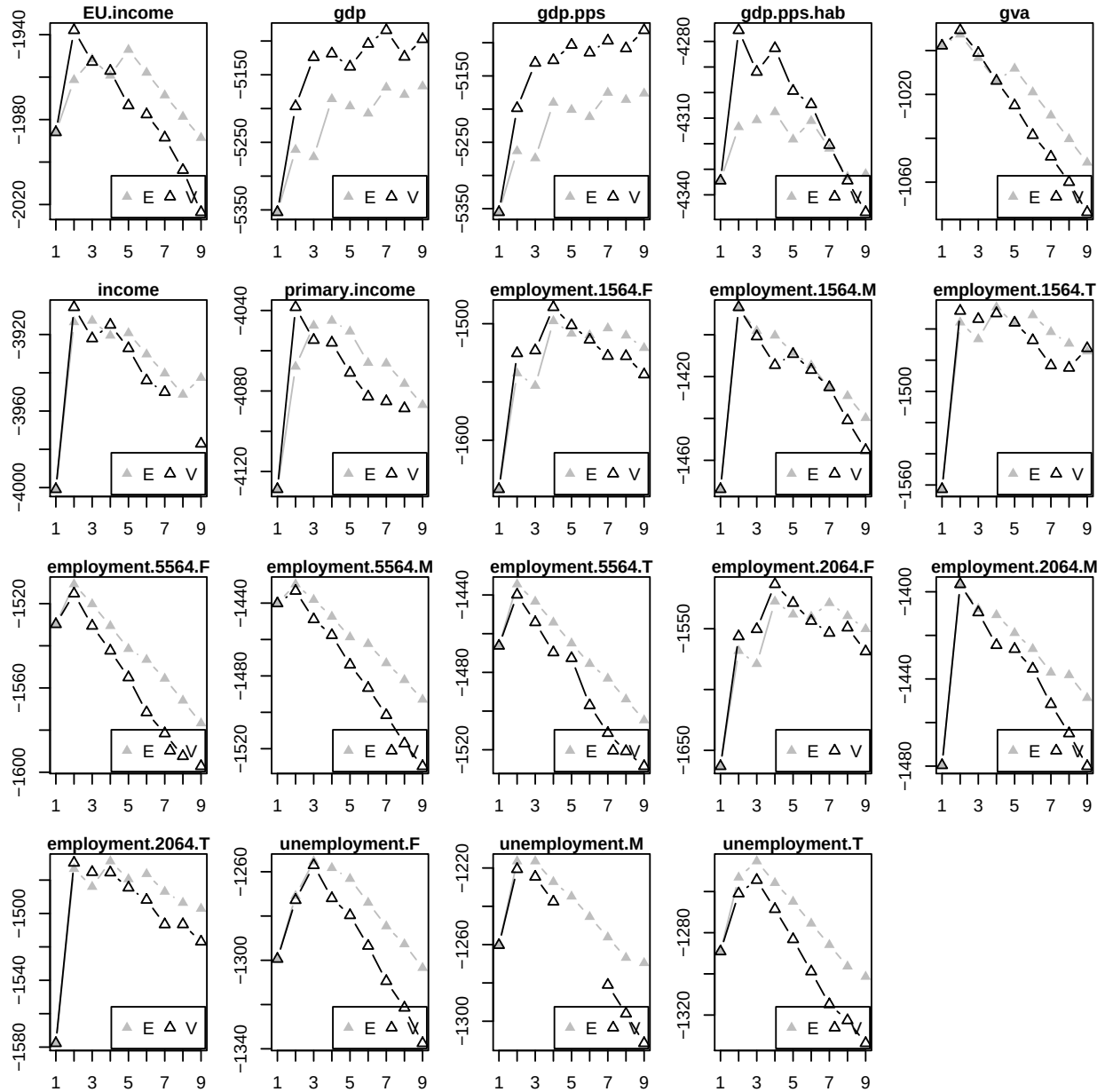


Figure 3.3: **BIC Score For Different Cluster Numbers, Economic Indicators** For most indicators, there is a stable peak between 2-4 components, which suggests a more complex distribution than strictly bi-modal.

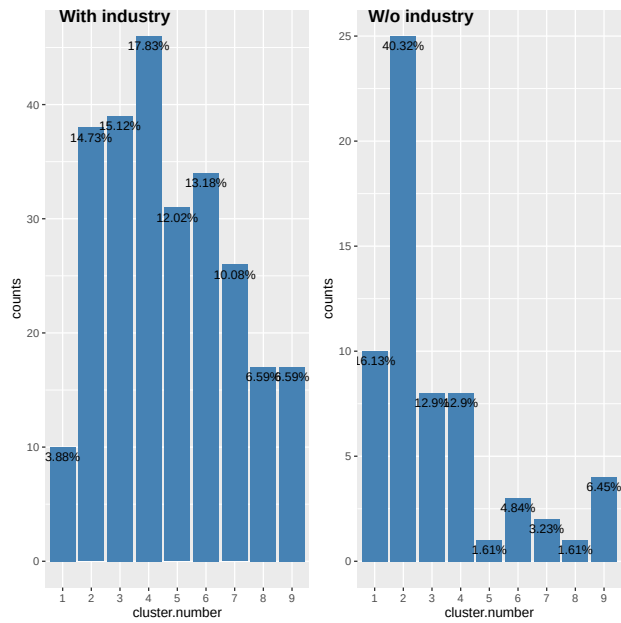


Figure 3.4: Geographical Variation: Optimal Number of Clusters

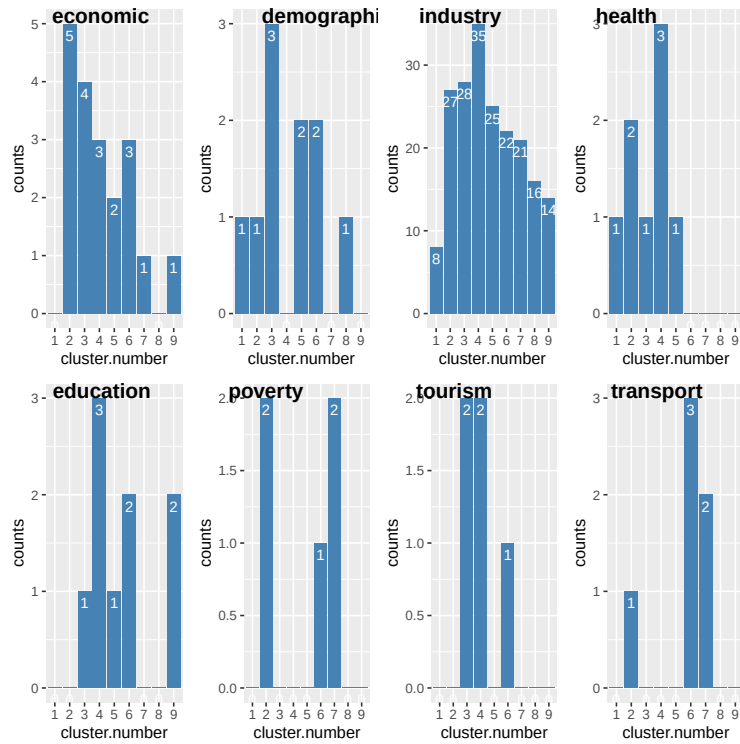


Figure 3.5: Optimal Number of Clusters

Table 3.3: **Table of Indicators with 1 as Optimal Cluster Number**

id	theme	description
accidents	health	Death due to accidents
fertility	demographics	Total fertility rate
hospital.beds	health	Available beds in hospitals
ischaemic	health	Death due to ischaemic heart diseases
transport.accidents	health	Death due to transport accidents
pop.change.natgrowrt	demographics	Population change, Natural
expectancy.M	demographics	Life expectancy at birth, Male
expectancy.T	demographics	Life expectancy at birth, Total
tertiary.F	education	Tertiary educational attainment, Female
tertiary.T	education	Tertiary educational attainment, Total

Table 3.4: **BIC Score Differences between Optimal and 1 Number of Clusters** The strength of the evidence against the model with the higher BIC value can be summarized as follows: 0-2, not significant; 2-6, positive; 6-10, strong; >10, very strong. [Schwarz, 1978]

	FALSE	TRUE
> -2	248	10
> -6	245	13
> -10	242	16

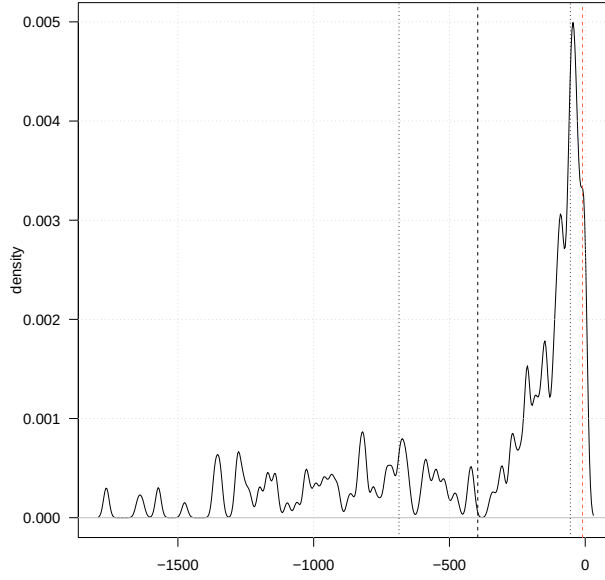


Figure 3.6: **Distribution of BIC Score Differences between Optimal and 1 Number of Clusters**

variances (V) for each component.

Finding the optimal number of clusters K in a systematic and quantifiable way, following the Bayesian Information Criterion (BIC, or Schwarz Information Criterion) is the purpose of Gaussian mixture modeling (GMM). The Bayesian Information Criterion is a criterion for model selection among a finite set of models that introduces a penalty term for the number of parameters in the model in order to avoid overfitting:

$$\text{BIC} = k \ln n - 2 \ln L \quad (3.3)$$

where L is the likelihood of the model, n is the number of observations, and k is the number of parameters [Schwarz, 1978]. Hence, equal-variance models enjoy of an advantage in the BIC score over unequal-variance models, since they use considerably less parameters, although they sacrifice explanatory power in the process.

This econometric technique is a model-based approach to clustering identification in the domain of unsupervised machine learning, whereby each observation is assumed to have arisen from an underlying mixture of a finite number of Gaussian distributions. The specific software package used is *mclust 5* [?]. The number of components in the mixture model (3.2) corresponds to the number of clusters to be imposed on the data [McLachlan and Peel, 2004]. Hence, this econometric technique may be very useful to identify existing geographical dispersion of indicators and properly inform policy to correct existing geographical imbalances within the social division of labor. Further, Gaussian mixture modeling helps characterize systematically the geographical core-periphery divisions within a national or international division of labor in the context of the literature on uneven development and

world-systems theory [Wallerstein, 1974; Arrighi and Drangel, 1986]. GMM also provides an alternative method to identify the existing convergence clubs found by the literature in economic convergence in a way that is fundamentally multi-dimensional. Finally, it also provides a systematic method of evaluation of Arnaldo Bagnasco’s hypothesis of a ‘Third Italy’ in the Italian context.

McLachlan and Basford [1988] highlighted the usefulness of mixture models as a way of providing an effective clustering of various data sets under a variety of experimental designs, which explains its wide applicability across the scientific disciplines¹ [McLachlan and Basford, 1988; McLachlan and Peel, 2004]. It was in the 1990s that the extent and the potential of the application of finite mixture models widened considerably — fields in which mixture models have been successfully applied include astronomy, biology, genetics, medicine, psychiatry, economics, engineering, finance, and marketing, among many other fields in the biological, physical, and social sciences [McLachlan and Peel, 2004]. More recently, mixtures of normal distributions have been increasingly applied in macro- and micro-economics (i.e. regime-switching models of economic time series, or analysis of dynamics of educational attainment), marketing science (structured representation of market information and forecasting of new product sales), and empirical finance (modeling stock returns, stochastic volatility models) [Khorunzhina and Richard, 2016]. Just before the advent of the euro, cluster analysis was applied to the European Monetary Union in the domain of optimum currency area theory to find a robust core-periphery structure in business cycle synchronization and real interest rate volatility [Artis and Zhang, 1997, 2002]. Hierarchical clustering has also been applied on the European Union to analyze industry structures, although only from a uni-dimensional, state-level perspective [Gräbner et al., 2020].

3.5 Statistical Analysis of the Italian Economic Geography

In particular, Gaussian mixture modeling allows to evaluate quantitatively Bagnasco’s hypothesis on the existence of a Third Italy beyond the stark dual polarization between an Italian North and South that is conventionally assumed. GMM is applied on a dataset of 257 Italian indicators extracted from the Eurostat database that are disaggregated regionally by NUTS2 regions within the time window of 10 years between 2008 and 2017, which amounts to 51,670 observations [tables 3.1 and 3.2]. Some of these indicators are also disaggregated by gender, while the indicators that belong to the theme ‘industry’ are disaggregated by industrial sectors following the standard ISIC classification.

When Gaussian mixture modeling is applied to each indicator separately, the BIC score reports 1 as optimal number of clusters for only 3.89% of them [table 3.3]. Excluding the 196 industry indicators, only 16.39% of the remaining 61 are best described by a single Gaussian

¹In labor economics, mixtures have been used to control for person-specific differences in models of dynamic discrete choice. Nonparametric maximum likelihood estimation has been used to study the effect of family background on educational achievement.

distribution [figures 3.4 and 3.5]. This result implies that presenting statistics in the form of national averages, at least in the case of Italy, fails to properly represent the whole regional distribution due to geographical dispersion. Figure 3.1 shows the different real distributions of the indicators belonging to the ‘economic’ theme and the estimated optimal superposition of Gaussian distributions, including their corresponding averages: national averages, depicted with horizontal black lines, systematically diverge from the cluster averages, depicted with colored lines. Excluding industry, the majority (39.34%) of the indicators are best described by a two-component Gaussian model, which supports the conventional classification of Italian regions between North and South. Since Italian unification dates from 1871, one would have expected to find convergence on the basis of conventional economic theory. Instead, these results show that economic divergence, not convergence, systematically characterizes the Italian division of labor, more in agreement with economic theories that emphasize growth clustering and uneven development. However, the three-cluster hypothesis often scores the second highest for most indicators in terms of the BIC score, much higher than the one-cluster hypothesis – hence suggesting that there is still empirical evidence for the existence of a “Third Italy”. For most indicators, there is a peak in BIC scores between 2 and 4 number of components, suggesting a more complex geographical structure than strictly bi-modal [figure 3.3]. The bi-modal/tri-modal dichotomy is evaluated for a particular indicator, GDP per capita relative to EU average (i.e. “EU.income”), in the next section.

3.5.1 GDP per capita relative to the EU average

Let’s examine one single indicator, GDP per capita relative to the EU average, in order to clarify what the econometric method of Gaussian mixture modeling does and what economic significance can be extracted from its results – especially by assessing whether its distribution is better approximated as bi-modal or tri-modal. Figure 3.7 shows both the indicator as a time series (left) and as a distribution (right). As a time series, it is not that clear that the different regional distributions are in fact clustered in two groups, which correspond to the classic dual polarization of Italy between North and South. When visualized as a distribution, its double-peaked character clearly stands out. Gaussian mixture modeling sheds further light on the rich underlying geographic structure of Italian income [figure 3.7]. Figure 3.8 shows the BIC scores for models of different number of components (from 1 to 9) with equal (E) and unequal (V) variance. The optimal number of clusters is V2, with unequal variance. The second optimal model is E5 (5 components with equal variance) and, with a lower BIC score, both 3-component models with equal and unequal variance. The different models with the highest four BIC scores are visualized in figure 3.9.

Using the Eurostat data from 2004 to 2015, when sorting the GDP per capita data for the regions into two clusters, the Third Italy is robustly identified as North. When dividing Italy into three economic clusters on the basis of GDP per capita for the period ending 2015, one gets a very different picture from that proposed by Bagnasco forty years ago, underlying its political and economic evolution in relation to the European division of labor: Piemonte and Liguria fall out of the northern cluster and into the Third Italy, while Lazio, Trentino-Alto Adige, and Emilia-Romagna (to a lesser degree) take up residence in the northern cluster.

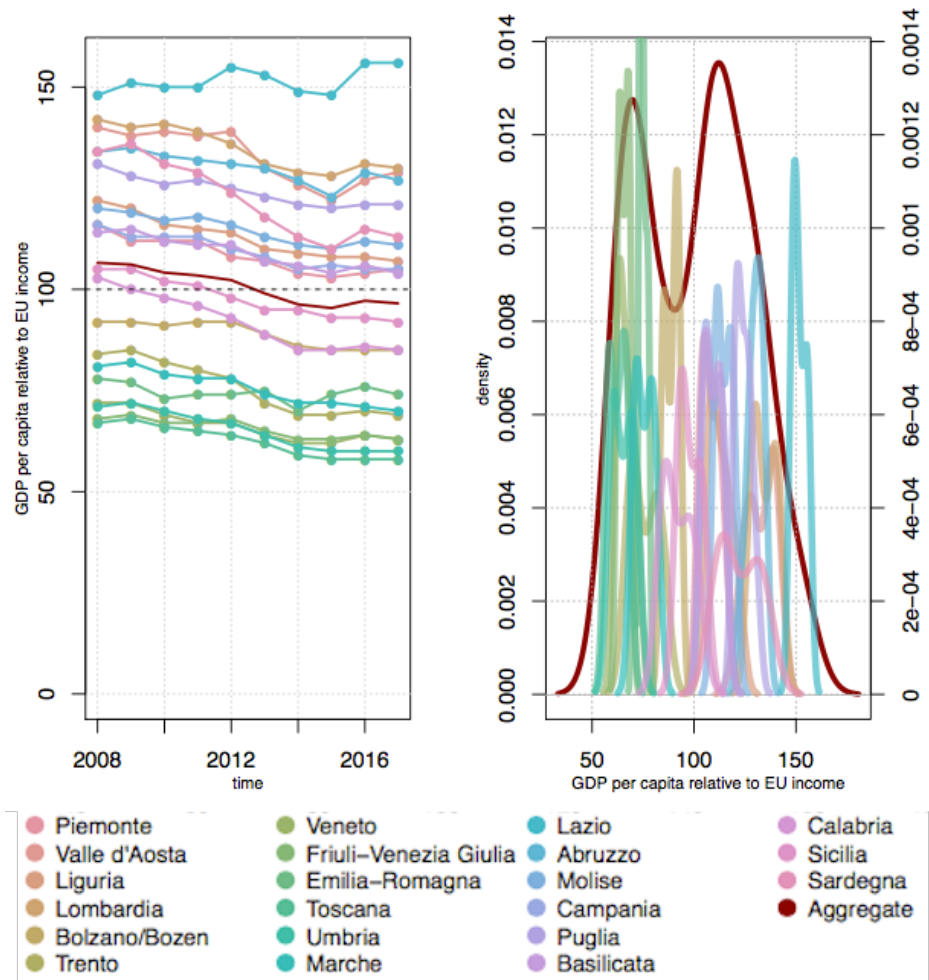


Figure 3.7: **Time Series of Italian Regional Income Data (EU Average = 100; left) and its corresponding frequency distributions (right)** For the right panel, the left axis indicates the density of the aggregate distribution, while the right axis shows the density of the different regional distributions.

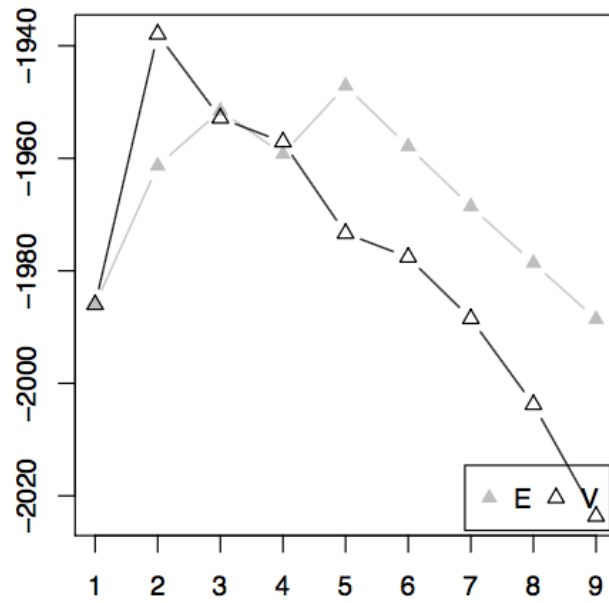


Figure 3.8: **BIC Scores for Gaussian Models (of GDP per capita relative to the EU average) of different number of components with Equal (E) and Unequal (V) variance** The optimal number of clusters is V2, with unequal variance. The second optimal model is E5 (5 components with equal variance) and, with a lower BIC score, both 3-component models with equal and unequal variance.

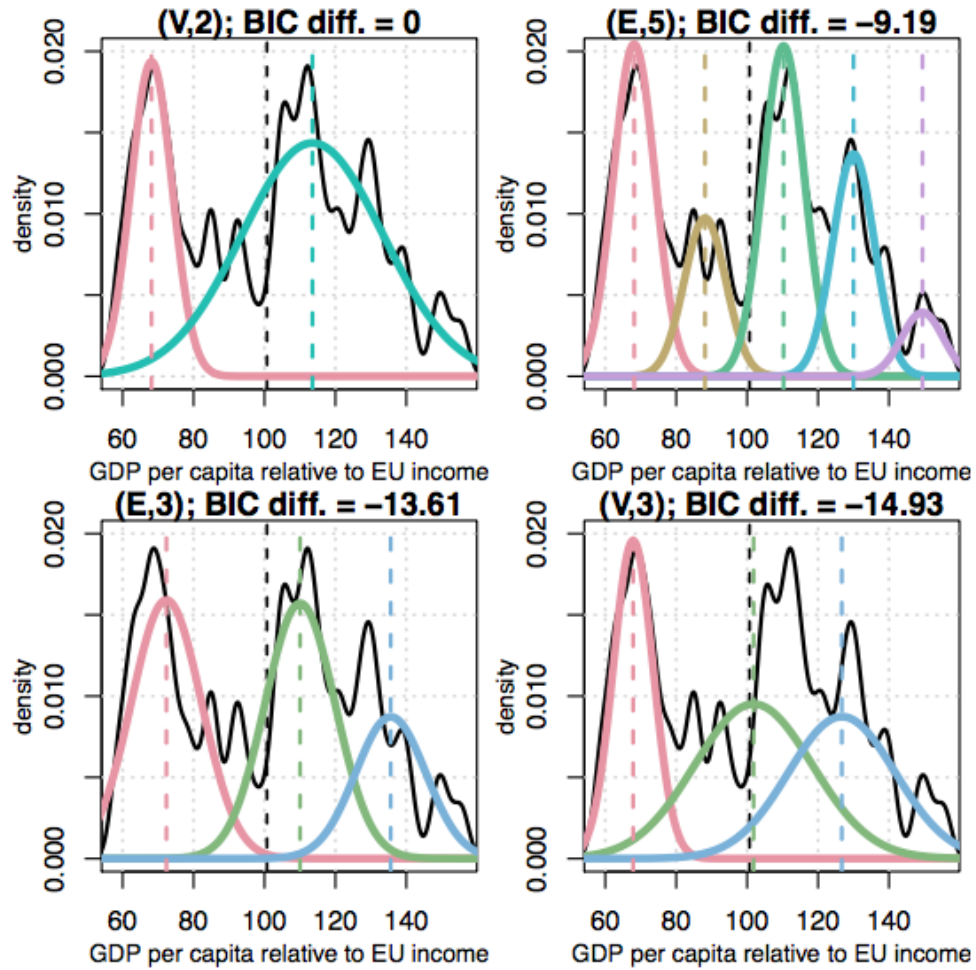


Figure 3.9: **Different Gaussian mixture models for GDP per capita relative to the EU average, with their corresponding BIC scores** The optimal number of clusters is 2, with unequal variance. The vertical dashed lines indicate the corresponding component averages, which systematically diverge from the total average.

The three-cluster classification allows a more nuanced understanding of the peripheralization of Italy, bearing in mind it is a complex process operating on many multiple dimensions affecting each region differently, yet following identifiable structural patterns. The Abruzzo, beginning in 2010, joins the Third Italy, less for its great strides than for its relatively stable GDP per capita during the general decline.

Piemonte, home of Fiat Automobiles, and Liguria, once part of the historical industrial triangle, now find themselves clustered with the “Third Italy,” while Trentino-Alto Adige and Emilia-Romagna, once the crown jewel of the Third Italy, now find themselves in good company with the wealthier northern cluster. This suggests the robustness of the economic fabric of Emilia-Romagna vis-a-vis other regions over a general dynamic of peripheralization. In agreement with Arrighi and Piselli, no specific economic organization has a privileged claim to economic success, but some economies are better at resisting secular demotion in the current macroeconomic environment. The key, as they stress, is the articulation of local economies with the demands and devices of a capitalist world-economy by definition more powerful than any particular local economy. Like Bagnasco, I find that Lazio is an anomaly for clustering purposes: though its GDP per capita would put it in the wealthier northern cluster, it might be revealing to examine internal differences in the capital region’s economy. Finally, the Mezzogiorno remains a solid and the poorest bloc, suffering neither regional desertions nor additions, save the borderline status of the Abruzzo.

The instability in both the northern and Third-Italy regional clusters may indicate that the three Italies schema upon which center-left economic development programs, have dined on for over 35 years may no longer obtain. Such programs, which went against the political constituency of a center-left in decline, had to be curbed under the Maastricht discipline of primary budget surpluses and inflation targeting that has drained the Italian economy of much-needed investment while contracting its consumption, a primary driver of Italian growth, typical of the European economies of the southern periphery. Instead, it seems the Italian core and periphery are being reshaped by a secular national economic decline within an evolving, interconnected European division of labor that is moving to the East through integrating its low-wage inputs into German industry. The former center-northeastern cluster, once an industrial competitor to Germany, is being pulled apart and pieces re-attached to either the classic Italian core or its own semi-periphery, while leaving its regional imbalances intact. Such economic dynamic happens to underlie the very political constituency of the former Italian government, a regional alliance between a North in demotion (the Lega Nord) and a South in stagnation (the 5-Star Movement), which is opposed to the contractionary economic plans of the European Commission.

Following the typical Polanyian double movement also under way in so many nations around the world, a far-right, populist government constitutes the political reaction of society to the social erosion mandated by the free-market forces of a neoliberal order in collapse. Even more than a Le Pen presidency in France, a Eurosceptic Lega Nord-Cinque Stelle majority in the Italian Parliament is one of the most probable pathways to the breakup of the euro in its current form, which may precipitate the greatest default in history. Maastricht discipline has imposed undue pain on Italy, which at moment and like many other European

Table 3.5: **Key Indicators on the Italian Economic Geography**

id	description
gva	Rate of Gross Value Added
gfcf	Gross fixed capital formation
labor	Labor Share (Compensation / GVA)
unemp	Unemployment Rate
income	Disposable Household Income
pr	Productivity (GVA / Employment Hours)

nations, requires better institutions at channeling and expanding investment to promote growth and employment appropriately over its regions.

3.5.2 Cluster Analysis of Key Indicators on the Italian Economic Geography

In this section, I explore the geographical and sectoral structure of the Italian economy by deploying the Gaussian mixture modeling approach. In particular, I study the regional structure of gross fixed capital formation as % of GDP, the wage share of income (employee compensation as % of GDP), the unemployment rate, the rate of gross value added (GVA), household disposable income, and labor productivity (GDP over labor-hours). Since employee compensation, gross value added, and employee hours are variables for which sectoral disaggregation is available in Eurostat, I evaluate the geographical and sectoral structure of the wage share of income (as % of GVA) and labor productivity (as GVA over labor-hours). In order to complete the picture of the Italian economic geography, capital intensity and the capital-to-labor ratio could be computed if regional estimates of capital stocks were available in the Eurostat database. Derbyshire, Gardiner, and Waights propose a simple procedure to compute such estimates using the perpetual-inventory method, which is improved by the corrections of Anwar Shaikh [Shaikh, 2016, pp.807-821].

The results are very interesting to the extent that they provide robust, rigorous evidence that confirms the already-known stark regional polarization of the Italian economy, although in general they provide a more accurate representation of the classical phenomenon of the spatial concentration and centralization of capital. At a geographical level, only the wage share of income can be approximated to a single Gaussian distribution (e.g. there is a single cluster of data), while unemployment, the rate of GVA, labor productivity, and household income can be approximated by two well-defined clusters in the form of Gaussian distributions. Further, gross fixed capital formation as % of GDP and the unemployment rate feature three clusters, which suggest that the hypothesis of a Third Italy may still be relevant after all.

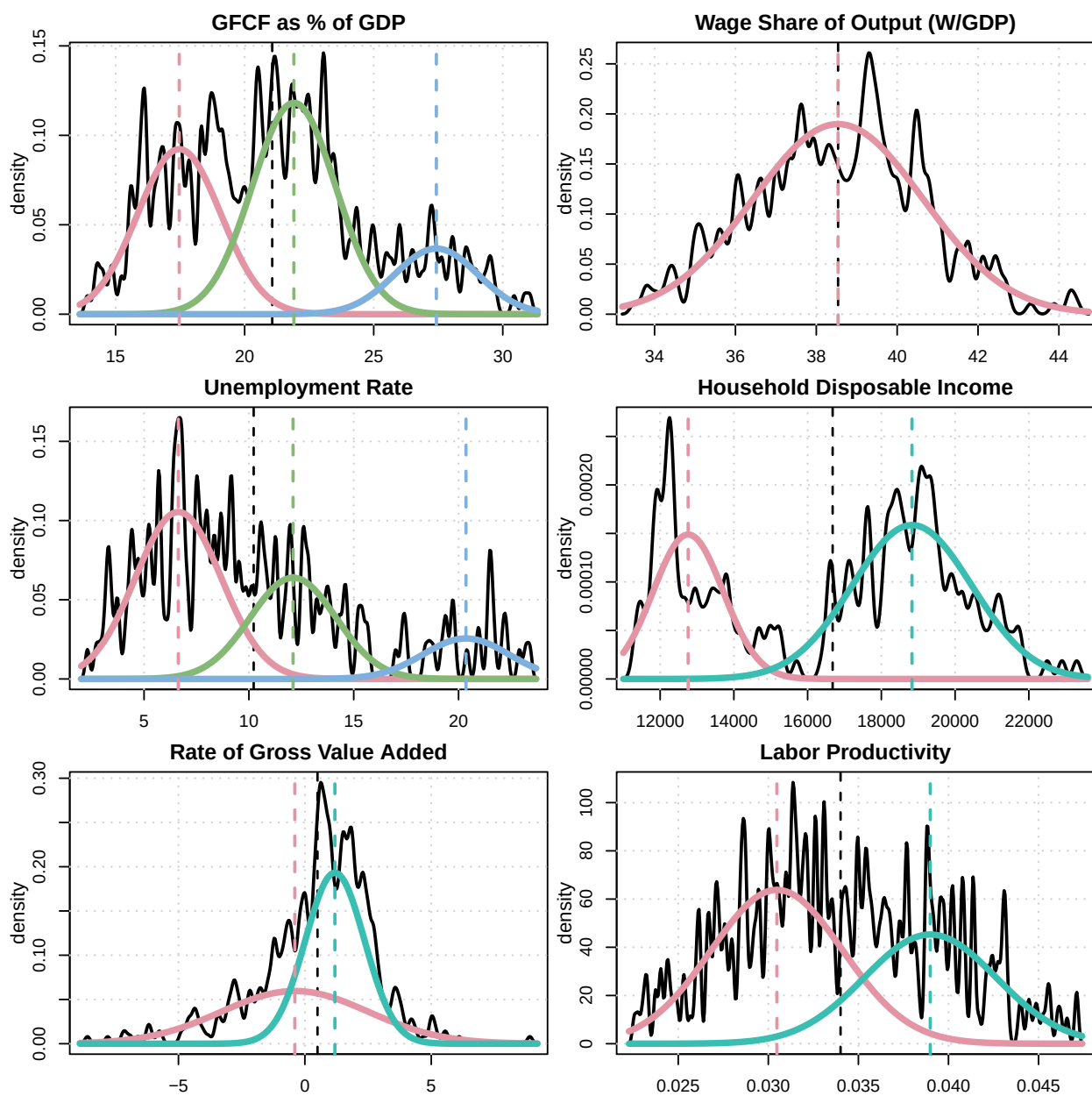


Figure 3.10: **Model-Based Clustering for Economic Indicators** The optimal number of clusters varies with the indicator from 1 (uni-modal distribution) to 3 (tri-modal distribution)

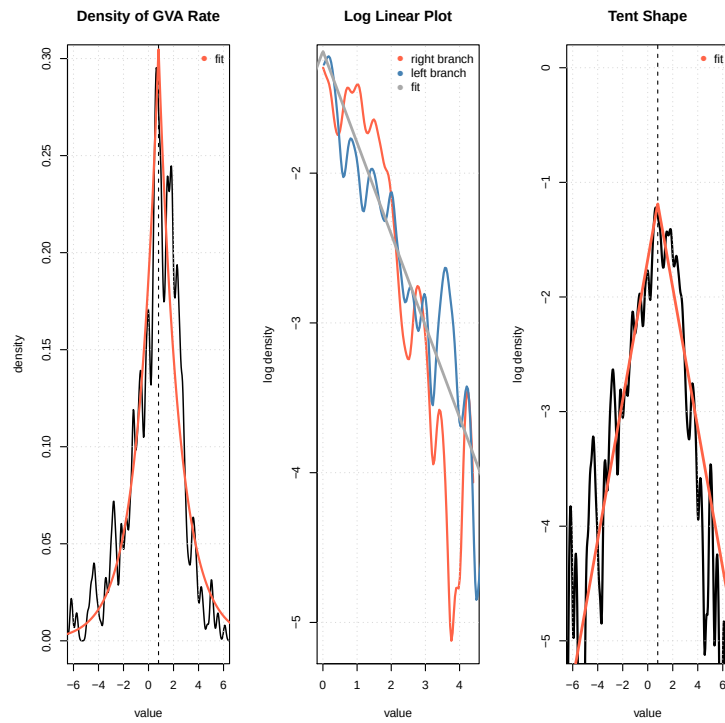


Figure 3.11: Laplace Distribution for Regional Rates of Aggregate Gross Value Added

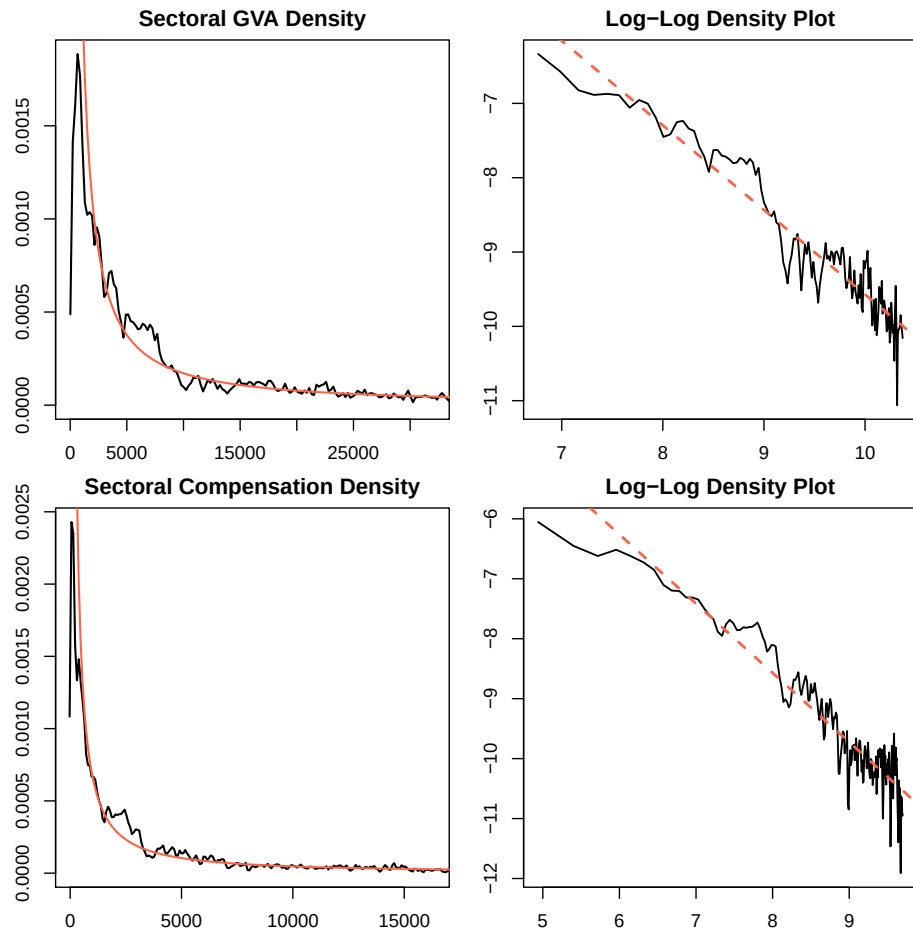


Figure 3.12: Power Law Distribution for Gross Value Added and Compensation

The rate of gross value added follows a Laplace or double-exponential distribution,

$$p(x|\mu, b) = \frac{1}{2b} \exp -\frac{|x - \mu|}{b} \quad (3.4)$$

where μ is the location parameter and b is the scale parameter. The maximum likelihood estimator of location is the median (which is also the mean), while the maximum likelihood estimator of the scale is the mean absolute deviation from the location,

$$b_{MLE} = \frac{1}{N} \sum_i |x_i - \mu| \quad (3.5)$$

The maximum likelihood estimation gives a very good fit of the distribution of the rate of gross value added.

The sectoral disaggregation of key economic indicators shows a more revealing picture of the statistical structure of the recent Italian economic geography. Both distributions of gross value added and compensation can be characterized by a power law, for which the BIC scores of different number of clusters seem to show a definite pattern,

$$p(x|k, A) = Ax^{-k} \quad (3.6)$$

Power-law distributions are ubiquitous around nature, in the form of occurrence of earthquakes, wars, city ranks, and word frequency (ie Zipf's Law); in economics, the observed Pareto distribution of income is a power law [Gabaix, 2009]. Power laws have the scale-free property in the sense that they are scale invariant and thus possess no meaningful scale [Barabási, 2009]. Given that both of the power exponents k of compensation and gross value added are less than 2, neither of these economic variables have a well-defined average and variance and thus show black-swan behavior. Like the top 1% of income in the Pareto power law, extreme events such as high and low GVA regional sectors are overrepresented in the power law in relation to Gaussian or even exponential distributions, further indicating the highly polarized character of the Italian division of labor, e.g. the existence of agglomeration economies.

Since compensation is a share of gross value added, it may be expected that both feature the same distributional functionals – gross profits, defined as gross value added minus compensation, also show a power law. However, the fact that sectoral gross value added follows a power law while its rate follows a double-exponential is quite striking. In the context of maximum entropy, power laws are associated with data-generating mechanisms that operate under the constraint on the average of the logarithm of the quantity in question – which in economics is related to a growth rate, ie the rate of gross value added. Laplace distributions are associated with the constraint on the average absolute deviation from the median, which is its maximum likelihood estimator for the scale of the distribution.

3.5.3 Regional Classification

In this section, I show the classification results from the application of the uni-variate GMM algorithm on the preceding economic indicators of the different regions of Italy, which yields a complex picture beyond the conventional dual classification between North and South. The GMM algorithm classifies each data-point of the time series for each regional value as belonging to a particular cluster. In order to classify a particular region, I average the classification results of its whole time series and thus the time dimension vanishes. While this often returns unambiguous results especially for the rate of gross value added, some regions can be classified in two different clusters, which adds complexity to the cluster analysis. For the variables that exhibit 3 clusters (gross fixed capital formation as % of GDP and the unemployment rate), the corresponding map shows the cluster that encompasses the maximum data points of each time series and thus is painted in a discrete scale. For the remaining indicators, the GMM algorithm yields a 2-cluster classification that can be depicted using a continuous scale, showing the percentage of data points of a particular regional time series that are classified as belonging to one of the clusters. For the sake of comparison, average values are also plotted in the adjacent map. Since the labor share of income is well represented by a single cluster, no classification analysis is required.

Out of all the economic variables, only the regional classification for disposable household income follows the conventional North-South division. Labor productivity follows a very similar classification to household income, but diverges for Toscana, Umbria, and Marche, which are classified along with the South. For the unemployment rate, the North cluster remains, while the South is divided into two regions. For the rate of gross value added, the time dimension becomes much more relevant: each year most regions belong to the cluster with a positive mean, except for 2000, 2008-2009, and 2012-14, years that were associated with recessions. However, its classification value from 0 to 1 still bears similarity to the classifications of income and labor productivity, which is expected.

3.5.4 Multi-Variate Cluster Analysis

Multi-variate cluster analysis can be performed in order to further understand the inter-relation of the economic indicators of the Italian geography. According to the Integrated Complete-data Likelihood (ICL) criterion, the EVE model (with ellipsoidal, equal volume and orientation) with 4 clusters is the second best performing after 9 clusters with ellipsoidal, equal volume, shape, and orientation (EEE). The information criteria tend to score better likelihoods for EEE models with a high number of clusters due to its low number of parameters, but they give very complex classification results to interpret.

For the 6 economic dimensions, we scale them for the sake of comparison and apply the multi-variate cluster analysis of an EVE model with 4 clusters. The results are plotted for the correlation of the economic indicators, the density of the clustering results, its classification, and a bar plot that assigns a maximum of 9 counts for each cluster (1 for each data point in the time series for which all indicators are available). Many of the correlation values are expected: labor productivity and disposable household income show the highest positive

correlation, while income and the unemployment rate show the highest negative correlation. While there is no correlation between income and investment, there is a negative correlation between unemployment rate and investment.

3.5.5 Sectoral Cluster Analysis

In order to shed further light on the geographical and sectoral structure of the Italian division of labor, cluster analysis is performed on sectoral wage shares of income (here defined as compensation over gross value added) and labor productivity (gross value added over labour hours).

The BIC scores for the sectoral labor share reveal a pointed two-layered structure that can be explained using 3 and 6 clusters grouped around varying averages. The 3-cluster classification clearly divides the Italian economy into 3 groups, irrespective of geographical location: cluster 1 only encompasses real estate (L), cluster 3 covers public administration (including education, health, and social work), and cluster 2 encompasses the rest of the economy. This can be explained due to the over- and undervaluation of real estate (a sector characterized by speculation) and public administration activities (a sector whose value added is difficult to estimate [Mazzucato, 2015]) in national economic accounts, respectively.

When disaggregating by 6 clusters, a more complex picture of the Italian economy arises, where cluster 2 is divided into 3 sectors: agriculture (A) and (partially) professional, scientific and technical activities (M-N) are grouped into cluster B; construction (F), wholesale and retail trade (G-J) and information and communication (J) belong to cluster C; and mining and quarrying (B), manufacturing (C), Electricity and Water and Gas Supply (E), and Financial and Insurance Activities (K) are grouped into cluster D. While real estate and public administration exhibit no regional variation, geographical structure is more clearly observed in the remaining clusters. For instance, the sectors of construction or manufacturing can be actually divided between clusters C and D; construction in the Italian South belongs to cluster C, while in the Italian North belongs to cluster D (with a higher labor share of income).

In terms of labor productivity (gross value added over labor hours), the best classification yields 4 clusters. Agriculture (A) and arts and entertainment (R-U) belong to the least productive cluster; information and communication (J) and financial and insurance activities (K) belong to the most productive cluster, while the remaining economic activities belong to the rest and show some geographical structure. For instance, manufacturing (C) or professional and scientific activities (M-N) are divided between North in the less productive cluster C while manufacturing in the South belongs to the more productive cluster D.

3.6 European Economic Integration

Italian economic decline since the introduction of the Euro in 2002 is by now proverbial, as highlighted by Stiglitz in his latest book on the common currency [Stiglitz, 2016]. Its economy has shrunk since the onset of the economic crisis by 12%, many of its national

corporate champions have been sold off to larger, more successful foreign firms, and even Fiat, the Italian marquee brand, has decamped to run its world operations among the Dutch. Its banks groan under the greatest percentage of bad loans, those gently referred to as *in sofferenza*, of any major northern country, the historic Monte Paschi di Siena teeters on insolvency, and the others are finding it very difficult to raise capital to surmount their sorry states [Blim and Vallès Codina, 2017]. Foreign investment has been all but non-existent for decades, though this trend has been interrupted to some extent by significant Chinese moves into ownership or participation in firms in Italian power, food, and machinery.

From a standard neoclassical exogenous growth framework, several studies have addressed Italy's decline [Parisi et al., 2006; Rabellotti et al., 2009; Ramazzotti, 2010; Bagnai, 2016], reaching a consensus on the fact that Italy's decline depends on a fall in labour productivity growth. Bagnai [2016] argues that these studies fail to provide a convincing explanation of the sudden stop of labour productivity growth experienced in Italy around 1996, after three decades of relatively good performance, and examines whether this fact can be explained by adopting the cumulative growth model of Kaldor, Dixon and Thirlwall. Such a stagnant economic scenario is the blueprint of a growing peripheralization of the Italian economy within the Eurozone [Cesaroni and De Santis, 2015; Gambarotto and Solari, 2015], which has triggered a strong populist backlash against EU economic integration and the common currency, led specially by the far-right *Lega*. According to my own calculations, a comparison of regional Lega results in the last two elections shows no correlation with regional income, but the highest correlation with *decreases* in income per capita. Further, decreases in income per capita are also very highly correlated with deaths by cancer (in an extreme way), deaths by ischaemic heart disease, and deaths in general, as well as tourist establishments.

What follows is a cursory exploration of regional inequality at a European level based on our analysis of the Italian economic geography of uneven development with the goal of applying Gaussian mixture modeling to detect its clusters in future research. The data shows that income divergence is a fundamentally regional phenomenon, not only at the state level of analysis, which is the exclusive focus of most studies on European uneven development [Storm and Naastepad, 2015; Regan, 2017; Gräbner et al., 2020]. In figure 3.21, the distributions of NUTS2 regional values are plotted based on a conventional classification of three major clusters of EU countries into core/North and its two distinct peripheries within the EU, South and the post-socialist East. At a first glance, there is no evidence of income convergence after three decades of the Maastricht Treaty, given how distinct the EU countries' regional distributions remain grouped by the three major divisions of North/Core, South and East; and how spread the EU aggregate distribution is. Within the European Monetary Union (i.e. excluding the East), there is no sign of convergence either. Instead, the statistical analysis at the regional level shows a marked economic geography of uneven development.

The dominant view on the crisis of the Eurozone is that current uneven development is driven by differentials in labor-cost competitiveness, which underlie the existing trade imbalances especially between Germany and the South and hence foreign indebtedness between surplus and deficit countries [Lapavistas, 2012; Sinn, 2014]. Thus, the mainstream solution is

a deflationary policy of ‘internal devaluation’ (i.e. cutting the wage share of income as much as 30%) in the deficit countries in order to restore labor-cost competitiveness [Sinn, 2014]. While in agreement with the fact that Southern competitiveness deteriorated, other authors dispute the mainstream view by suggesting that causality is in fact reversed, highlighting that it was the inflow of capital to Southern countries precipitated by the monetary union that relaxed their borrowing constraints and thus deteriorated their competitiveness. At the same time, they contend that German export-led prowess is not due to its lower labor costs, but rising foreign demand by fast-growing countries outside of the EU (especially China) [Gabrisch and Staehr, 2013; Storm and Naastepad, 2015]. However, it is important to note that the increase in German cost competitiveness does not necessarily need to be explained by the 2004 Hartz IV austerity reforms, but by the Eastern expansion of the European Union, which allowed it to repress its labor costs by outsourcing the manufacturing of its industry inputs to the low-wage post-socialist East, weaving global value chains in intermediate goods within the EU: an average of 25 percent of the total exports of the Czech Republic, Austria, Poland, Hungary, Slovenia, Slovakia, and Romania, with a high of 32 percent for the Czechs and a low of 19 percent for the Romanians, go straight to Germany. Of those totals, an average of 10 percent of these countries’ exports are in one category: parts and accessories of motor vehicles [Blyth, 2016].

This paper argues that both views are not necessarily incompatible but instead that they emphasize two aspects of the same phenomenon that are self-reinforcing: considering the role of trade in intermediate goods, the observed pattern of uneven development between European North and South can still be driven by absolute-cost advantage, in line with the classical theory of free trade [Shaikh, 1980]. Free trade will lead to persistent trade imbalances if there are structural differences in international competitiveness, while linkages between trade imbalances and international finance liquidity implies that persistent trade balances are compatible with balance of payments through countervailing short-term capital flows. In the context of the classical theory of free trade, the massive flow of capital from North to South in the first decade of the monetary union can still be explained as responding to the existing labor-cost differentials underlying the imbalances in trade and further deteriorating them in self-reinforcing, unsustainable fashion until an inexorable sovereign debt crisis comes up. Such debt crisis came in 2010-12, forcing Southern indebted countries into fiscal consolidation (i.e. austerity) and wage repression. This was already the intended outcome of the architecture of European monetary unification all along: by not building a supra-national fiscal authority to regulate fiscal transfers from surplus to deficit regions, the burden of the adjustment was to fall on wage repression in line with the interests of European capitalists [Rathgeb and Tassinari, 2020]. In this sense, one can consider the Maastricht Treaty as the European version of the neoliberal Washington Consensus.

Remarkably, the two-pronged Italian distribution of household income reveals that the Italian North is at income levels akin to the Core, while the Italian South is at the level of the South, which gives the South distribution of income the same two-pronged shape of Italy [figure 3.21]. Italy’s unification dates from the 1870s; no income convergence has been attained since under its own monetary union and no income convergence seems to

be expected within the European monetary union. In addition, in the last two decades the Italian North has been experiencing a protracted peripheralization, when before it was well integrated into the high-income “European banana” that ranges from Milan to London going through Austria, Germany, and the BENELUX. Such protracted peripheralization, explained by decades of fiscal consolidation, stagnant GDP and starved investment, underlies the current Euroskeptic political insurgency in Italy, where the wage share is already very low with respect to other Eurozone members. In this context, Italy is another loser in the process of [EU-led] globalization [Villaverde and Maza, 2011; Stiglitz, 2002; Williamson, 2005; Milanovic, 2012; Case and Deaton, 2020].

In figure 3.22, three maps are plotted based on a simple k-means classification of household income, the unemployment rate, and investment, which clearly shows the so-called “European banana” in terms of household income and the unemployment rate. In contrast, investment (in the European context of free mobility of capital) flooded Spain and Ireland in search of higher interest rates, fueling real estate booms at an unprecedented scale. One can note the high investment levels (in tandem with low unemployment) in Eastern Europe, product of German outsourcing of its manufacturing inputs, as well as very low levels of investment for the UK, which may help explain its recent support for Brexit. In absence of massive investment to correct its regional imbalances, the economic geography of the European Union will still be marred by protracted uneven development and will move towards disintegration, not integration [Gräbner et al., 2020]. Yet how such disintegration may occur?

Debates by European policy-makers on crisis management mimic the conundrum, on a regional dimension, detailed by Kalecki in ‘Political Aspects of Full Employment’ [Kalecki, 1943]. Often these debates are coached in terms of Northern versus Southern countries, but such framing serves to obfuscate the actual class war waged by European elites on its workers. In other words, trade wars –as they are occurring between North and South– are in fact class wars. At the European level, surplus Northern regions (where most European capitalists live, after all) play the role of capitalists in advocating for fiscal austerity, while deficit Southern regions play the role of the working class in their advocacy of fiscal expansion. As Kalecki pointed out, if working-class demands for fiscal expansion are not met by capitalists, those demands will be met by authoritarian Fascist leaders by providing employment through an expansion of the military-industrial complex, policing, and border security. As an advisor to Roosevelt noted while discussing 1930s deflationary European politics, “one either observes a surge in prices, or a surge in dictators”. By simple arithmetic, the Northern emphasis on a trade surplus strategy exports deflationary pressures and imposes systematic economic contraction not only on the European South, but also, on the rest of the world, especially the Global South. Following the Polanyian double movement, Neo-Fascist authoritarianism emerges as an alternative to elites to quell any political opposition to the subsequent wage repression that follows the economic contraction imposed by European capitalists. For instance, the 2011 Arab Spring, the 2016 coup in Turkey, or the 2016 coup in Brazil against Dilma Rousseff can be thought of as political responses to the trade war the European North has been waging against the rest of the world through the operation of the Single Market.

3.7 Conclusions

This paper presents a multi-dimensional, statistical characterization of Italian regional inequality and suggests that the observed “twin-peaks” polarization into North and South should be treated as the natural outcome of political and economic integration rather than an exceptional result. The machine-learning method of Gaussian mixture modeling for cluster identification can be very useful in order to characterize the economic geography of the European division of labor by distinguishing the regional elements that compose core and periphery, as well as to inform policy aimed at correcting its imbalances. Most studies on European uneven development focus on the convergence of state and regional time series in income per capita, which grossly underestimates the level of geographical unevenness in the economic development of Europe. I find evidence of a tent-shaped distribution for regional rates of gross value added, as well as evidence of power-law distributions in the distribution of sectoral compensation and gross value added. In line with the econophysics literature, the existence of such well-defined stable distributions indicates that robust economic processes are at play, rather than being exceptional outcomes of imperfect competition [Shaikh, 2020]. Instead, the statistical analysis gives evidence of economic competition on classical lines rather than neoclassical, driven by increasing returns to scale and in the process yielding the agglomeration economies that lie at the basis of uneven development.

The next step in this research is to implement the algorithm in a more systematic way for the whole of the Europe in order to provide further light on the statistical structure of its economic geography, and estimate regional capital stocks in order to add regional capital intensity and capital-to-labor ratio as relevant variables. Finally, a more rigorous analysis could be complemented by PCA analysis of the indicators in order to clarify the interaction between the bi-modal and tri-modal hypothesis.

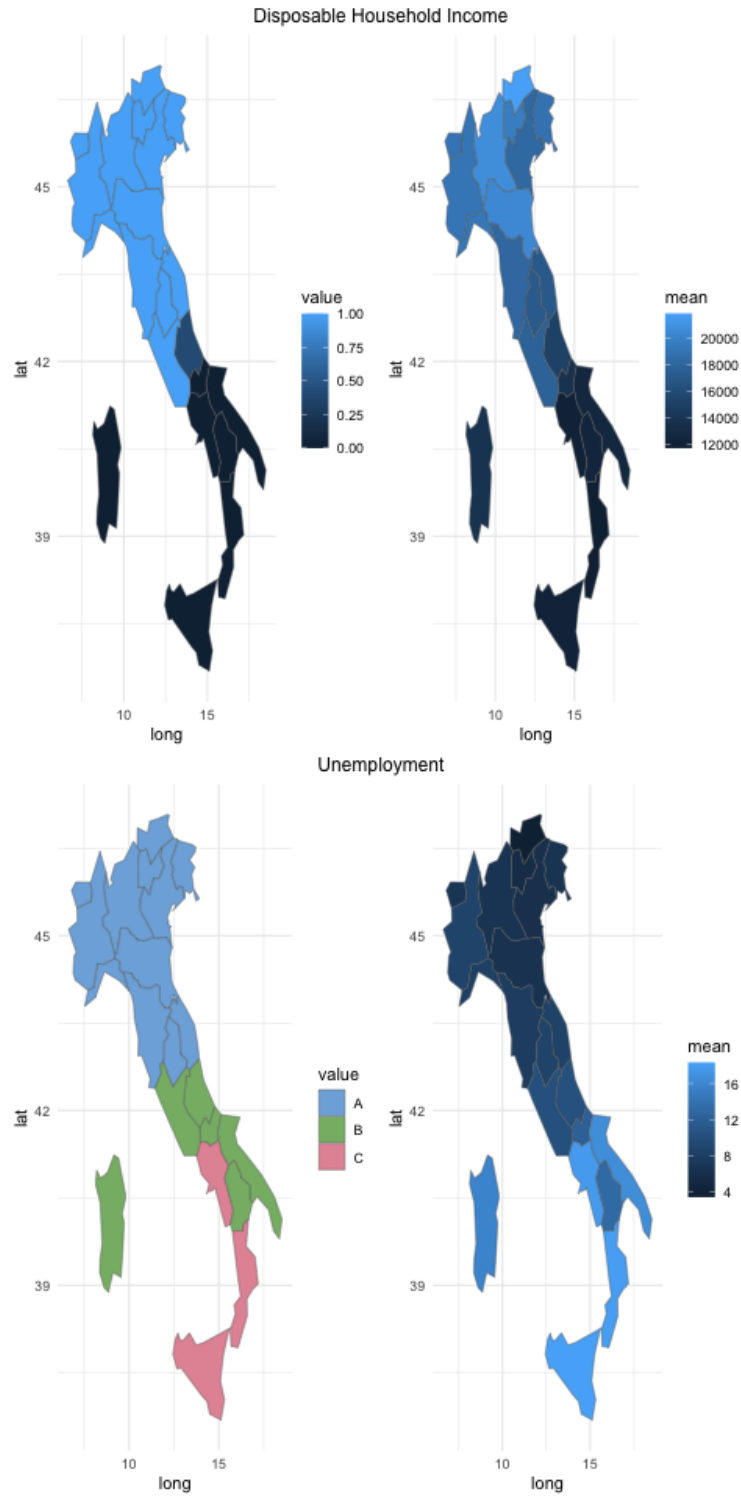


Figure 3.13: Household disposable income and unemployment rate, maps including cluster identification of each Italian region and their corresponding time average

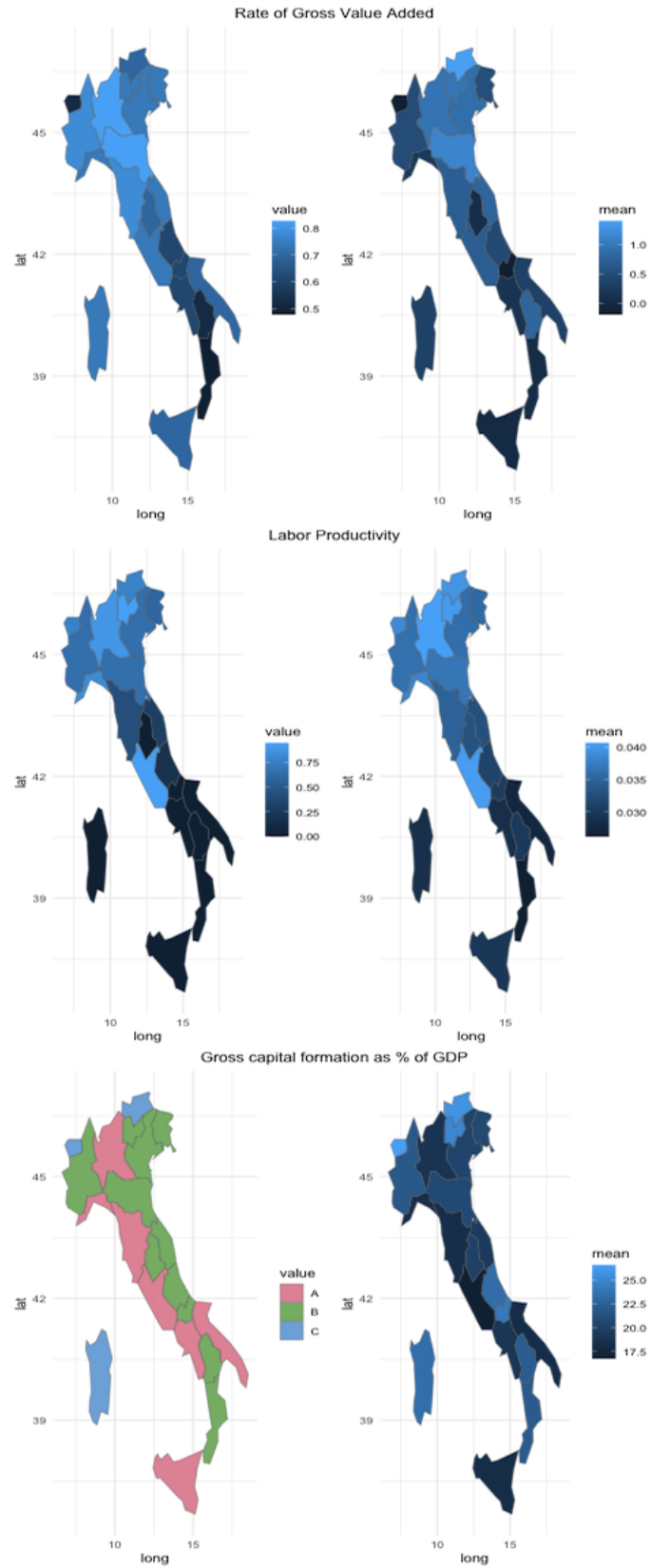


Figure 3.14: Rate of gross value added, labor productivity, and gross fixed capital formation as % of GDP, maps including cluster identification of each Italian region and their corresponding time average

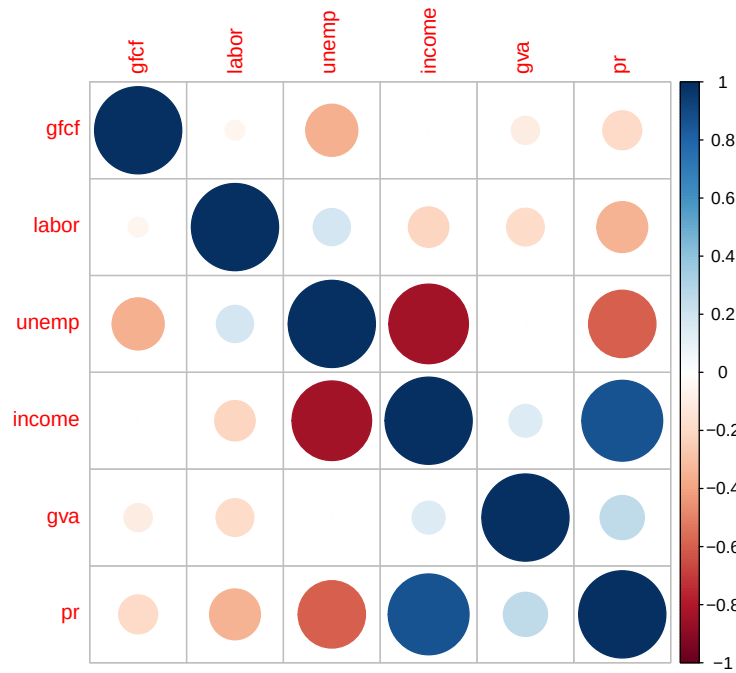


Figure 3.15: **Correlation between Economic Indicators** investment (gross fixed capital formation as percentage of GDP), labor share of income, unemployment rate, household disposable income, rate of gross value added, and labor productivity.

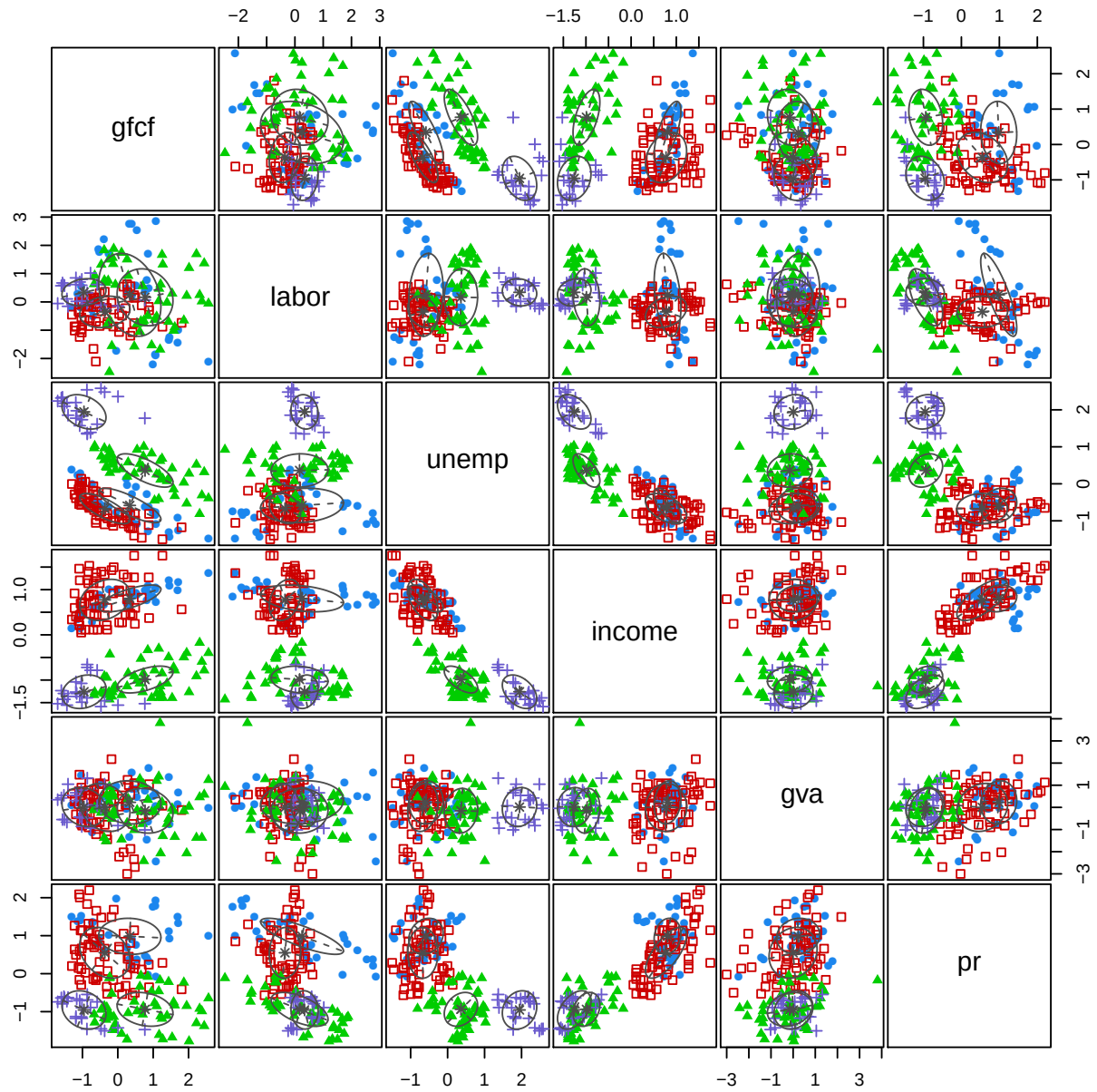


Figure 3.16: Multi-Variate Classification for Selected Indicators

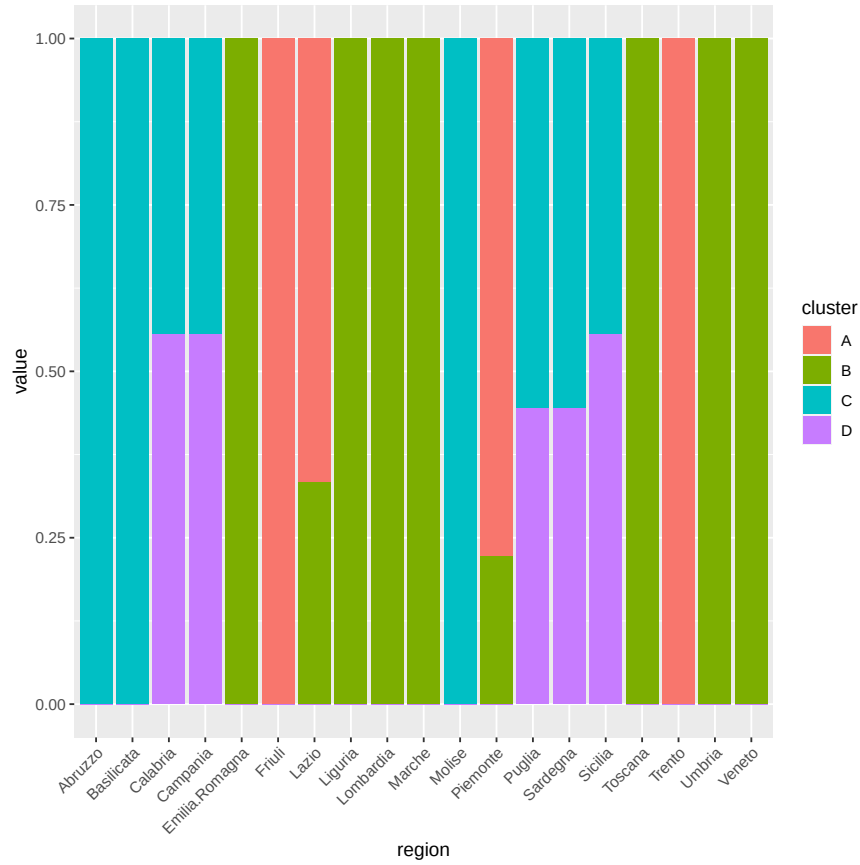


Figure 3.17: **Classification of Italian Regions into 4 Clusters** The classification is unambiguous (one cluster scores 1; the rest 0) for most regions analyzed (12 out of 19), while the remaining 7 occupy boundary regions between two clusters; Calabria, Campania, Puglia, Sardegna and Sicilia border C and D, while Piemonte and Lazio border A and B (the color scale is the same than in previous figure 3.16).

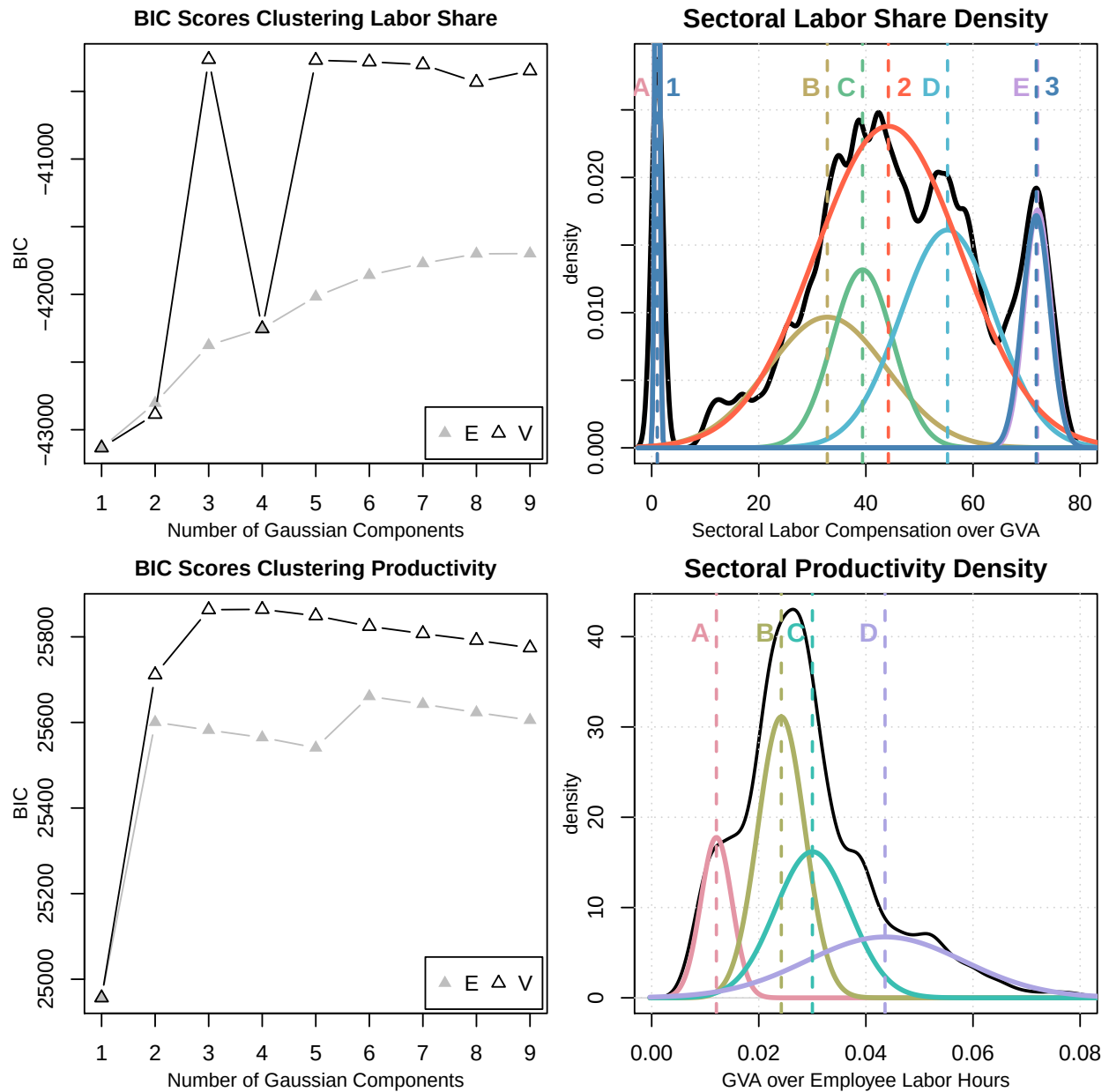


Figure 3.18: **Model-Based Clustering for Sectoral Indicators** labor shares (top) and labor productivity (bottom). The actual distribution is depicted in black. Sectoral labor shares can be either described by 3 or 5 clusters, while labor productivity can be accounted already by 3. In the top-right panel, numbers and letters refer to the 3- and 5-cluster classification, respectively. Vertical dashed lines indicate the corresponding averages.

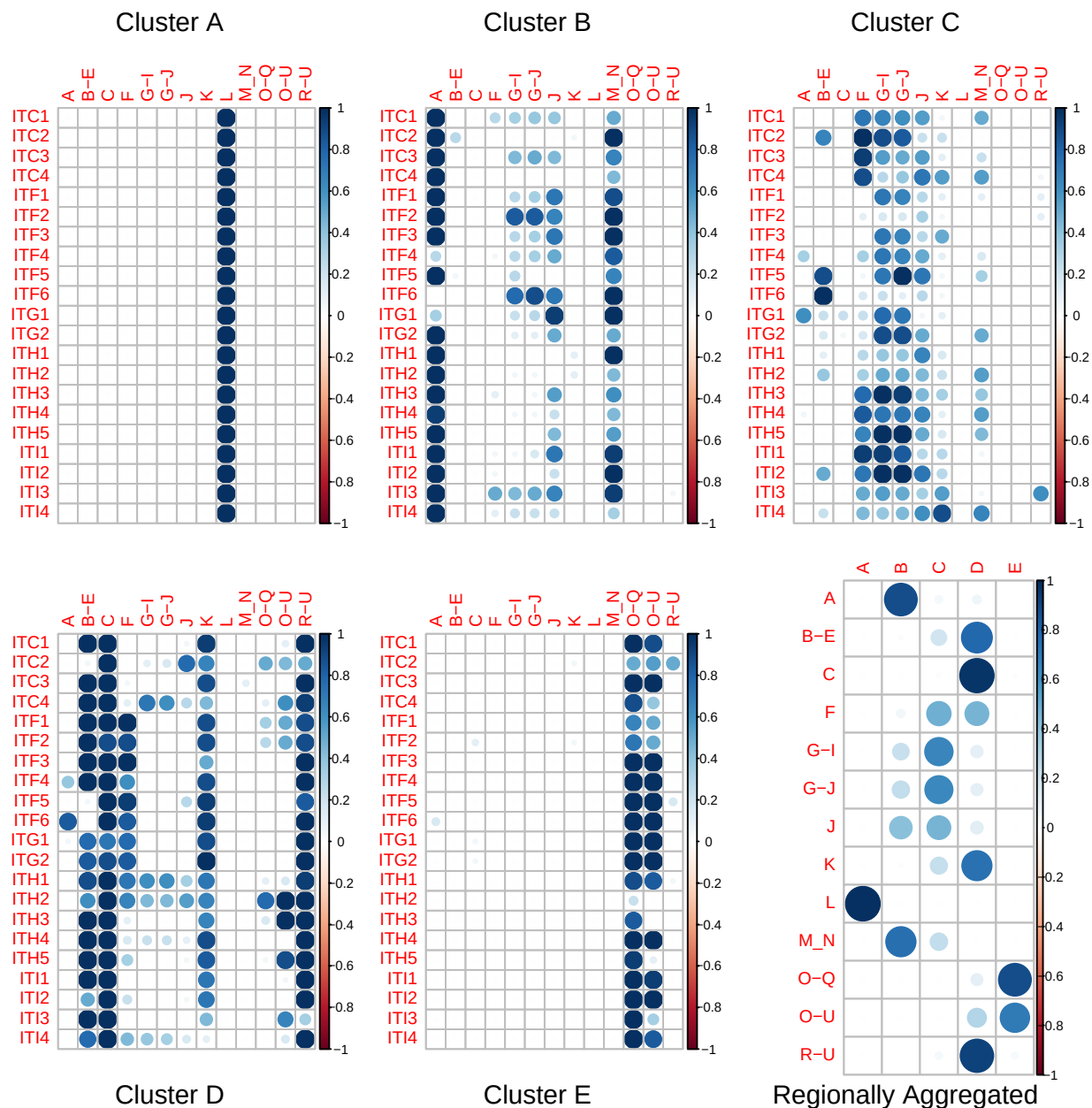


Figure 3.19: **Classification for Sectoral Labor Shares into 5 Clusters** Shade and size of the blue circle indicates the classification score of a particular region and sector into a particular cluster. The last panel shows the sectoral classification when values are regionally aggregated.

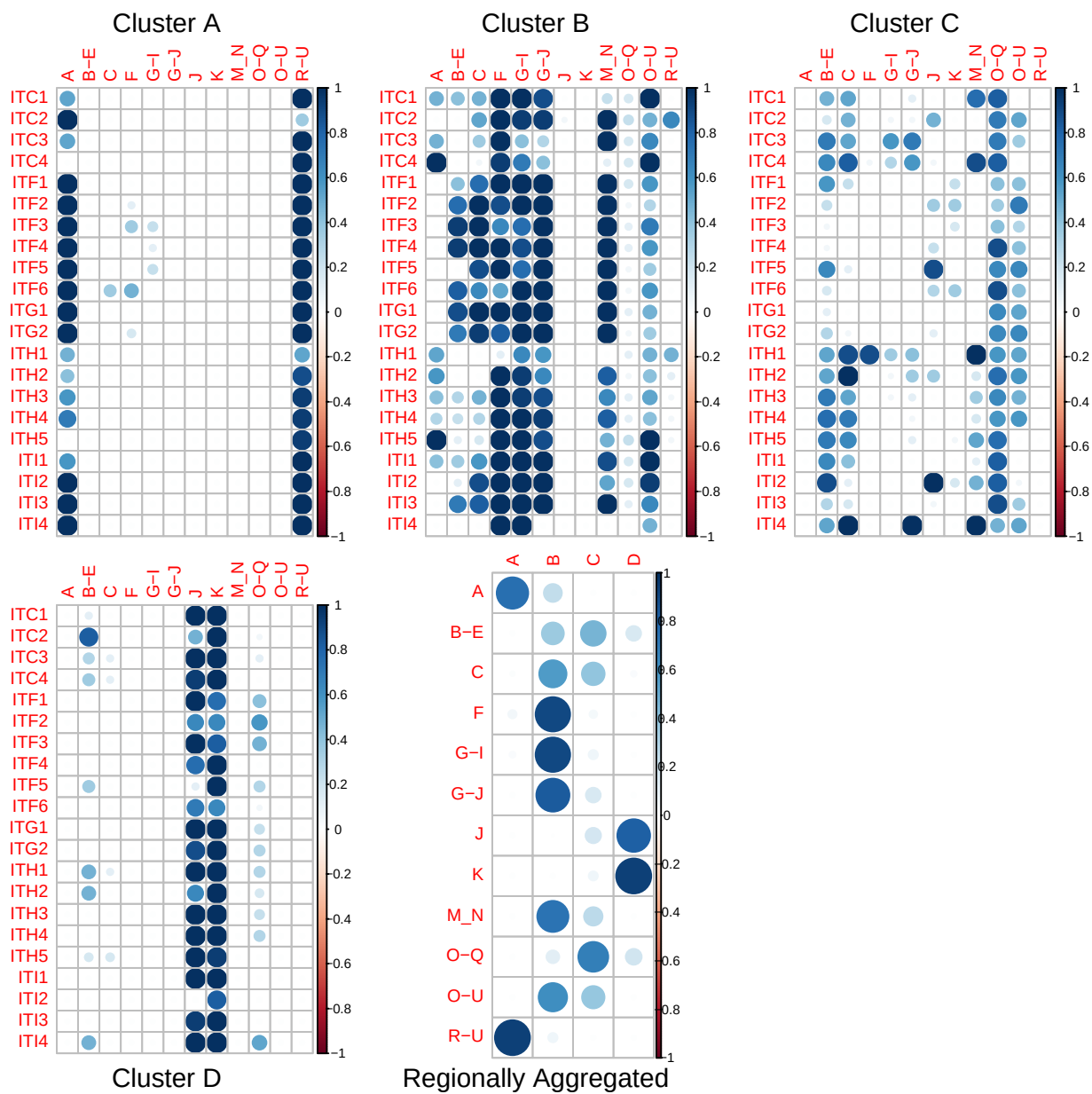


Figure 3.20: 4-Cluster Classification for Sectoral Labor Productivity

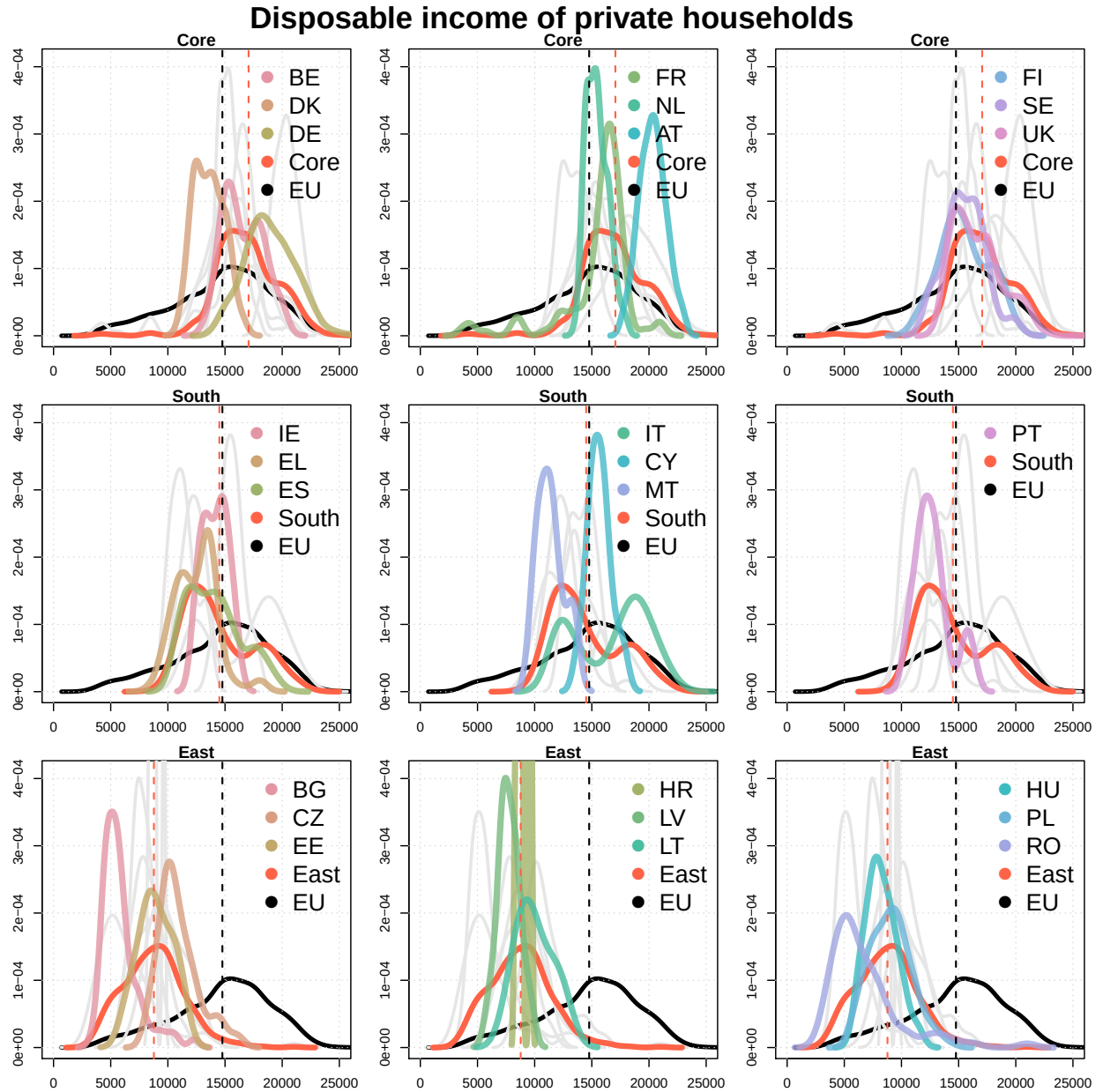


Figure 3.21: Distributions of NUTS2 regional values of household disposable income, by country within three major subgroups (Core/North, South, and East) the aggregate EU distribution is in black, subgroup distribution is in red, vertical black dashed line indicates EU average, red vertical dashed line indicates subgroup average.

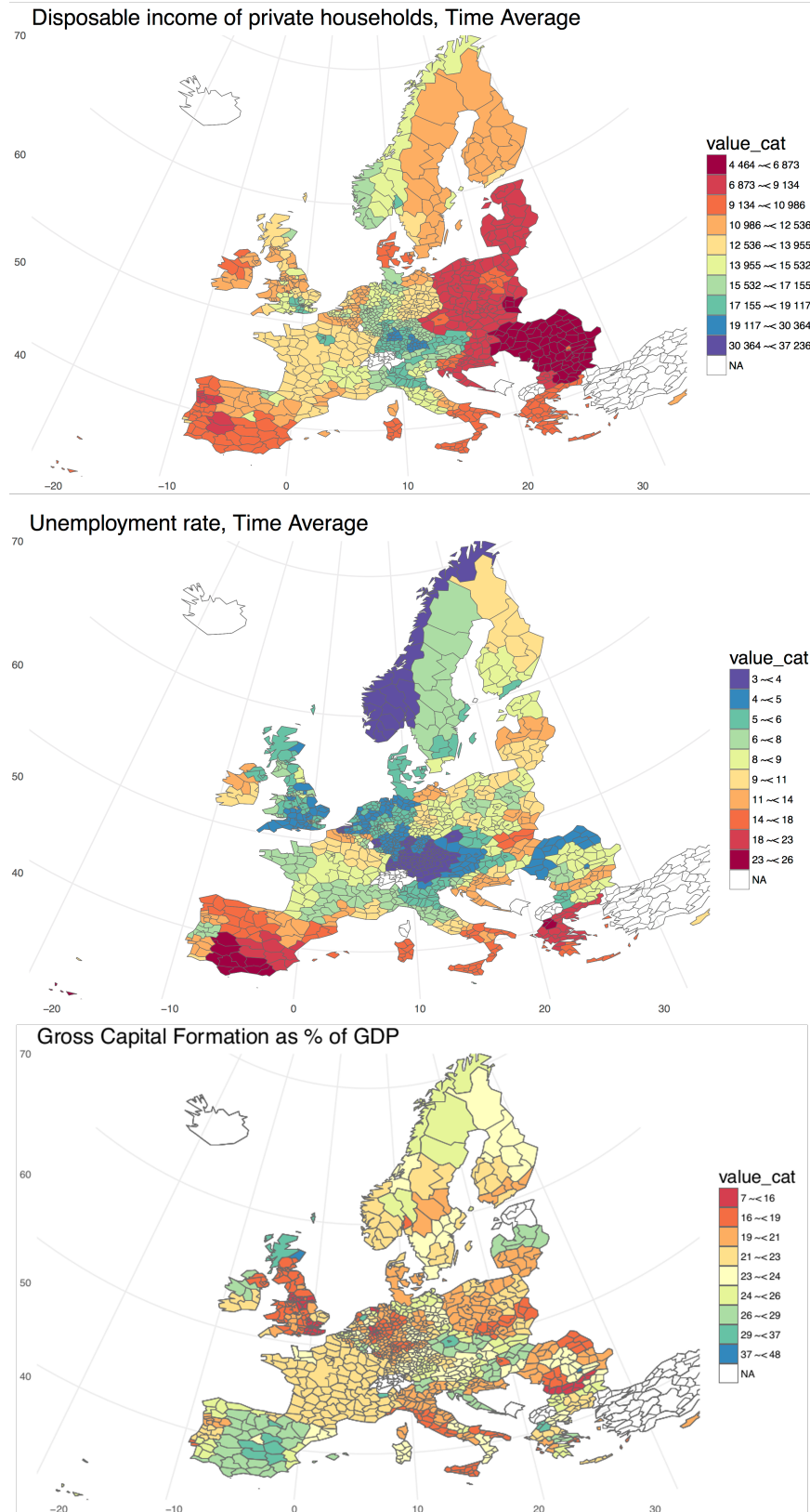


Figure 3.22: European maps with NUTS2 regions showing the time averages of household disposable income, unemployment rate, and gross fixed capital formation as % of GDP the clustering algorithm is simple K-means.

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